Did you know prompt engineering is one of the most in demand skills in the world of generative AI right now? Hey everyone and welcome to our prompt engineering full course by simpler. Now as AI becomes a bigger part of our lives. Being able to craft the right prompts is the key to unlock its full potential.

So, if you're just starting out or looking to take your skills to the next level, this course is designed to teach you everything you need to know from the basics to more advanced techniques. Throughout this course, you will get hands-on demo real examples that show you exactly how to apply what you're learning.

Now, by the end, you'll be able to harness the power of AI with confidence. Here's a quick look at what we'll cover. First, we'll have a look at introduction to prompt engineering. We'll kick things off with the basics and explain why prompt engineering is so important in the world of AI. Next, we'll have a look at the elements of prompt engineering.

We'll dive into the key components of prompt and see how they come together. Prompt patterns in chart GBT. We'll learn how to craft prompts that get the best results from Chad GBT. Advanced techniques and applications. We will take your skills further with some advanced techniques that will make your prompts more powerful.

Next, we'll understand large language models, what LLMs are, and how they shape the future of AI. You'll also learn how to fine-tune your prompts and use charging task. And then we'll check out GitHub copilot and some no code app builders, which can make your workflow much more smoother. And let's not forget agentic AI, multimodel prompting, and context engineering.

We'll get into advanced techniques and explore how to use context in your prompts. You'll also learn top tools like OpenAI Codeex, Land Chain and other tools for job

interviews. Along the way, you'll also understand MCP, search GPT, even Google Flow. And by the end of this course, you'll have the skills to apply prompt engineering to practically do anything.

So, let's jump in and start exploring the exciting world of generative AI. Now, before we move on, here's a quick quiz question for you. What is the primary purpose of prompt engineering? Your options are to write code for AI models, to create effective instructions that guide AI to produce desired outputs, to build AI models from scratch, or to manage AI data pipelines.

Let me know your answers in the comment section below. Also, if you are interested in mastering the future of technology, then the professional certificate course in generative AI and machine learning is the perfect opportunity for you. This is offered in collaboration with ENIC Academy IT Cardboard and it's an 11-month live interactive program which will provide you hands-on expertise in cutting edge areas like generative AI, machine learning and tools like chat GPT, DL2 and hugging face. You also gain practical experience through 15 plus projects, integrated labs and live master classes delivered by esteemed IT Kpur faculty alongside earning a prestigious certificate from IIT Kpur. You will receive official Microsoft badges for Azure AI courses and career support through Simply's Job Assist program. So what are you waiting for? Hurry up and enroll now and you can find the course link below. Hey there. Let's start with a quick thought. Have you ever tried to talk to a machine and it gave you the wrong answer. Maybe you asked Siri, "What's the weather today?"

and you got something like, "I'm sorry, I didn't catch that." Or even worse, it gave you the wrong city's weather forecast. This might sound frustrating right now. What if I told you that this happens because you didn't give it a good question? That's where prompt engineering comes in.

So, in simple terms, it's the art and science of crafting the right questions or prompts to get the best answers from AI. In today's video, we are going to cover the basics of prompt

engineering, which is not just about asking AI questions, but about asking them in the right way. And by the end of this course, you will understand what is prompt engineering.

I'll break it down for you. So, it's very super simple to understand why prompts matter. You'll see how much the quality of your prompt can change your AI's responses. We'll then understand how AI processes prompts. I'll explain you how AI understands your questions and it's easier than you think.

Then we'll understand tokenization. This might sound a bit technical but I'll make it an easy way to grasp. Don't worry, I have got you covered. Then we have context window and memory. So how does AI remember things? We will dive into this in a very simple way. The anatomy of good prompt. What makes a good prompt? We will break it down for you and show you what exactly to include. Weak versus strong prompt.

Now, small changes can definitely make a big difference and I'll show you how to do that. Now, after all of that, I'll show you a real life demo on how to use prompt engineering with chat GPD. You'll see how to apply everything we learned today in real world situations. So, are you ready to see prompt engineering in action? Let's get started now.

Meet David, a marketing manager at a growing e-commerce company. David's team is launching a new product and his task is to come up with catchy, attentiongrabbing ad copy for their digital ads. He decides to use an AI tool to help him generate some creative ideas. David types a very basic prompt into the AI.

He says, "Give me some ad copy for a new product." The AI generates a few generic lines like buy a product today and new product now available. After seeing this answer, David scratches his head because these lines are overly broad and it fails to capture the essence of the new product. He needs something more specific, something that resonates with their target audience.

So, David refineses his prompt. He types, "Write three creative ad copy ideas for a new eco-friendly water bottle targeted at young professionals who care about sustainability." Now, the AI gives David more targeted and unique ideas. It gives like stay hydrated, stay green, drink sustainably, work sustainably, eco-conscious, stylish, and practical.

Now, after seeing this, David smiled because this is exactly what he needed. By refining his prompt, he got the right tone, the right message, and the perfect ad copy. So, here's the lesson. Just like David, when you're working with AI, the more specific and detailed your prompt is, the better will be your results.

A vague prompt will give you vague results, but a well ststructured prompt leads to clear, relevant, and targeted response. Next, let's understand what exactly is prompt engineering. So prompt engineering is the process of designing and crafting the right questions or instructions for AI so that the AI can generate the most accurate and useful responses.

You can think of it as giving clear directions to a GPS. The clearer the destination is, the more accurate the route will be. Let's say for example instead of saying tell me about climate change you could ask explain climate change in three bullet points in simple language for high school students.

Now this level of specificity help the AI understand exactly what you want. So it can generate a response that fits your needs. So now that you know what prompt engineering is, let's see some real world applications of prompt engineering. So where can we actually apply prompt engineering? Here are a few examples. First one is customer support.

AI can be used to respond to customer queries. The more precise your prompt is, the better the AI can assist you. It also helps in coding. Now, developers use AI to generate code, debug or offer solutions to coding problems. Let's say for example, write Python code to sort a list of numbers in ascending order.

Now in education, teachers use AI to generate quiz questions, explain complex concept or even create lesson plans. Then we have marketing. Marketers like David use AI to generate ad copy, create social media content or even come up with product names. For example, create three catchy slogans for a new organic skincare product.

It is also useful in data analytics. Now AI can be used to interpret and summarize large data sets. It can automate your reports. It can generate insights based on your structured data making data analysis faster and much more accurate. For example, you can say summarize these key insights from this report.

Now, the more specific you're with your prompt, the better the AI can assist you in these real world scenarios. All right. So, how does AI even understand your prompt? Well, it doesn't understand like we do. It basically predicts. AI doesn't truly understand your question. It just predicts the most likely next words based on what you have typed.

It works like your phone's predictive text. Just like when your phone guesses the next word in a sentence. In the same way AI guesses the best next word to keep the response flowing. It's not magic. It's pure pattern and mathematics. AI uses pattern it learned from tons of data during training.

No emotions, no reasoning, just super smart pattern matching. You can think of this like playing the game fill in the blanks. You're given a sentence with some words missing and you have to guess the missing words based on what you know. The more you practice, the better you get at filling those blanks.

In the same way, AI guesses the next word in a sentence based on all the word it has seen before, but without any true understanding. So the clearer your prompt is, the better the AI's guess will be. Now let's dive into the key concept tokenization. You can simply put tokenization which is the process of breaking down your sentence into smaller pieces or tokens.

Now these tokens could be whole words, parts of word or even punctuation marks. Basically the building blocks that the AI works with to help AI understand text better. It breaks down sentences into smaller pieces called tokens. A token could be a word, part of a word or even a punctuation. Let's say for example, if you type chat GP is amazing, the AI breaks it into these tokens.

Chat plus G plus PT plus is amazing. Understanding how AI tokenizes text is important because the more efficient the tokens are, the clearer the AI understanding will be. When writing prompts, keeping them concise and avoiding unnecessary words help AI generates better responses. AI doesn't see the full picture immediately, just smaller puzzle pieces which are the tokens and then it works to assemble those into response.

The better the pieces which are the tokens, the clearer the final picture will be. Next, we will understand context window and memory. Now, AI has a context window which acts like its working memory. The context window determines how much of the conversation the AI can remember at any given time. Now if you ask AI a long question and the context window gets full, it may forget parts of the earlier conversation.

Let's say for example in simple terms, AI has a memory box called the context window. For GPT3, this box can hold up only up to 8,000 tokens which are the words or parts of the word. Once it gets filled up, it starts forgetting earlier parts of the conversation which can affect its responses. GPT4 on the other hand has a larger memory box.

So it can hold more tokens and remember more details without even forgetting. The image here shows how GPT4 can handle longer conversations while fewer memory limits

compared to GPT3 but struggles to keep track when the conversation gets too long. Now we'll dive into the some important part which is understanding anatomy of a good prompt.

Now that you know how AI works, let's break down the anatomy of a good prompt. A well-constructed prompt is just like a recipe. It needs the right ingredients in the right order. And that recipe includes a clear task which is what do you want the AI to do? Your target audience who is the answer for. You can think of it like giving directions to a friend.

If you can just say go right, they'll have no idea where they being headed. But if you say take a right at the second stoplight and then a left at a coffee shop, you are being much more specific and now they will know exactly where to go. This is how you should approach your prompt. Clear, specific, and with all the right details. Then we have context or role.

That is what context should the AI take into account. Formatting or constants. How do you want the answer to be formatted? So this is the anatomy of a good prompt. Now let's understand the difference between weak and strong prompt. So at first we have got weak prompt. For example, tell me about AI. Now this can be vague which is it doesn't specify what aspect of AI is needed. It's too broad.

It AI can't give a general unclear response. It's also unfocused because no guidance is given on how to explain AI. Now let's understand the example of a strong prompt. Let's suppose you type in three bullet points explain AI to a 10 yearear-old in simple terms. Now this tells AI exactly what to do which is explaining in three points.

It's also audience focused which is it tells AI who it's explaining to which is 10year-old kid. It requests a specific format which is in bullet points. Now we can see from here that strong prompt is much more effective because it gives clear direction to the AI making its response more useful and focused.

You can think of this like asking someone to explain a movie to you. If you just say explain it, you might get a full analysis of the film which might be way more detailed than you need. But if you ask, can you explain this movie to a 5-year-old? You'll get a much simpler childfriendly summary.

That's the power of strong prompting. Now I'll show you a demo using charge GBT. The difference between weak prompt and strong prompt. So now let's start with the demo part. I'll show you a clear difference of how you use the vague prompt and how we can improve it using a specific prompt. Let's say my first prompt is very normal.

Uh so I type something like uh tell me about the climate change. So this is my prompt to GPT and you see what type of answer will it give to me. So yeah so you can see here it has given me this particular set of answer. It says climate change refers to significant and lasting uh changes of the earth and the key causes the consequences and everything.

But uh I feel like uh this prompt is mostly broad and it's open-ended. The charge responds with a general explanation covering multiple aspects like the causes, effect and the solution. So while this information is accurate but still I feel that this is not tailored for my uh particular target audience or any purpose which makes it less useful if you have a specific goal in your mind.

So now I'll be giving a better improved version of my prompt. So here is my prompt the improved one and I have said GPT that explain the key causes of climate change in under 100 words using simple language suitable for high school students. So you can see that uh this here this model it gives a clear boundary a specific topic focusing on the number of words and a defined audience.

Now the response is maybe more shorter. Let's see look at the response. What type of response does Chad GPT give us? Yeah. So you can see that uh it has actually given me

the accurate the only information which is needed that to in under 100 words suitable for high school student. It might be more shorter direct and written in accessible language.

Now this version would work well for teaching purpose summarizing or even including in a school newsletter. Now we'll look at the second example which is the missing context versus clear instruction. So let me show you the first simple uh normal prompt. So my prompt is write a post about productivity.

Okay this is the my first prompt a normal prompt like how you guys usually do. So now let's see how what chat jeopardy generates. So you can see here that uh this is a very common kind of prompt that people give it to chat GBT. Now the model will try to fill in the blanks but since it doesn't know who the audience is and you know who are my target audience, what style should to use, it ends up writing something very uh bland and unfocused.

So now we'll be giving the second prompt which is the improved version. So here is my prompt. It says you're a productivity coach. Write a LinkedIn post on how to manage distractions while working from home. Keep the tone conversational limited to three short paragraphs.

So let's see what charge GPT generates for me. So you can see the answer that the GPT has given us. It's um accurate and also suitable for my target audience. So in this version we tell the model what persona are to adopt like who the audience is, what topic to cover and how to structure the response. So that level of instruction results is much more specific, engaging and also well organized post that's likely to perform better on LinkedIn or maybe your own personal blog as well.

So this was for a second example. I'll show you a few more couple of examples. So let's move on to a third example now. So third example is the flat request versus stepbystep prompting. So first we will be giving a very normal basic prompt like uh write a blog post about AI in education. Okay. So I've given my prompt here.

Write a blog post about AI in education. All right. So you can see here that u this is the answer which GPT has given to me. Now you can see that this prompt gives no direction on what angle to take, how long the post should be or how it should be structured. Now the model might generate a long unstructured post that tries to cover everything and it ends up doing nothing particularly well.

Now we will be giving a better improved version of this prompt. So now as you can see I've given chart GPT uh three step. The first step is to generate three blog title ideas about AI and education. The step two is pick the most engaging one and write a 200 word blog intro. The third step is suggesting an outline for my full blog.

Now this approach breaks the task into smaller manageable parts. It gives the model room to brainstorm and select the strongest direction before writing. So by structuring the task step by step, you're not just only getting better results but also maintaining creative control at every stage.

So we'll just give this uh prompt now. And you can see it has given me the answer for the first step two which is the chosen title. And you can see my title is also there. The blog intro is under 200 words and it has given me step by step the blog outline, introduction, the full story line. Yeah, it looks good to me.

So here you get a clear difference of how you convey your message to chat GPT AI. First, if you give a normal prompt like just a normal oneliner prompt, you see the answer here. And when you give a stepbystep detailed explanation of how you want the blog to be your outline, engaging 200 blog intro, the blog title ideas.

So this is how charges answer for you. Now the fourth example which we'll be looking on is the unfocused request and the difference between role based prompting. So I will give

a prompt like summarize this article. Now this is a very normal basic prompt. So u yeah you have to provide the article which you which needs to be summarized.

Yeah. So I have an article here of uh about talking about generative AI. So I've given this to charge GPT and I write here that summarize this article. Okay. Okay. Now it has given me a full summary of the article. You see it's a bit uh short. So it might look a bit clear but then you know it lacks details about what kind of summary you actually want.

Is it for academic purpose, for social media, for a quick internal update. Now without that context the model makes assumptions and the result may not match what you actually need. Let's look at the improved prompt to give to charge GPT. So my prompt here is you are a journalist writing a tweet thread summarizing this article for tech professionals.

Keep it uh keep this point under 280 characters and focus on the actionable insights. Now while I give this prompt to GPT let's look at the answer how it generates for us. So as you can see here the model has actually got a clear person and a platform in mind journalist writing for Twitter.

It also knows who the summaries for and what kind of information to highlight. Now this particularly makes the output snappy, concise and tailored for busy professionals who just want quick takeaways. Here you can have a look of all our um key points addressed in this article. So there is a major difference you know it depends on how you actually give prompts to GPT.

Using such improved prompts can increase the productivity of your task. Let's now look at the fifth example which is generic questions versus goal oriented prompt. A normal prompt would be uh what are some marketing strategies? Let's type this through chat. So I want to know more about the marketing strategy.

So this is how I give prompt. So the answer it has given me in bullet pointers 1 2 3 4 5. Okay. So it's given me basically 12 pointers. You can see here it might respond with a generic list as you can see the content marketing email campaign social media usage generated content referral marketing influencer marketing.

But you know they're not filtered or prioritized for real world scenario. Now I'll show you how you can improve it using the improved version of this prompt. So here I've given again the improved version of the prompt that is you're a digital marketing consultant advising a startup with a limited budget.

Suggest five cost effective marketing strategies that can implement in the next 30 days. Let's look at the answer now. So you can see it has given me a clear detailed uh answer for my question with the bonus tip actionable tip what you can take with all the clear five pointers with the tips as well you know you can just look from here that here now the model has got a clear context it's advising a startup there's a budget constraint there's a timeline the result is a list of more actionable and realistic strategies such as organic social media referral programs or even collaborations which are tailored for your business needs or your startup needs. So guys now you can see I have clearly shown you how you can optimize your prompting using a clear well structured specified prompt. It's very important that you stop using such vague prompts

and start giving much more specifications details and use a proper prompt if you want to utilize charge GPT. I hope with this video you all must have understood the fundamentals of prompt engineering and what you can do to develop a basic prompt into a more precise and high quality prompt. >> So guys let's get started and let's understand what is prompt engineering.

So prompt engineering is like directing AI models such as the advanced GPD4 to ensure they perform their best based on how you ask your questions. And now we'll see why it's crucial. So imagine you are seeking restaurant recommendations. If you ask where should I eat tonight, you might get random suggestion.

But if you specify, I need a cozy Italian place for a date night. Within walking distance, you will receive much more relevant advice. That's prompt engineering. Shaping your questions to fetch the most useful answers. So this was about why it's useful. Now we'll see the crafting effective prompts. So crafting effective prompts.

So the number one reason is be specific because detail is key. Asking an AI what are some easy vegetarian dinners that is better than just asking for dinner ideas. The next is provide context. Adding context helps AI tailor its responses like telling a friend a story with enough background so they understand. The next is focus attention.

Highlight crucial details to keep the AI focused on what matters most for your question. And then comes iterate as needed. Refine your prompts based on the responses. Similar to adjusting a recipe to get it just right. So this was about crafting effective prompts. So these are the basic ones.

Moving forward in this course, we'll see the most prominent things that we can add in the prompt that will come in the next 4 to 5 minutes. So let's move to the next one and we'll see a practical example for a prompt. So the example is suppose you are using AI to plan a birthday party. A vague prompt might be that how do I plan a party and this could lead to a generic checklist.

However, a well-crafted prompt can be like what are some creative themes for a 10-year-old's outdoor birthday party in summer and what games would you recommend? So this prompt will likely result in more specific and actionable ideas. So this is how you can generate a prompt. So prompt engineering is essentially about making AI work smarter for you.

Transforming complex task into simple enjoyable activities. It's a skill that enhances your interactions with technology making every AI encounter more effective and

engaging. So having explored what prompt engineering is and how to craft effective prompts, let's now dive into the various ways this skill can be applied.

So prompt engineering is not just a technical skill for AI specialist. It has practical uses in nearly every industry imaginable from enhancing customer interactions to streamlining software development. The applications are vast and varied. So let's see some of the key use cases.

So the number one use case is content creation. So in digital marketing and blogging, prompt engineering helps generate targeted content such as articles, social media post and marketing copy that resonates with specific audiences. The next is customer support. AI can be used to automate responses in customer service. Well-crafted prompts ensure that the responses are accurate, helpful, and contextually appropriate.

Then comes software development. Developers use prompt engineering to generate code snippets, debug programs, or even use AI to conceptualize new software solutions. Then comes education and training. EI can tailor educational content to students learning levels or answer specific academic queries making learning more personalized and efficient.

And then comes market research and data analysis. By directing AI to analyze large data set with specific prompts, businesses can extract meaningful insights about market trends, customer preferences, and operational efficiencies. And then comes healthcare. In medical settings, AI can assist with diagnosing from symptoms described in prompts or help in researching treatment options by processing medical literature.

And then comes legal and compliance. That is the most used case for the AI. AI can help pass through vast amounts of legal documents to find relevant precedents or compliance issues based on prompts tailored to specific legal questions or topics. These use cases illustrate the versatility of prompt engineering highlighting its potential to

enhance productivity and creativity across a wide range of industries. So these were the use cases.

Now we'll see the flow of AI technologies from where this LLM models or the large language models or GPD4 that's an example come into action. So let's start with the flow. So AI is the overarching category that defines the goal of creating machines capable of performing task that would require intelligence. if done by humans.

And then comes ML. So ML is a method within AI focused on giving machines the ability to learn from data. Then comes deep learning. So deep learning is a technique within ML that uses layered neural networks to analyze various factors of the data. And then comes LLMs that are large language models that are specialized application of deep learning focused on understanding and generalizing and generating human language.

This hierarchy moves from broad general techniques and applications down to more specialized and sophisticated systems showing how foundational concepts in AI lead to more advanced implementations. So this was all about a conceptual or the context of the prompt engineering. Now moving to the applications of prompt engineering and we'll be using GPT4 for this purpose and we'll be writing prompts in the GP4 and asking the GPT4 model to provide us the relevant answers.

So let's move to GPT4. So as you search on on any of your browser uh that would be openai open.com you would be directed to this website and here are their products that is chat GPT that is for everyone for teams for enterprising and there has been a pricing listed here. So you could come here and click on chat JPY login and after proceeding with your credentials you can login into chat GPT and start writing your prompts.

So coming back to the openAI website. So you could see here that's a research and the latest advancements that is GPT4 DL3 Sora. So GPT4 that's a model that has been developed by OpenAI that can use the camera or the vision technology and can tell you

that what object it is and if you show him a code snippet that will tell you what that code snippet has been written for.

And if you use that to just scan writing on any of your pages or any of your copy, it will just scan it and translate to you in what language it has been written and you can translate to any other language. And then comes Dal3 that is used to create images. And then we have Sora that is used to create videos.

So now moving to the next that is products and you could see that the chat GP these are the version that are for everyone for teams for enterprises and then we have the pricing for that and here we have the chat GPT API or open a API so you could click on that and see so before that before going to the API we'll move to documentation and let's have an overview of all the things so here's an introduction about the key concepts that is text generation models that is GPT4 and GPT 3.

5 then there are assistants that GPT4 can be act as a assistant to anyone and then we have the embeddings that is it's a vector representation of a piece of data that is meant to preserve aspects of its content or its meaning and then comes the tokens so there are tokens so you could see here that text generation and embedding models process text in chunks that are called tokens And as a rough rule of thumb, one token is approximately four characters or 0.

75 words for English text. So these are tokens. And now moving to models. So we have GPT4, GP4 Turbo, 3.5 Turbo. And here are all the models that are listed by OpenAI. So here we'll be talking about GPT4. So GPD4 that's a large language multimodel model and it has multilingual capabilities that has multilingual capabilities and you can ask him any question in any language and then we have Dal that is used to create images then we have TTS whisper and then the embeddings so let's move to chat GPD and before that let's have a API reference that is if you want to use open AI API or you want to integrate to create a chatboard or we have seen a use case that is to create a customer service representation. So for that you could use this openAI API. So for that

you have to install the OpenAI module with the command pip install OpenAI and after that you could use the npm and here are the API keys from where you can generate it and they have provided you all the steps how you can use it for streaming purpose for the audio to create speech.

This is how you could use the API to create your own models. So this is the interface of chat GPD or GPT4. So this is the audio of chat GPT and here we write the prompts that is the message box and this is the login person that has login and if you click here you could see my plan and we have purchased the plus plan that is $420 per month and there other features also we will come to that.

So let's see one of those that is customized chat GBD. So here you could write custom instructions that what response you want from chat GPT. So you could mention here that I want the tone to be specific, mild, not to be too loud. And you can ask him don't use uh advanced English. I want the answer to be prompt and in a simple English manner.

So you could write instructions for the responses you want to have. And this is the window. If you click on that you could see explore GPT section and here is the history section that what you have written and what prompts and what responses you have created till date. So if you click on this you could get a new window for chat GP4 and these are the models that are listed here that is chat GPT 40 chat GPT4 GPT 3.

5 and this is the temporary chat section. We'll discuss about all this but currently we'll start with the types of prompts or generating the prompts and what things to consider while generating the prompts. So moving back. So let's see how you can create a prompt. So you have to have six things in prompt to make it more precise.

Number one is context. So this sets the scene or provides background information necessary for understanding the prompt. For example, if you're writing a prompt that in a

world where artificial intelligence has replaced most jobs, describe a day in the life of a human worker.

So you have asked him and provided the context here. And then we have task. So the specific action or question that the responder needs to address. So if you are writing a prompt, number one is context that needs to be included. Then comes the task. So what task you want the GPT to act and provide the response.

So for example write an essay on the effects of global warming on agriculture. So you have provided the task here that is write an essay. So now the next thing is persona. So persona specifies the identity or role the responder should assume while answering. For example, as a medical professional advice a patient on managing type 2 diabetes without medication.

So this is the persona. Then comes format. So define how the response should be structured or presented. For example, list five tips for improving personal productivity in bullet points. So the format you have asked him to address the personal productivity in bullet points. So that's the format you have asked him and then comes exampler.

So if you want to give the example to GPT that we have a sample here, use that and provide us a response according to that. So for example, sometimes a sample answer or part of an answer is given to illustrate the expected style or content. So we will have a prompt example here like in the example with a protagonist overcomes fear.

Write a story about overcoming a personal challenge. So this is the example that we can give to an LLM model. And then comes the tone in which tone you need the answer. So indicate the mood or attitude the response should convey. For example, write a humorous blog post about the trials of parenting toddlers.

So you have mentioned what humorous blog you want the tone as a humorous blog. So with the right technique you can craft prompts that are not only precise but also versatile suitable for any learning management system. So this approach ensures that your prompts will engage students, encourage critical thinking and drive meaningful discussions no matter what the platform is as we are using the GPT4 here.

So you could use cloud or anthropic like there are many platforms you can use any of them with these type of crafting the prompts techniques. So embrace these strategies and you will be equipped to create prompts that resonate across various educational environments enhancing the learning experience for all.

Each component plays a crucial role in guiding the response to ensure it meets the desired objective and quality. So let's see some of the examples and we will use all the context or all the types that can be used to create a prompt. So our first example is as a nutritionist speaking at a high school career day, create a presentation outlining the importance of healthy eating habits for teenagers and use a friendly and engaging tone and include real life success stories to illustrate your points.

So the context here is high school career day. You have given a context and the task you want is create a presentation on healthy eating habits. And then comes the persona that is nutritionist. You have asked him that you act as a nutritionist here. And then the format you want all the response in a presentation that is presentation with real life stories.

And then comes the exampler. So you want real life success stories here. And you have set the tone here that is friendly and engaging. So if you write this prompt in GPD4. So let's ask this prompt from the GPT4 and let's see what he answers. So we are mentioning as a nutritionist. Uh I don't remember the spelling but uh we will write.

So, speaking at a high school career day, create a presentation outlining the importance and importance of what we are writing importance of healthy heating emits.

for teenagers. And what we want here is use a friendly tone and include real life success stories to illustrate

your points. So let's see uh how does the chat gypy respond. As we have mentioned all the types of prompt or all the components of prompt that can be used to create a prompt that is context, task, persona, format, exampler and tone. So you could see that he has started providing the response with the slides that is slide one introduction, slide two nutrition matters, slide three the teenage plate and slide four the success story.

So this is how you could write a prompt and you could get a fully structured response as you want. If you want to like moderate this response or alter this response as you are not satisfied with it, you can go on and write some more prompts to get it more precise. So we'll see another example for that.

Imagine you are a sci-fi author writing a short story set in a future where water is scare, where water is scars. So craft a narrative that explores the daily challenges faced by a family using a dramatic and suspenseful tone. So what we'll see here is context that we have mentioned that is future world with water scarcity and the task we have asked him for the GPD model is write a short story.

The persona that we have given to the LLM model that is GPT is that you are a sci-fi author and the format we want is narrative and the example we have given is the daily challenges of a family and the tone is dramatic and suspenseful. So we have mentioned all the things for a prompt that could be created a better response.

So similarly we'll see another example. So this example is as a financial advisor prepare a guide for young adults on managing finances after college. So use a conversational tone including actionable steps and start with the story of a recent graduate to illustrate common financial pitfalls.

So we have mentioned the context here that is financial management post college. The task is to prepare a financial guide. The persona is financial advisor and the format we need is guide with actionable steps and the example we have given is story of a recent graduate. The tone is conversational and this was all about this prompt.

Similarly, we'll see another example that is as a war correspondent draft a report on the effects of the conflict on civilian life focusing on a particular city. use a serious tone and provide interviews as examplers to underscore the human aspect of the story. So the context here is effects of war on a particular city's civilian life and the task is draft a report and the persona is war correspondent.

The format is report with interviews and the example we have mentioned here is interviews with civilians and the tone we have set here is serious and impactful. So we have seen some of the examples for the prompt creation. Now we'll see the examples for writing prompts for a particular field. So we'll start with number one field that is content creation.

So we have mentioned some of the use cases for prompt engineering. So starting with number one that is content creation and here we'll write a prompt and that could be that as a marketing manager draft a blog post aimed a new entrepreneurs on the importance of branding. So use an authoritative yet approachable tone including examples of successful brands to illustrate key points.

So let's write this. So, as a marketing manager, draft a blog post aimed at new entrepreneurs on the importance of branding. Use an authoritative yet approachable tone, including examples of successful brands to illustrate key points. So this prompt we have written for the

content creation. So similarly you can uh ask him to write a story or draft a blog post or write any content that you have asked him a persona here that is marketing manager. So

let's simplify this prompt. We have marked the context here that is blogging for new entrepreneurs. And the task we have asked him is draft a blog post on branding.

And the persona that we are asking the GP to act as a marketing manager. And the format is blog post with examples. An example that we have given him is case studies of successful brands. And the tone we want is authoritative and approachable. So similarly you can write prompts for content creation that I want to create a blog post or I want to create a YouTube video or I want to create an article.

So provide me a story line. How can I approach a particular topic that could be what is LLM? And as we can write it here that act as EI specialist and help me write an article on topic what is LLM and and keep the tone in engaging manner. So you could see that chat deputy has started generating the response that is he is creating an article on what is a large language model and he has decided the title here and providing all the content for your article.

So what we have provided here is context that is we want an article on what is LLM and the persona that act as a EI specialist and the format we want an article and we have set the tone here that is in an engaging manner. So similarly you can draft other prompts and you could help them with your content creation journey.

So moving to next example that is for SEO purpose. So there's another use case that is SEO and for that we can write prompts that imagine you are an SEO expert running a workshop. So create a presentation that explains the basics of SEO including practical tips on keyword research and link building and use a professional yet engaging tone to keep your audience interested.

So similarly you could have a prompt in your large language model and ask him with all the components of prompt and we can wait for the answer and let's see what it responds. So imagine you are an SEO expert. Here we are asking the GPT to act as a SEO persona or SEO expert and we are asking him that you are running a workshop also.

So we are setting a context and asking him to create a presentation that explains the basics of SEO including practical tips on keyword research and link building. And now we'll set a tone here that is use a professional. Use a professional yet engaging. And now we'll set the tone here that is use a professional yet engaging tone to keep your audience interested.

So we have crafted a prompt here that is for the SEO purpose and similarly you can create your own but so let's simplify this prompt and see what context task or persona we have mentioned here. So the context here is SEO workshop and the task we have assigned to LLM is create a presentation of SEO basics and the persona that we have asked the GPT is to act as a SEO expert and the format we want the responses in slide presentation with tips and the example we have given him is screenshots of SEO tools and the tone we have asked him is professional and engaging. So you could see the response that has been generated by the GPD4 that he has provided us that in slide one you could have a introduction slide with a title and the opening remarks and then in slide two you could have the title that is what is SEO you could mention the definition here and what would be the goal for this presentation and similarly we have the slide three that

is for how do search engines work and then we have slide four slide five and similarly all the slides And you could just mention how many slides you want and you can also mention how many slides you want as a response. So this was about the SEO thing. Now moving to the next use case that is for developers.

So for developers we can draft a promp that as a software engineer write a tutorial for beginners on writing their first web application using React. Include step-by-step instructions and code snippets. Make your tutorial detailed yet easy to follow. So this is one of the prompts.

Similarly, you could ask him to debug any code. You can provide a code snippet to GPT and he will debug the code and provide all the necessary changes that can be made to the code snippet. So for this we will write an example that so we will write the prompt here that as a software engineer and we are asking him to write a tutorial for beginners on building your first web application using React.

using React and include step-by-step instructions and code snippets. So we'll write here that include step-by-step instructions and code snippets and make your tutorial detailed yet easy to follow. So you could see that GPD4 has started generating the response and similarly you could ask him to generate some code for a particular application or you could also create a website by just asking him the HTML file, the CSS file and the JavaScript file and what options or the applications you want a website should have. So first see the response

that Chad Gypty has created. So you could see that he is writing a tutorial for beginners on building their first web application. So first we will see the simplification of this prompt that is we have set the context here that is tutorial for building web applications and we have assigned a task to the GPT that is write a tutorial on react and the persona we have given him is as a software engineer and the format we want the response is in tutorial with code snippets and the example we have given is example project and the tone is informative and clear. So you could see here that he is providing the code snippets for setting up your project then navigating into your project directory and starting the development server and then creating a list component. As you go along, you could see that he has created the whole tutorial. And if you want the tutorial to be specific that in the first tutorial, you want just setting up your

project, you could ask him and he will create that for you. And in the second one, you want it to navigate into your project directory or you want to create your first web app that GPT will help you create that. And similarly if I ask him that as a software developer or act as a software developer we are giving a persona to him.

Help me create a travel website and make it engaging and easy to handle and userfriendly. So you could see that GBD4 has started getting the response and you could see that he's providing us the steps that is to define the scope and features of the website and if you want the code snippets you can ask him that I want the HTML file for the current website.

So he will provide you a HTML file. If you want any modifications or want to alter that, you could provide him the prompts that I want a specific navigation bar or the search functionality or the visuals. So similarly, the GPT4 will act upon the prompts and provide you the code snippets that would be helpful in creating your website.

So now moving to the next use case that is data analysis. So for data analysis, so you could see a pin bar here. Here you could upload the documents from computer that could be your text files, xlsx files or you could connect it to the Google drive or the Microsoft one drive and have your documents here.

So we will paste a xlx file that would be for the data analysis purpose. So here's the excel data. Uh here we are uploading it and providing it to chat GPD and I will open for you guys so you could also see. So I have this data for a particular company. So you could see that they have the order ids and the quarterly order date, the shipping date and the shipment mode.

So we will use this data and ask the chat GPT to simplify this data or analyze the data and provide us some that could be pivot tables or creating bar graphs or bar charts or provide us the KPIs for the particular data. So starting with we will provide a simple prompt to chat GPD that is you are a data analyst conducting a workshop.

Prepare a guide that teaches how to create effective dashboards in Excel and include practical examples and visual aids to enhance understanding and use a direct and instructional tone. So starting here we will ask him that you are a data analyst and we'll

start with that file only. We will upload that file that is Excel data and we have the file open here.

So we will ask him to create a pivot table. So we'll write a prompt that you are a data analyst and I have provided you a sample data. Create a pivot table. So let's move to the table and see that we can create a pivot table with sales and order date. So we'll ask him that create a pivot table and a corresponding chart to analyze sales performance by order date.

So let's wait for the response. And you could see here that he has started analyzing it. And I have something more for you. If you go to explore GPD section, you could see here that these are the GPDs provided by OpenAI and other creators. These are the training ones that is image generator, scholar GPT and these are provided by chat GP that is DL data analyst, creative writing, hot modes, coloring, book hero.

So you can use this. I will show you guys. But before that, but before that we'll move back and see what our prompt has generated the response. So you could see that chat JP is asking that it appears that the Excel file contains several columns related to sales data including some duplicate or similar columns with variations in names.

So to simplify the analysis, I will focus on the main columns like order date and customer name. So we will ask him to proceed with that only. Yes, proceed with that. And if you click here, this is the analysis. This is the code that you could use to do the analysis. So you could see that it has started generating the response and he is generating the chart here.

So here is the chart and he is provided the description that this line chart showcase the sales performance over time based on the provided order dates and you can see how sales fluctuate on different dates which help in identifying trends, seasons, impacts or specific dates with high sales volumes. So if you need further analysis or adjustment of the chart, feel free to let me know.

And if you click here, you could see all the analysis and all the code snippets that are used by the chat JD to do and create a bar graph here. So you could use this code and you could use any ID that could be visual studio code or any ID that you have hands on and you could do the similar analysis there.

So this was about the data analysis use case. Now moving to the next one that is for educational purpose. So if you want to learn something or you want a road map to learn any programming language, you could use these chat GPT or the LLM models for a particular road map. So for that you could write a prompt that as a experienced educator write a road map for learning Python programming and provide a road map that should cater to beginners and include resources, practical exercises and milestones and use an encouraging tone to motivate learners. So let's see what this LLM or GP4 provides a response to this prompt. So we'll ask him that as a experienced educator write a road map for learning Python programming

and the road map should cater to beginners. and include resources, practical exercises and milestones. Use an encouraging tone to motivate learners. So let's provide this prompt and see what our GPT will respond to that. So let's simplify this prompt. So we have set the context here that is we are learning a Python programming and we have assigned a task to the LLM model that is write a learning road map and the persona we have asked him to be as a

educator and the format we want is a road map with resources and the example we have provided him is step-by-step progression and we set the tone as encouraging and supportive. So let's see what response the chip has provided to us. So you could see that he has provided a step one that is understanding the basics and the goal here is get familiar with Python syntax and basic programming concepts and the resources he provided is documentation for Python that is python.

org OG code academyy's python course and then we have exercises milestone and now moving to the step two that is diving deeper and the duration is 3 to 4 weeks and the goal is to explore more complex programming concepts like data structures and loops and we have the exercises here and he has mentioned the resources and the milestones.

Similarly, you could see that he has provided the full road map with step three that is with applying projects in Python and then we have step four exploring advanced topics and then we have joining the Python community and then the conclusions. So similarly you could have a road map with him and ask him that you act as a educator and a guide to me and I will start with this road map and provide the road map to a day wise columns and you start with the day one and ask him to judge me or examine me with the knowledge or the basics that I have concurred through the day one. So he will act as a educator here and he will examine all the information or all the skills you have gained and ask you questions and analyze them and help you get through the road map easily. So this is all about the educator use case. Now

moving to the next use case that is legal and compliance. So for the legal and compliance use case, we could have an example as a prompt that is as a legal adviser specializing in data privacy. Create a compliance checklist for small businesses managing online customer data. Use a formal yet accessessible tone and include examples of common compliance pitfalls and how to avoid them.

So you could ask the charg as a legal guider or as a legal advisor that will guide you for the particular acts or the compliances that are aligned with judicial custodies of your country. So we have a prompt here and we'll ask the chat GPD that as a legal adviser specializing in data privacy. So create a compliance checklist here for small businesses.

Managing online customer data and use a formal tone and include examples of common compliance of common compliance pitfalls and how to avoid them. So we will simplify the prompt here that is we have provided the context that is compliance with data privacy laws for small businesses and we have assigned a task that is create a compliance checklist and the personas we have asking the GPD to act as a legal adviser

and the format we want the response is in checklist with examples and the example we have provided is case scenario of non-compliance and the tone we want is formal and accessible. So

similarly you could have other legal advis JPT and what you need to do is if you want to read any act or you want to analyze any act you could provide the documents to ch and he will analyze that and provide what rules or regulations you need to follow or with what compliances you should move forward with that act.

So you could see here that he's drafting a document with all the compliance efforts for small businesses. So with that we have done with the legal and compliance use case. Now moving to the next use case that is healthcare. So in healthcare we can have an example prompt that you are a medical researcher presenting at a conference on the advances in teley medicine.

So prepare a detailed paper discussing the impact of teley medicine on patient care during the pandemic using clinical studies as references and maintain a scholarly get engaging tone to captivate your professional audience. Or in the healthcare section you could also have a diet plan or a chart plan and you could ask him recipes for a particular diet and you could also mention if you have any allergies.

So we will act with a prompt here. So let's write here that you are a dietician. So provide me a recipe that is healthy and include carbs, protein content, and a small amount of fat. And remember that I am allergic to peanuts. So you have set the context here and the persona that you are a dietician and you are asking a prompt that you want a recipe that is healthy and include carbs. So you could see that

Chad Gibbri has started generating the response and he's suggesting a kuna chicken salad. So you can also mention that you are a vegetarian or non-vegetarian and similarly chair deputy will act upon that and he has provided all the instructions and all the

ingredients for the recipe. So similarly you could use the prompts for the healthcare use case.

Now coming to the next use case that is customer support. So we could have an example for that that is as a customer service trainer we will design a training module for new agents that focuses on handling difficult customer interactions and include role-play scenarios, key phrases to use and tips for maintaining professionalism.

So use an instructive and supportive tone to encourage learning and confidence among trainees. So let's draft this prompt and let's see what does Chad Jeop respond to that. So we'll ask him that as a customer service trainer design a training module for new agents that focuses on handling difficult customer interactions.

include role-play scenarios. key phrases to use and tips for maintaining professionalism. So use an instructive and supportive tone to encourage learning and confidence among trainees. So we are setting the tone here that is use an instructive and supportive and setting the tone to encourage learning and confidence among trainees.

So let's proceed with the prompt. And now we can see that we have set the context that is training for customer service agents. And the task we have assigned is designer training module. And the persona we have set here is customer service trainer. And the format we want the response is in training module with role plays.

And the example we have set here is scripts and responses for role plays. And the tone we want the responses in instructive and supportive. So you could see that chat JD has started generating the response and given a model overview that you could have understanding difficult interactions then communication skills and key phrases roleplay scenarios maintaining professionalism and module review and assessment.

So he has drafted a whole new training module for the new agents that can handle difficult customer interactions with this module. So this was all for the customer support use case. Now moving to the use case where you could create PowerPoint presentations using the VBA code that is provided by JPD.

So let's ask him to write a VBA code. So we'll write a prompt that act as a presentation specialist and write a VBA code to create a presentation on topic modise LLM and provide the steps where we can use the VPA code to create one and you could also set the tone here but let's see what chat dep respond to that and we will open a presentation here so

we'll open a blank presentation and for using the VBA code to create a presentation. What you need here is developer options in your PowerPoint as I've already enabled them. You could enable them by just right clicking on the ribbon and clicking on customize the ribbon and you could see here the developer options as I've already enabled it.

I will checkbox the tick and apply it. So after applying it, move to the developer option. Click on visual basic and after that click on insert user form and click on module section here. Now coming back to our GPT and he has created the VBA code. We will just copy it and paste in the module section. Getting back to the module section, we'll paste it here and click on this play button that is run sub or user form.

So you could see that this runtime error that is an error occurred while PowerPoint was saving the file. Let's debug it and we'll run it again as we're getting the error again here. So we'll try to copy the error and provide it to chat JPT. So let's ask him that we're getting an error on this. I have encountered an error and the error is and we'll ask him that the error is while saving the file.

So let's see how does he provide the response to this query. So he's writing the modified VBA code. We'll copy that and move back to module and paste it here. And let's see if it

works or not. Currently we are having the same error. And now moving back. Let's see what it provides here. So you could see that we are not saving the presentation now.

And he has generated the PPT for us. And this is the basic PPT. You can customize it. You can ask the chat JPT to create dialogue boxes or insert shapes or you could just choose a design from here and make your PPT or the presentation a good-looking one. So now moving back to GPT model. So this is all about the use cases.

We have hands on the prompts now. Now we'll see the key features of chat GPT and some are newly introduced. we will have a hands on the memory feature that has been latest introduction by the open AI. So if you click on the settings and move to personalization section, we have a memory feature here that is you could store the memory that we entered as a prompt in chat GP.

So the chat GP memory section works by capturing and summarizing key information you provide using your interactions. So unlike one-time commands that need to be repeated in every session, this memory is designed to retain useful details and automatically incorporate them into future interactions. So this means chat gy can adapt over time improving its response based on the history and context it has accumulated almost like it's getting to know you better with each chat.

And if you want to delete some chats, you could move into the manage section. And these are the memories that has been created with chat JPD. If you want to delete one, you could delete it from here. Or you could just write a prompt or you could just write a prompt that I want to delete a memory and you can mention some keywords there.

So this was about the chat JP memory section and if we go to settings we have data controls that if you want to export data you could export the whole chat and send it to someone else. And similarly you have the security feature. If you want to have multiffactor authentication enabled on your GBT you could do that. And you can have some connected apps that could be Google Drive, Microsoft One Drive.

And we have a builder profile section also. That is if you want to build your profile here, you could place your LinkedIn mail ID and the GitHub section. And you could also link your X account. And then we have speech module here. That is you want to listen something by chat GBT, you could have a voice assistant here and listen to a particular voice.

So this is all about the features of chat GBT. Now moving to the chat GPD or export GPD section. Here I have shown you that these are the GPDs created by creators and some are created by chat GPD own. And here you have an option to create your own chat GPD or the GPT model. So to create that you just have to write prompts here that I want to create a data analysis GPT and you could write more prompts here and you will having a preview here how your GPT looks and here's a configure section where you could name your GPT provide a description to it and the instructions and you could have the capabilities of web browser or the daily image generation and you can add an icon for your GPT here. So you could see the preview here that he has provided a

sample prompts with the data analysis insights and visualization support. So this is how you can create a GPT on your own. Then we have a feature that is temporary chat as we have discussed about the chat GP memory. So it stores memory as you ask him or write a prompt that remember I want all my responses to be a tone specific and they should be emphasizing.

So he will support this and store this memory into chat jupdy memory section and use that as a response for the next upcoming responses. And if you don't want chat GPD to store this as a memory you could use a temporary chat section here. So you just have to click on chat GP4 drop-down and here we have temporary chat.

You just enable it and here you can have a chat with the LLM model or GP4 model here and he won't be storing any feature or any memory regarding this chat. >> Have you ever wished you could just tell your computer what you want and bam, it builds it for you.

Well, guess what? We're pretty much there. Hey everyone, today I'm going to show you something called wipe coding.

And before you roll your eyes, hear me out. This isn't some fancy buzz word. It's literally coding that just flows. You know that feeling when everything clicks, when you're in the zone and ideas turn into working code without the usual headaches? That's called vibe coding. Here's what's crazy. AI tools have gotten so good that you can basically have a conversation with them and walk away with a real app.

No more staring at blank screens wondering where to start. No more getting stuck on syntax, just you hear ideas and tools that actually get it. Today, I'm testing three AI coding tools that everyone's talking about. At first, we have Bold. Now, this thing is insanely fast. You describe what you want and it builds it while you watch it.

Then we have Lovable, which is perfect if you care about making things look good. It's like having a designer and developer role into one. Then we have Pythagora. The coolest part is you just chat with it like explaining your idea to a friend and it builds complex app. I'm going to show you how exactly each one works the good stuff and not the good stuff.

Real demos, real code and at the end we are doing a showdown which tool actually delivers on this whole VIP coding promise. So let's find it out. So let's start with our first AI tool for VIP coding which is Pythagora. So from here you can ask Pythagora to build anything. For example, it's mentioned here that ask Pythagora to build a CRM app.

So we will ask the same build a simple interactive CRM app. And I'll just hit enter from here. You have to choose any of your account you're using. So as you can see it's setting up your Pythagora workspace. So as you can see our workspace is being created and now from here you can just uh simply describe.

So as you can see from here I've given a prompt to create a simple customer relationship management app. The goal is to keep the track of customer information manage relationship communication and uh help with the sales and support. And here's what I have in mind. I've given the app features how I want it to be.

The task management note section simple dashboard communication logos. I've also given the UIX design element. Now before this you can just fill out this essential uh questions ask. So I'm using it for some other purpose. How big is your company? You can just mention anything. And then after I've given the prompt I'll just hit continue.

We'll wait for few seconds. So now as you can see it is giving us to write some specification. So this is the overview of the specifications which I have mentioned. So the Pythagora is now generating a detailed specification for app based on your input. So it's basically creating a documentation of the overview which we had given.

So it's given that right specification. So are you satisfied with the project description? So I'll just hit on yes. So it's building all the components and the APIs. uh it takes a bit of time. So now as you can see that our app is ready and I'll show you the preview of our app. So we'll open this in our browser.

So you can see from here you can just enter your email and then you can enter your password or if you don't have an account you can just sign up and then you can create your account. So as you can see this is our dashboard that has created for us. I mean just look at it. This is just awesome. Look at the features it has added and can you believe this AI has made it completely.

This is the number of customers, the active leads we have, the upcoming tasks. I mean the quality is just so premium and that you get it in free version without even paying spending any bucks and I'm pretty impressed by this. I would rate it 10 out of 10. This is absolutely best.

Now let's move on to our second AI tool on our list which is uh bolt. So from here you can just uh click on the first link. So here in bold you can just share your ideas. So I've mentioned that build a simple ecommerce website with good UI UX and then I've mentioned more app features and the shopping cart I want to be added and I'll just hit on enter.

So you can just sign in with your Google account. So now from here you can just give the prompt I've already given. So we'll wait for a few minutes till our preview will appear here. So it's first creating the overview of our prompt which I've given. And now you can see it's installing the dependencies from here. And now it's written that I have created a comprehensive production ready e-commerce website with beautiful design and excellent user experience.

The application features a modern clean interface with sophisticated color schemes, smooth animations and intuitive navigations. So we'll just have a look at once we are done with creating this. It takes a bit of time but uh yeah I think it's ready now. So we can just open in a separate tab.

So now from here you can see that our e-commerce website is ready and yes this also looks amazing. The name is given shop hub and this is the new collection here and these are the products and you can just uh see to this you know hover animation that it has given to us and this also looks really premium. Just look at this.

And this is the uh drop-down list where we have all the quick links, customer service and the contact as well. I'm pretty impressed by this bold new because it has created such an amazing website and the minimalism and the sophisticated look that it has given to my website. It's really really good.

The color layout also it's blue white and it has kept very minimalistic. So for this also I would rate it maybe 9 out of 10 or maybe 10 out of 10 because both of them are really good. Just look at this Pythagora and here we have the Bolt new. Both are amazing. I mean uh I can't actually compare both but then again let's check out the third AI tool on our list. We have lovable AI.

So we'll be checking out this AI tool and how we can just create amazing applications within seconds. So I'll ask this AI to create an amazing weather app with features and good UI UX design. So let's hit enter. Okay, you can just log in with your account. So now we'll just wait a few seconds till our application is being built.

So now as you can see just working on the progress and I must say the interface of this AI looks pretty good. You can also upload images as reference and you can preview your changes as well. So now it's basically working. So from here you can see that it's working on the index. This is the code which you can see from here.

So now you can see that our app is ready and we'll just preview it and again just look at the features it has added. This also looks really good. This is the uh weather application named weatherly and here you can just search your city. You can just enter your city name.

Let's suppose it's Bangalore and it will give you the weather updated report. So you can see from here no it's UK just enter. So I don't know why is it giving UK but then yeah the weather information is actually correct. You also have the weather details the humidity wind speed visibility plus you get the 7day forecast today tomorrow and this is again amazing and I would rate this maybe this is also good.

So I'll rate this 9 out of 10. So as to compare all the three AI tools which we created our applications. So the first one we had is the uh CRM application CRM Pro using Pythagora. So here is our application. The second one we built using Bold and e-commerce website named at shophub. So you can have a look at this.

And the third one we created a weather app using lovable. And here you have this. Now let me know in the comment section which one you actually like the most. For me personally I think I would go for bold number one. Second I would rate this Pythagora and third I would rate this lovable.

Then let me know which one you liked the most. There was a time just a few months ago when we thought we cracked it. You could literally say hey AI build me a to-do app and just like that code appeared. No setup, no planning, just vibes. It felt like magic. We called it vibe coding. But here's the thing about magic. It's impressive until it breaks.

People started realizing the code didn't scale. The APIs were hallucinated. Tests were missing. And once you moved past a simple prototype, everything felt apart. Why? Because the AI wasn't actually thinking. It was just guessing. And that's where context engineering comes in.

Instead of just saying build a to-do app, you tell the AI who is it for, what features it needs, how the code should be structured, which tools to use, and even what good output looks like. It's not about shouting instructions. It's about setting the stage. And once you do that, the results are on a different level.

Today, you'll learn what context engineering is, how it works, and how to use it with a real hands-on demo. Here's what we'll cover. Feel free to skip ahead, but it all connects. First, we'll go through what context engineering is, where VIP coding broke our hearts, prompt engineering versus context engineering, the full ingredient list of good context, biggest challenges and proven fixes, and live demo.

So, let's get started. So, first let's start by understanding what is context engineering. Now, I'll give you an analogy of a picky chef. Imagine hiring a chef and only saying, "Make dinner for me." No ingredients list, no dietary notes, no guest count. The meal outcome

is pure roulette. Now here, context engineering gives the chef everything needed that is the ingredients in the pantry, the dietary notes which includes no nuts, vegan friendly, past dinner so that we don't repeat, the preferred plating style. So basically in AI terms, context is equal to rules plus data. memory tools and the desired output. Now, it's the engineered environment that lets enlarge language model reason and not guess. So, context engineering in AI involves organizing and managing all the elements needed for a task such

as rules, data, memory, tools and the desired output which I already told you. Now this helped the AI make clear reliable decision instead of just guessing by providing all the necessary backgrounds information. Context engineering ensures that AI can perform well in complex situations. To sum it up, it's the art of providing all the context so that the task is plausibly solvable by the model.

But why did we even need this term? Because we tried the opposite. So now we'll understand why vibe coding failed and what we have learned. Let's rewind to early 2024. The AI scene was booming. Tools were dropping every week and developers everywhere were obsessed with something called vibe coding.

The idea was to just tell your AI assistant vaguely what you want like build a to-do app, create a landing page for my startup, give me a chartboard that replies like Shakespeare and boom, instant code. It felt magical. No setup was needed, no thinking, no planning, just proper vibes. This was especially fun for hackathons, weekend projects, quick prototypes, that wow moment in front of friends or colleagues.

And to be honest, it was addictive, of course. But that dopamine hit from watching the AI generate full code blocks with zero effort. incredible but really hit hard once people tried to ship those projects or use them in real production. Let's now understand the problems with vibe coding. Now while vibe coding felt like cheating the system, it turned out we were mostly cheating ourselves. Here's why.

So the first problem faced was the hallucinated APIs. AI would confidently use functions, libraries or endpoints that didn't exist. you would fetch a data function that was literally made up. Second was no scalability. Now AI didn't use designed a codebase for future use. No modularity, poor file structure, zero comments or documentation.

Then we had brittle test or none at all. Now most AI generated tests either didn't match the code logic, skipped edge cases or didn't exist at all. So once the code needed to grow or evolve, it obviously collapsed. Now let's look at what the data said. Now this wasn't just a gut feeling. Codo released a major industry report, the state of AI code quality.

And here's the stat that stood out. 76% of developers said they don't trust AI generated code without human review. Why? Because vibe coded projects often look at the first glance. It would break under pressure and it would require more time to fix than to build from scratch. The core issue with vibe coding was based on intuition and not intention.

You hope the AI gets it right. You assume the structure is okay. You skip the hard thinking. As a famous person once brilliantly said that intuition doesn't scale, structure does. So VIP coding is basically AI plus guesswork and context engineering is AI plus planning clarity and with structure.

So if VIP coding is just winging it and context engineering is all about building smart then where does prompt engineering fit in? Let's clear that up next. Now let's understand the difference between prompt engineering versus context engineering. We break down in the simplest way possible with examples, stories, and a comparison that actually sticks.

Let's first look at the core difference. Think of prompt engineering like asking someone for a favor in one sentence. And context engineering, that's like handing them a folder with everything they need to do that favor well, not just once, but repeatedly. Let's look at an example analogy, which is making a sandwich.

Let's say you want someone to make you a sandwich. So in the case of prompt engineering, you would say, "Make me a sandwich." That's it. No details needed. They don't know if you're allergic to peanuts, if you like mayo, or if you're vegan. So what do you get? Probably a sandwich, but maybe not the one you wanted.

Next, in the case of context engineering, you would hand them a sticky note that says, "I'm a vegan. No onions needed. Toast the bread. Use the sauce from the top shelf. This is how I like it cut. Now, they're not just making a sandwich. They're making your sandwich exactly the way you want it. Now, let's put that in AI terms.

So, first we look at the feature. Now, in case of prompt engineering, the focus is more on like have you asked the question. On the other hand, in case of context engineering, the focus is on everything around the question that includes data, rules, tools and memory. The typical size of prompt engineering is like basically one to three line.

And in case of context engineering, multiple files are there, settings, examples, and proper instructions. Now, the main goal of prompt engineering is one decent response. But in the case of context engineering, it's a reliable system that works across step or task. Prompt engineering is useful for casual use like for example chat GPT or question and answer.

But then context engineering is useful in real applications, automation and in AI agents. For example, in prompt engineering, you would say something like write a clean Python to app. But in case of context engineering you would say add system rule, use TypeScript, include API docs, provide sample code, define JSON output format.

The reliability of prompt engineering sometimes hit or miss. But in case of context engineering, it's more structured, predictable, and scalable. But why this matters? Because prompt engineering is great for one-time question, fast iteration. But then if

you're building an AI chatbot, creating a custom AI agent, you need context engineering because the AI needs more than just a question.

It needs the full story. So now that we know the difference, let's break down the actual ingredients of good context. Imagine you're giving an AI assistant a task. Now the AI is not human. It doesn't understand things the way people do. It only knows what you give it right now. So if you want it to do a good job, you have to give the right information in the right way.

That set of information you give the AI, that's what we call context. And just like cooking a recipe, good context needs few specific ingredients. If you forget even one, the AI might mess up. So what makes up good context? Let's look at each ingredients in simple beginner friendly terms. The first one is system instructions.

This is just like the basic rule book you give the AI. For example, you say always write a clean code. Use British English. Start every response with hi there. These are the universal rules that the AI should always follow no matter what the task you give it. The second one is the user input.

This is your actual prompt, the question or command you give. Let's say for example, you say summarize this news article or build a weather app in Python. Simple direct. And this is what kicks off the task. Third one is shortterm memory which is the chat history. Think of it like the conversation you had so far.

If you're chatting with AI, you could say something like, "Can you write a report?" Then you say, "Make it shorter." The AI needs to remember the first request to understand the second. That memory of what you said, that's short-term memory or called chat history. Next, we have long-term memory. This is the memory from older sessions or saved preferences.

Let's say for example, you always want the AI to avoid using certain words or you have already told it your name and the job title. Now, if you have shared this before and it remembers, that's long-term memory. Not all AI tools have this yet, but the best ones like advanced agents do have. Next is knowledge bases.

These are the external sources of information the AI can use like documents, website or APIs. For say example, you're building a health app. You link the AI to a medical database or PDF guide. The AI will search through the material and use it to give smarter answers. Next, we'll talk about the workflow state.

This means where are we right now in a bigger process. Let's say you're building an app with AI. This first step would be planning. Second step would be writing code and the third step would include testing. Fourth step will be fixing the bugs. Now if the AI knows which step it's on, it can focus better.

Otherwise, it might try to do everything at once and that often fails. One small problem is that the AI can't hold unlimited information. It has a limit which is called the context window. So how do we fit in all of this without confusing it? That's exactly what we will talk about next, which is the problems with context and the smart ways to fix them.

When working with AI, giving it the right information called context is super important, but doing that isn't always easy. Here are a few common problem and simple ways to fix them. First is too much of information, not enough space. Of course, the has a limit to how much it can read at once.

Now, if you give it too much, it forgets or it gets confused. You can fix it by saying like summarize older or less important information to make space for what matters. Second is information overload. Dumping large chunks of unstructured text can overwhelm the AI. Then is wrong order of information. Now if the most important information is hidden, the AI might miss it.

Next is multiple sources which confuses the AI. If your project uses different databases or tools, the AI might not know which one to use. Then is messy memory. When too much random information is stored, the AI gets lost. So you can fix it by saying use memory blocks to organize what the AI remembers like facts, past charts or fixed rules.

Now these simple fixes help your AI give better, more accurate responses every time. Now enough of the slides. I'll show you a demo with Chptd. So first I've given a prompt to chat GPT to create a project plan for launching a new website. So as you can see that uh the GPT has given me this answer the project plan the project overview the goal timeline team and the phases you can see the phases they have directly given me the phases not it has not explained in depth and detail. Now uh we will use the custom GPT to add more context to this. So we can just head over to here and we can just select customize GPT and then you can just name project and just enter this. I have entered that you are a project manager with expertise in website launches. Create a detailed project plan with deadlines for each

phase considering the following. All right. And again you can just add more information from here if needed. Now this is the advanced attributity capability. So I have also selected that and I'll click on save from here. Now I'll click new chart and again I'll ask the same question that create a project plan for me for launching new website.

Let's see what it responds. So now as you can see it has given me project plan for launching a new website and this is the project goal and this is the phase one objective task and it has also given me the day wise uh time needed for doing each task and this is the phase two it has also given me the milestone from here.

So as you can see it has divided the phases into many parts and from it can just look at the task and the number of day the milestone achieved. So I'll just show you the difference between the um charge GPT customized using context and the normal charg.

So this was a normal charge GPT generic answer which had given me the phases task and just the normal things without proper date timing.

And when I used the customized chart GP it gave me the objective the task needed to do each phases with the days as well. So this is the kind of um answer you can expect from charge when you use context engineering here. Are you feeling like every other video, article or tweet you see these days is about generative AI? You're not wrong.

It's the tech world's current obsession. But instead of just nodding along wondering what it all means, what if you could actually master it? In this video, we are cutting through the hype, giving you a practical step you need to truly understand. I will be breaking down the entire journey into manageable phases from fundamental knowledge to building your own gen AI projects.

Before we begin, let me tell you what generative AI or Gen AI means. As the name suggest, it is generating data. It's a technology where it uses existing data for creating new ones. It might be in the form of text, images, audio, video, code or even 3D models. For better understanding, I will be dividing the entire road map into four phases plus one additional phase as a bonus phase for you to excel in this field.

Phase one is all about laying the foundation which includes the basics of AI and machine learning. Think of artificial intelligence as making computers smart enough to do things which people can do. Machine learning is just one of the way to do it. allowing computers to learn from example instead of being told exactly what to do.

It's a learning by seeing and doing it not just by a rule book. There are three main types of machine learning. Supervised learning, unsupervised learning and reinforcement learning. Supervised learning is where the computers learn with correct answers. Whereas unsupervised learning involves finding patterns on its own and reinforcement learning is about learning through rewards. You will need Python 2.

Key tools include numpy for numerical operations, pandas for organizing data and mattplot lib and seaborn for data visualization. These help you to work with and understand data. You need to understand data preparation which involves cleaning and organizing data. Also learn about the feature engineering which is the process of selecting important parts of the data of the model. Gen AI uses deep learning.

You can think of deep learning as a computer network with many layers. understand the basic idea of how these networks learn from data. This is the key for understanding more complex gen AI models. Moving on to phase two where you can start exploring the gen AI models. Some of Gen AI models includes GANs which are two networks competing, one making fake data and other tries to spot it.

This model works well for creating realistic outputs. The second model is VAEs. This model learns a hidden representation of data to create new or similar data. The next model is transformers which are excellent at understanding context in sequence like text, audio and even images using attention. The last model is diffusion model which can create highquality data by reversing or noising process.

This model is very popular for images. Apart from gen AI models, you need to cover text generation model which falls under natural language processing which is NLP. These models power text creation from writing articles to translating languages. And lastly, you will cover image generation using GANs and diffusion models.

In phase three, we will focus on developing our practical skills. You can start with small projects such as generating basic images or text with small data set. Do experiment and learn the workflow. Join platforms like hugging face, Reddit, Stack Overflow and Kaggle forums to ask question and learn from others to showcase your work.

You can use GitHub to share your projects and code. Phase 4 is truly about staying ahead. Gen AI is always changing, so you need to be aware of it. New architectures are constantly being built. So briefly be aware that more advanced models are continuously being developed beyond the basics that we have covered.

I would also recommend learning about deploying models on cloud platforms like AWS, Google Cloud or Azure which will be useful for larger projects. Additionally, you can read research papers, follow tech expert online whether on Twitter or LinkedIn and stay curious about this field. Continuous learning is the key in this field.

Now that you have built a strong foundation in generative AI, phase 5 is all about mastering the future of technology. A certification course provides formal validation of specialized knowledge and skills, making individuals stand out to employees and demonstrating a commitment to professional development. This recognition often leads to enhanced credibility that improves career opportunities with specific field.

Wondering which certification to go with? Don't worry, I've got you covered here as well. There are so many certificate courses available, but I would strongly suggest you to check on applied generative AI specialization course by simply learn in partnership with Perdue University and Microsoft. This will be 16week course which will give you hands-on learning experience on building AI and agentic AI apps through hands-on industry relevant projects.

You will also learn popular gen AI tools which includes gaining in exposure to copilot longchain hugging face Azure AI studio open AI and other tools. Additionally to this you will provide a certificate from Purdue University and course completion certificate from Microsoft. Hurry and enroll now. Find the course link in the description box and pin comments.

That's your road map to understanding generative AI. Start with the basics and get practical experience and never stop learning. The power of Gen AI is within your reach.

>> Have you ever wondered how AI models can understand and process multiple types of inputs like text, images, and audio simultaneously? Hello everyone.

Welcome to our deep dive into multimodal prompting, a transformative approach that's reshaping the landscape of artificial intelligence. My name is Bassel Dlala and in this video we'll explore what multimodal prompting is, why it's essential, and how it's applied across various industries. Let's embark on this journey to uncover the future of AI together.

Here's our agenda. Introduction to multimodal prompting. Evolution of prompting. Core concepts of multimodal prompting. tools and platforms that support multimodal prompting, use cases and applications, getting started with multimodal tools, and then we'll take a look into simple multimodal prompts in action, and we'll have a deep dive or step-by-step prompt building.

Last but not least, we'll look into experimenting with different use cases. You might be wondering, what is multimodal prompting? Multimodal prompting is a technique that allows AI models to interpret and respond to input from more than one modality like combining text, images, audio, or video. Instead of just reading text, the model can basically analyze visuals, sounds, or other forms of data to generate richer, more contextaware responses.

It basically mimics how humans process information. Now, why it matters and where is it used? Multimodal prompting matters because the real world isn't just textbased. We live in a sensory rich environment and multimodal AI helps machines understand that it's used in applications like image captioning, video analysis, interactive chat bots with visual context, and even healthcare diagnostics where images and notes are combined.

In addition, multimodal prompting is very useful for content creation and recommendation. Let's talk about the evolution of prompting. Let's learn about the

evolution of prompting. From textonly models to multimodal models, prompting has evolved from simple text inputs to complex multimodal interactions.

Today's models can understand and respond to combinations of text, images, audio, and more for deeper context. Prompting began with textonly models like GPT2 which responded solely to written input. They basically leveraged NLP which stands for natural language processing. Over time models evolved to process images, audio and even video giving rise to truly multimodal systems that can interpret and connect across different types of information.

How models understand and process multiple input types. Modern multimodal models have specialized architectures such as vision transformers, multimodal neural networks to process different inputs. They are also trained on paired data like images with captions or videos with transcripts. These models learn how different modalities relate to each other, enabling them to take in multiple types of inputs and generate informed outputs that bridge those sensory channels.

Crossmodel attention helps models link relevant features across inputs, enhancing context and accuracy. At the core, these models use shared embedding spaces to map inputs like text and images into a common understanding. For example, a picture of a dog and the word dog are brought close together in this space allowing the model to reason across formats in a unified way.

Let's talk about core concepts of multimodal prompting. Modalities are different types of information we can give to AI like text, pictures, sound or video. They help the model understand things the way humans do through multiple senses. These senses are text with written language input like prompts or documents, image, visual content such as photos, charts or drawings, audio, soundbased input like speech or music and video timebased visual and audio content combined.

In multimodal prompting, in unimodal prompting, the model receives one type of input, typically text. In contrast, multimodal prompting involves multiple types, like combining a question with an image or a diagram. So, input modalities include one or more types. This requires the model to integrate different streams of information and make connections between them.

So let's recap the key difference between unimodal and multimodal prompting. First, we've got the input type. Unimodal prompting has one single type and typically is text. Multimodal prompting has multiple types and it could be text, images, audio, etc. Second, context depth. Unimodal prompting is limited to one form of input.

On the other hand, multimodal prompting is richer in context and has combined inputs. Three use cases. Unimodal prompting can only be used on textbased tasks such as writing. Multimodal prompting can be used in multiple tasks and complex ones like image analysis cut like image analysis and speech to text four understanding.

Unimodal prompting is linear and it's language focused. Multimodal prompting on the other hand is crossmodal and humanlike. Despite its potential, multimodal prompting comes with challenges. Models can misalign inputs, generate hallucinations, or be biased towards dominant modalities. Input format issues from the model misinterpreting poorly formatted inputs.

Two, limited modal support. Not all models handle every modality. Three, ambiguity inputs. Unclear inputs can lead to inaccurate responses. Four, high computational load. Multiple modalities require more resources. There's also lack of standard benchmarks and combining training data from different formats can be resource intensive.

Let's discuss tools and platforms that support multimodal prompting. First, we've got one of the most popular generative AI tools, Chat GPT. Specifically, Chat GPT with vision.

This version of Chat GPT can understand and analyze both text and images. It's useful for tasks like generating captions or answering questions about visual content.

If you're not familiar with chat GPT, it's basically an AI chatbot based on OpenAI's GPT, which stands for generative pre-trained transformer models. It can understand and generate humanlike text for tasks like writing, coding, summarizing, tutoring, and casual conversations.

And now it has multimodal capability. It's widely used in education, business, and creative work. Chat GPT can handle multiple cut. Chat GPT can handle multimodal inputs by combining image recognition with text generation, allowing users to ask questions or generate text based on images. It processes the image and merges it with text for a complete response.

The second next popular tool is Gemini by Google. Gemini is Google's family of advanced AI models designed to rival GPT. It's multimodal, meaning it can handle text, images, code, and other inputs. Gemini powers tools like Bards and Google Workspace AI, aiming to provide accurate integrated assistance across Google's ecosystem.

It helps with complex interactions like visualbased search and multimodal content generation. It's designed for broader contextaware AI applications. It handles multimodal inputs very quickly and uses advanced fusion models to integrate text, image, and video inputs, understanding relationships between them for more contextaware responses and generation.

Next, CLAG is a conventional AI developed by Enthropic focused on safety, helpfulness, and transparency. It's known for its thoughtful responses, and is designed to be steerable and aligned with user intentions. It can also integrate text, images, and documents, allowing for in-depth analysis and interaction with various input formats.

Claude is a strong competitor to chat GPT and excels at tasks like PDF creation, Q&A, and content summarization. Claude also does a good job of breaking down each modality for text, images, and documents separately. It processes them and then combines them for tasks like document analysis or multissource content generation.

Dolly by OpenAI is an AI model that generates high quality images from text descriptions. It's known for its detailed creative outputs and is integrated into chat GPT's interface for users on the plus plan. It basically accepts textual prompts and translates them into relevant images, seamlessly merging descriptive text with visual output for creative image generation.

It's got a lot of tools that help you build a scene and will turn it into an image from realistic to fantastical. Let's talk about use cases and applications in multimodal prompting. We have users that can upload images such as diagrams and charts and then ask questions about them with AI providing contextaware answers.

Next, AI creates articles, social media posts, or marketing copy based on both text and image inputs. In terms of content generation applications, AI can create articles, social media posts, and marketing copy based on both text and image inputs. for product tagging e-commerce platform. Use images and descriptions to generate accurate product tags and recommendations.

Multimodal prompting can also be effective for interactive education. AI helps in learning by analyzing images, videos, and text to provide interactive educational content. It's also useful for accessibility. It assists visually impaired users by converting visual content such as photos and infographics into descriptive text.

Let's see how we can access and set up a basic multimodal prompting environment. First, choose a platform. You can sign up for access to platforms like chat GPT or Gemini. Next, install required tools and install necessary libraries. Then obtain API keys for the platform and configure your environment.

Ensure inputs are properly formatted and compatible for the application. Lastly, test with basic prompts or run simple multimodal tasks. Let's talk about the required tools and input formats. First, you need an API access. Second, install some Python libraries and have a Jupyter notebook. Also, you can use cloud platforms, although some of these applications can be done using your local machine.

Here's a list of the input formats that you can leverage. Got text, image, audio, and video. Let's jump into a demo to show how LLMs can be used for multimodal tasks. In this demo, we'll be using chat GPT to showcase its powerful multimodal capabilities. Keep in mind that most LLM tools like Gemini and clogged can also provide similar capabilities that I'm about to show. for Chat GPT.

Many of the capabilities we're about to showcase are only available in Chat GPT Pro. It's available for a limited number of prompt in the free version. If you want to leverage the full capabilities, you'll need to subscribe to Chat GPT Pro. First, after you log in to Chat GPT, you need to make sure that you have GPT4 activated.

Next, you'll see that there's a plus button that is popping up where I can add media. But before we do this, let's generate media using text prompts. Write a three sentence summary of the plot of inception like you are explaining to a 10year-old. You can see chat GPT gave us a response with a concise three-s sentence summary of the plot of Inception.

So I'll read the first sentence. Inception is about a man named Cobb who goes into people's dreams to steal their secrets like a spy in their sleep. Let's take things to next step. I'm going to ask chat GPT to generate a poster based on the description in cartoon style. Let's make sure that it's pointing to the description above.

Go ahead and run this and we'll give it a few seconds to generate the image. Keep in mind that chat GPT sometimes runs under a heavy load of requests and sometimes image generation takes time. Image generation is done. You can see chat GPT generated an output according to my description.

A cartoon like poster of inception. Next, we'll show chat GPT's ability to analyze images. I find this task very useful for analyzing complex visuals such as stock charts. I'll click on the plus sign and then add photos and files. Select your photo and then insert the following prompt. Analyze the support and resistance levels of the attached 4hour interval chart of Google.

You can see that chat GPT was able to analyze the chart for me in an instance and was able to give me multiple support levels and resistance levels with their descriptions. Let's do another example of image analysis. I'm going to click the plus sign here and upload a handdrawn chart. And here is my prompt. Explain the attached image of a diagram step by step.

For reference, here's the image and here's the explanation. Start of the process. The program begins execution. Step one, input value. Gives me check for stack overflow. And then it explains the rest of the steps according to the diagram. It was also able to analyze the gist of the diagram. It is a standard stack push operation with overflow checking where it prevents adding data when the stack is full and confirms insertion when space is available.

It can also offer an equivalent code in Python or C. That was a good example for visual and text reasoning. Let's increase the level of difficulty of this type of task by giving it a mathematical problem. I'll click on the plus sign again, add photos, and select my image. And I'll use the following prompt.

What is the answer to the math problem in the attached image? Here's the image for reference. Find the sum of the series. And you can see that chat GPT is generating latex

or markdown type of syntax in the background so it can display the mathematical syntax and Greek symbols properly.

Not only was it able to solve for these problems, but also was able to give me a stepbystep instructions on how to solve for it. Plus, it recognized that the we have two problems, not just one to solve for. In this last example, we're going to do the opposite of multimodal prompting. Instead of providing an image, we're going to ask chatgpt to generate a complex image based on my description.

Provide a detailed process for data science and feature engineering for or the diabetes healthcare data set. The diabetes healthcare data set is a popular data set available online. So chat GPT will be able to understand exactly what I'm looking for. Here we can see step-by-step instructions of the process along with Python code on how to run the analysis such as EDA, outlier detection, multivaried analysis and then feature engineering as requested all the way to modeling and providing proper evaluation metrics. After getting the steps in text format, I want chat GPT to build a visual diagram of those steps. In its early stages, artificial intelligence resembled a young child discovering the

world. At first, it could only observe and label. Much like a baby learning colors and identifying objects. This is blue, and that's a dog. It could recognize and repeat patterns, but lack imagination, the spark to create something new. As it matured, AI learned to think deeper to find connection, predict outcomes and understand meaning.

It became intelligent, yes, but still confirmed within the boundaries humans set for it. It was a remarkable student, but yet not an artist. Then something extraordinary happened. AI evolved not just to understand, but to create. It crossed a new threshold and became what we call it now generative AI.

an intelligence that can write stories, compose music, generate images, and analyze data and even code entire programs. AI had become a creator capable of imagination

rooted in knowledge. And this is where our journey begins. If you are enjoying learning how generative AI works and want to take this knowledge to a professional level, there's a fantastic opportunity waiting for you.

The applied generative AI specialization by Purdue University and simply learn is a live hands-on program where you can actually build real gen AI and agentic AI applications using tools like lang chain Azure AI studio and hugging face. It's a 16-week live online certification taught by industry experts and Purdue faculty and you even get a perurdue alumni association access for global networking.

By the end, you'll have seven plus projects under your belt from AI powered HR assistants to rag based BI dashboards. So if you're ready to turn your learning into real career edge, check the link given in the description box below and in the comment section and explore the program. Before we move ahead, here's a quick quiz for you to get started with.

Which of the following best describes generative AI? Is it analyze data only, creates new content, or predicts outcomes or stores data? Let us know your answers in the comment section below. Now, let's see why prompts matter. If your instructions are vague, the AI can give you a random or useless answer.

But a clear, structured prompts give you exactly what you want. We'll explore the origin and the essence of this transformation. We'll uncover how generative AI differs from traditional AI, how it learns to create rather than just classify, and the key breakthroughs from GBT models to diffusion systems that shaped its rise.

We look at the world through AI's eyes, understanding the power of large language models and neural networks that allows machines to think, reason, and create like never before. We shift from understanding to interaction. Here we learn how to speak language of AI through prompting.

Every grade output begins with a great question and every question can be shaped to unlock creativity, precision, and insight. You will learn how to design effective prompts, refine them through practice, and experience first and how small changes in wondering can transforms AI's response.

This is where creativity meets control, and you become the guide. We'll see how generative AI becomes a powerful alley in creation and learning. Whether it's researching complex topics, summarization, vast information or creating content from articles and visuals to fully presentation, Geni becomes your collaborative partner.

You will explore how to use AI tools responsible maintaining originality, accuracy, and ethical awareness. Here you will begin to realize that AI doesn't replace human creativity. It amplifies it. We'll bring everything together, learning how to apply generative AI in real world. We'll uncover how AI turns data into stories, helping us find meaning hidden within numbers.

We use it to brainstorm, innovate, and design new solution faster and smarter. You will see how AI integrates into workflows, boosts productivity, and inspires new forms of storytelling, analytics, and innovation. As we look ahead, we glimpse the future of agentic AI and autonomous systems that can reason, remember, and collaborate alongside us.

From learning what AI is to mastering how to communicate with it to discover how to create and apply with it. This course takes you through every stage of generative AI journey. Because the future of intelligence is no longer about machines that follow, it's about the machines that co-create. and you are about to learn how to shape the future.

Not long ago, the ability to create art, music, or stories belong solemnly to humans. Today, algorithms can draft poems, paint lifelike portraits, compose melodies, and even

hold natural conversations, reshaping our notation of creativity. uncover how generative AI evolved from basic pattern recognition to a groundbreaking systems powering creativity at scale.

We will reveal how these technologies learn not just to analyze but to imagine and generate new content that touches business, education, entertainment and daily life. Generative AI or Gen AI is a special branch of artificial intelligence that does not just analyze information. It actually creates new things from it.

It can write essays, generate realistic images, produce videos, compose music, and even help design new products. You might have already experienced it without realizing like when you use chat GPT to help you writing something or when an app turns your selfie into digital painting using AI.

Now, how do we define generative AI? Generative AI is a branch of artificial intelligence that focus on creating new content or data based on patterns and knowledge it has learned from existing data. Unlike traditional AI which primarily analyzes classifiers or predicts outcomes, generative AI can generate original output such as text, images, audio, video, or even code.

Generative AI learns from lots of data to see patterns and relationships. Using this knowledge, it can create new content that makes sense, is relevant to the context, and feels original. Similar to how a chef studies countless recipes to invent a new dish, or how a musician listens to many songs to compose a unique melody, Gen AI can write stories, generate artwork, compose music, or even produce realistic images, all based on what it has learned.

In simple terms, generative AI is an AI system capable of imagining and creating rather than just recognizing or analyzing. Okay, now you have an idea of what generative AI can do. But how do you think this is possible? Imagine AI as a magical robot with brain that

can learn from everything it sees and hears. Just like we humans learn by reading books, listening to music or even watching videos, AI learns from huge amount of data.

It notices patterns, figures out relationships and remembers what works together. Now this robot has different superpowers for different tasks. The first power will be the text power which writes stories, emails or even code. For example, we have chat chibeti. These kind of models are called text generative models. The second superpower is image power which creates drawing illustrations or even realistic photos.

And some examples are Dal E and midjourney and these models are called image generation models. The third superpower is audio power which composes music, voices or even sound effects. These models are called audio generation models. The next superpower is video power which makes animations or even short films. And these kind of models are called video generation models.

And the final power that the robot has is the code power which helps the programmers write or complete code quickly. And some of the examples are GitHub copilot. These models are called code generation models. But powers alone aren't enough. Every superhero needs tools. AI tools are the technologies that makes this power work.

Let's see which tools generative AI uses along with its power. The first one being neural networks. They are like the robots brain cells. They help in recognizing patterns and connect ideas. The second one is large language models which are LLM. They are kind of language brain that understand text context and can generate coherent writing.

And the third one is generative adversarial networks which is GANs. They are the artistic eye that help it create realistic or imaginative images. The next one is variational autoenccoders which are VAEs which help AI handle complicated structure and organize complex information. And the last one is foundational models.

They are the generalpurposebased models that can be trained for many tasks from text to image to data analysis. So whether AI creates something like a story, a picture, music, video or even code, it's using its power with these tools. The powers decide what to create and the tools make sure it can learn, understand and produce something original and meaningful.

In short, generative AI is like a magical robot friend. It learns from the world, uses its powers, applies its tools, and brings ideas to life. Now that we understand how generative AI work, let's see what it can do for us and what we need to be careful about. Generative AI brings incredible opportunities, but it also comes with some risks.

On the bright side, it can save time, boost creativity, and help people solve problems faster. Businesses can generate reports, marketing content, or even product design in minutes. Students and educators can get help with research, learning, and personalized tutoring. Artists, writers, and musicians can experiment with ideas they might never have imagined.

But AI isn't perfect. Sometimes it can make mistakes like giving wrong information or creating content that does not make any sense. There are also ethical concerns like copying someone else's work, spreading false information, or even invading privacy. And that's why it's important to use AI responsibly, making sure it's fair, transparent, and respects people's rights.

Generative AI is already making a huge impact across many sectors such as business, which helps companies write emails, create ads, design products, and even automate customer services. In education, it's helping provide personalized tutoring, generating practical exercises, and helps explaining complex topics in simple ways.

While in entertainment, it is helping create music, movies, video games, and artwork, giving creators new ways to express themselves. And some of the other sectors include

healthcare, finance, research, use AI to analyze data, generate reports, or assist in discovering new solution. Generative AI isn't just a futuristic idea anymore.

It's a tool that's already shaping how we work, learn, and create. Have you ever been frustrated by AI's unexpected answers? What if the secret to getting better AI response lies not only in the machines, but how we ask? This is where prompt engineering comes in place. We'll unlock the power of effective prompting, which is the key to steering AI towards answers that are precise, creative, and truly useful.

We'll dive into the art and science of prompt engineering, which are the skills of crafting instructions that guide AI to produce accurate, relevant, and creative outputs. You'll also learn not just what prompts are, but also structure them effectively. Experiment with different approaches and troubleshoot when the results aren't quite good.

Through hands-on practice, you'll also get to see these techniques in action on platforms like Chat Chibeti, GitHub, Copilot, and other Gen AI tools that are gaining the confidence to interact with AI like a pro. A prompt is nothing but essentially a set of instruction you give to the AI. Think of it as a way you guide a student, an artist or a chef to create exactly what you want.

The better the instruction, the better the output. Every imprompt has a few key parts. The first part is instruction, which says what you want the AI to do. And the second one is context, which are any background or information that the AI needs. The third one is the input data which are nothing but the raw materials the AI should use.

And then we have the output format which is how you want the final results to look alike. Without these components, it's like asking someone to cook dinner without telling them the ingredients or what cuisine you want. Now, let's see why prompts matter. If your instructions are vague, the AI can give you a random or useless answer.

But a clear structured prompts give you exactly what you want. For example, if I give a very simple prompt which is write a story about a robot. So let's see what it generates. Notice it's very vague. The AI doesn't know what kind of a story we want, how long it should be, or what details to include. Now let's refine the prompt.

We will add instructions for the current prompt. We'll say write a story about a robot learning how to paint. Now what are the condition? I'll say make it five sentence long and include a funny name challenge that the robot faces while painting. Now in the third condition, let's say end it with a positive or surprising twist.

Now here in this prompt we have added instructions. The robot is learning paint. We want exactly five sentence, a funny challenge and a positive twist by the end. By giving more context and structure, we guide the AI to create something much more specific and engaging. And check out the results.

It's much better. By refining our prompt with context and instructions, we had a story that interesting, funny, and exactly the length we wanted. Prompt engineering is all about asking better questions. Here are some strategies. The first one is to be specific. Instead of write an email, say write a professional email to a client explaining a delayed shipment.

And the second strategy is set the role. Ask the AI to act as a teacher, coach, developer, or a designer. For example, you're a marketing expert. Write a catchy slogan for a vegan snack brand. And the third strategy is use few short examples. Show the AI few examples of what you want before asking it to generate something new.

So for example, we have fast food logo which can be done with red and yellow burger icon. And for a tech company logo, you can add blue and white abstract design. Now try creating a logo for a fitness brand yourself. The fourth strategy is system messages. Explain highlevel instructions to guide the AI's behavior.

For example, in chat GBD, you can give always respond in a friendly tone. Now, let's apply whatever we have learned and create a simple chatbot that can answer questions about movies. So let's give some clear instructions. You can use chat gypy or any other LLM model for this. So first let's specify the role. Let's say you are a movie guide and uh the instructions being instructions being you should provide uh three sentence summary and one fun fact and then we have to provide the context.

So let's do that. Uh let's say the user might ask let's specify how our output should be which is keep it concise and engaging. So as you can see it has updated the memory. So it's asking us to feel free to ask about any movie. So it has automatically changed to a chatbot that acts like a movie guide.

So let's say I ask about a movie. Can you describe the movie Inception? Let's ask about the movie Inception. So as you can see it is providing me a concise explanation of what the inception movie which is shot in 2010 and it is giving me a fun fact also which is Christopher Nolan. The director used a unique visual effect in inception.

The folding city scene by creating a practical set allowed buildings to bend blending practical effects with CGI to create a sural dreamlike experience. So without any coding your chatbot is ready to use. By giving a clear instruction, a clear prompt, the GPD understood its assignment. Even experts don't get it perfect results on their first try.

Sometimes the AI misunderstand what you mean, gives you incomplete answers or goes completely off track. And that's totally normal because working with AI is an iterative process. And for this you can experiment with different prompt versions, adjusting wording, adding examples and checking how each change influences the results.

This process helps you understand what makes the AI think better. Suppose you write a very vague prompt such as write about a climate change. So let's see what the chat GPT gives us. Observe that AI gives an explanation which is not wrong but it's not focused either. there is no clear goal or format.

This happens when our prompt lacks direction. Now let's consider another example which is explain AI. Now again this will give a vague answer cuz there is no proper instructions or any context given to this. When prompts are too general AI doesn't know what level of detail or tone to use. It is unclear if this is for a child, a researcher, or a business audience.

Now, let's alter one of the prompts and make it even more clear for the model to understand. Instead of writing about a climate change, let's give a more detailed prompt. For example, let's give explain climate change for a 10year-old student. Using simple language and one example here we have added context who the audience is and what level of language to use and the response is short and clear explanation.

Now let's give the same prompt with some structure like let's say explain the climate change for a 10-year-old student again. for a 10 yearear-old student in three short paragraph where the first one will be what is it? Second one will be what causes it and the third one will be what you can do about it. This kind of prompt help organize the response in a more readable way.

the response will be neatly formatted in three sections. Another way is to specify the output format. For example, again let's go with the same example. Let's say explain climate change for 10 year old student in bullet points. with emojis. Let's say with one emojis, at the start of each point, we are telling the AI exactly how to present the answer. bullet points,

emojis, and the tone. Watch how clearly it follows our instructions. Now, the output is neat and engaging with emojis. Now, compare this version to the first one which we have

received. Clearly, you can see the difference between the first prompt and the final version of it. The output is more clear, structured, and engaging.

Now, sometimes even when your prompt is clear, the AI still makes mistakes. Let's look at few common cases and how to fix them. The first scenario is where the output is too. For example, if you give the prompt suppose explain Python function, the prompt which we have given is clear prompt.

The answer would be too lengthy for you to go through. So in such cases keep it under four sentence for the summary which will give you a more concise version of it. So let's give the prompt. So here you go. It'll generate exactly four sentence to summarize the functions of a Python. The second scenario is where the answer is being off topic or irrelevant.

So consider the prompt to write a story about data. So let's just give write a story about data. This prompt will give you random topics which is irrelevant to the data. So in such cases you can write prompt like 100word story about the data scientist. Let's say about a data scientist discovering a pattern. Okay.

Now this will make sure that the topic is not irrelevant. The last scenario is missing detail or creativity. For example, let's give the prompt of describe a city. Now the answer is very plain description whereas you can modify the same prompt and say describe a futuristic city. So let's do that. Describe a futuristic city in 2050 focusing on technology and nature.

This will give us more creative answer of how 2050 city will look like purely from the imagination of the AI. In a world moving faster than ever, how we can learn, research and create content efficiently. AI as your partner to accurately discover, amplify creativity and craft high quality work in less time.

all while maintaining originality and ethics. That's exactly what generative AI can do when used for research, learning, and content creation. This module will show you how to use AI not just as a chatboard, but as a true research and creative partner. AI can help you discover information faster, learn better, and create smarter.

In this module, we'll cover four key areas. The first one is AI enhanced research. how to use Gen AI to search, summarize, and organize research materials or data. Second, we learn AI for learning. How AI can act as a study partner, explaining topics, answering questions, or creating personalized learning materials.

And the third one is AI for content creation. How to generate articles, scripts, images, and presentations using AI tools. And the fourth one is advanced prompting for content. How to design prompts for specific needs like SEO, social media or even academic writing. And the fifth one is tool comparison.

A quick look at some of the best AI tools for research, for creating like chat GPT, Bing, C-Pilot, Gemini, and others. Now let's compare two of the LLM models and see the difference. I'm here going to compare Chat GPT versus Perplexity. So let's give a prompt to chat GP which is summarize recent research paper let's say research on LLM memory architectures It will give you a clear overview covering techniques like retrieval, augmented generation,

episodic memory and vectar database. Now the same thing I'm going to compare it with perplexity pro. So I'm going to enter the same prompt. So let's see what's the difference. You'll see it provides the same content but live web citation making it more credible for academic references. Once we find research papers, the next challenge is to summarize and synthesize them.

There are two types of AI summarizations. The first one is extract summarization which selects key phrases or sentence directly from the text. While abstractive summarization rewrites the context in its own words, preserving the meaning. Let's consider an example.

Let's take an abstract of a research paper and paste it in the chat chip and ask to summarize this paper on transform efficiency in five bullet points.

You will see a science summary with methods, findings and limitation. Then ask it rephrase it in a structured note format with sections, objective, method, results and limitations. This turns unstructured text into usable research notes into perfect for academic or even professional documentation.

Now if you ask me how LLM models are able to do this, this is because they perform summarization using attention mechanism which is identifying the most contextual relevant tokens in the text. That's how they can understand which part matters the most. Research is not just about reading. It's about connecting ideas.

AI can help to organize knowledge into themes, tables, and concept maps. Now suppose if you have five research papers about AI in education. Let's ask Chat GPT to create a table summarization each paper with columns. So first you can upload all the five research papers and give this a prompt saying create a table summarizing each paper with columns and the columns would be author And then followed by topic followed by key insight

and then research gap. So let's give this So as you can see the model generates a structured comparison instantly. Here are some of the tools you can use. The first one is notion AI for turning notes into organized databases. Then you can use scholar C which is automatically summarizes PDFs and highlights important sections and reuse earlier notes for continuity. AI doesn't just store data.

It helps in knowledge structuring, a crucial skill for both academic and professional research. Large language models or LLMs are incredible versatile at generating written content. Whether it's an email, blog post, essay or video script. These models understand context, tone, and audience.

Which means you can customize output easily. For example, let's give a prompt which says write a 200word blog on AI in education in a professional tone. So let's see what it generates tone. Your output will be ready. If you want to refine it further, you can just ask it, make it sound more professional. Let's say persuasive.

Make it sound more persuasive. Add a call to action and format it in mult. Now here the LLM adjust vocabulary, tone and structure to match the prompt. The next essential part is visual storytelling which is another domain revolutionized by AI. Text to image models like midjourney, DLE and adob Firefly can convert text prompts into realistic or artistic visuals.

Now let's give a prompt to chat GPD which has inbuilt DLE for creation of images. Let's ask the chat GPD to create an image of a futuristic classroom where AI tutors, students Let's say digital holograms on walls. And let's add a cinematic lighting. In seconds, the AI generates stunning

images. You can modify your prompt to control. Say the style. You can set it to realistic, anime, or even watercolor. And lightning also you can change it for cinematic or even soft light. And also perspective also you can change by mentioning if it's a top view or a close-up. Now let's talk about multimodel tools like Canva AI, tome, gamma, and slides AI.

These platforms merge text, visuals, and design automatically generating complete presentations from single prompt. Now let's consider Canva AI. Now let's give a very simple prompt to Canva AI such as create a five slide presentation on say the topic the future of AI and healthcare. So let's just generate.

So here it is asking which type. So I let me go with minimalistic. And here you can see it is asking me what is the flow of the PPD presentation. So you can do changes here also

so that you don't face any issues further. So you can modify it. If everything is seems all right, you can just generate design.

Now, as you can see, within seconds, the AI generates a title slide, key insights, data charts, images aligned with the theme. Now, think about this. A presentation that might have taken hours of work, hours of creation, hours of ideas to align this kind of presentation just took about few minutes.

The effectiveness of your AI or code depends on how well you craft your prompts. Different formats need different prompting style. For example, the SEO prompt should be something like write a SEO blog on AI tools for students. with keywords. Let's add some AI learning assistant, let's say productivity, and even one more study tools.

And this is how easily you can create a blog. Whereas if you want any social media prompt, you can just say create three catchy. Let's create LinkedIn about AI in content creation. Now here you go. You can just copy and paste it in your LinkedIn. And here you have Viral 3 pros just created within seconds.

Or if you want an academic prompt, you can just say, "Summarize this paper in API format with a text citation." So you can just add a document and ask it to summarize this paper in APA format with text citation. Now what about if you want to generate a script you can just mention write a YouTube script explaining ethics

in generative AI in an engaging tone. And there you go. You have an entire YouTube script that you can just read it and get your content creation done in a matter of few seconds. Always specify audience length and format. This gives the model clear direction to take your results from good to great.

Use advanced prompting methods. Some of them are chain of thought prompting which is ask the model to think step by step. The second one is role prompting which is to give prompt as to act as a research assistant summarizing a scientific paper. And the third one is output format control which is to provide results in a markdown table or return output in JSON format.

These help in making the results structure, reusable and consistent. Now let's put all our knowledge that we gained and let's identify new trends in healthcare in India. How to do it with the help of chat chipity and perplexity. Let's say I need to identify new trends in healthcare in India.

So first I will choose perplexity because its AI is connected to a live web. So it's ideal for upto-date trend identification. In publicity AI, I will give a prompt which says what are the latest emerging trends in India healthcare sector. Let's mention the range also. Let's say from 2023 to 2025. Here you can see you get live news insights report from WH, NITI, AIOG and also Indian Ministry of Health.

You also get citation from trusted sources which will be mentioned. Let's say these are the general trends that we have obtained. Now to explore more deeper, give more prompts such as explain the growth of AI and digital health startups in India. So it is answering all the questions. It's giving role of AI, healthcare, digital health subsegments, modules, policy, infrastructure and demand driver challenges and future related things.

Now let me ask a questions. What are the current challenges? So what are the current challenges in teley medicine adoption in India again it'll give more and more information so here you can see it is about teley medicine clinical and trust related concern, data privacy, regulation, governance, operation and demand side issues.

And let me just summarize this. Summarize recent governmental initiatives for digital. healthcare in India. The more prompting questions you ask, the deeper

knowledge of each point. So these are the insights that we have received from Perplexity. Now you can just copy and paste this insights to chat GPT.

So let's just copy the entire insights that Perplexity has given. Let's open a new chat. Let's just paste it there. And let's just ask chat to organize these healthcare trends from India into three major categories. group them into let's say group them under headings such as technology policy infrastructure two and patient care

and also ask it to explain each trend briefly. Now this is asking chat GPT to summarize or tabulate trends for easier visualization. Now you want to create a table based on this information. So you can just mention over there as a prompt create a table summarizing key healthcare trends in India with columns.

Let's also name the columns so it is neat. The first one let it be trend. Then let's say description little impact and let it add an example and also I need the source. This will give you a clear comparative overview you can use in the report or even in a PPT. So let's just copy and paste it in Canva AI and mention create a five slide presentation on emerging

healthcare trends in India. Let's give it 2025 outlook. uh let's also provide with some instruction which is include key categories such as digital health AI diagnostics tele medicine and policy reforms. So again it's going to ask us which type

I'm going with minimalistic and submit. Here you can see style is minimalistic audience professional and length is 1 to5 and again we have the overview. If you want any changes, make sure to adjust it here. So we have introduction page where so much information will be there. Digital health and awan digital mission is given.

AI diagnostic transforming healthcare tele medicine policy reforms everything is mentioned. So I think it's good to go. So let's just generate the design. So as you can see Canva AI here have already given you a clear PPT for you to ready to present. This entire process will take few minutes and we have a list of top five to seven emerging healthcare trends in India.

A structured table summarizing key insights, impacts and examples. A short blog or report on emerging healthcare trends in India. Optionally a five slide Canva presentation for visuals. This would have taken weeks of research and analysis which was achieved in just minutes. And that's the power of generative AI. Now imagine transferring raw data into compelling stories that drive smarter decision and spark innovation.

Generative AI is not just about creating content. It is about being a powerful partner in analyzing information, uncovering insights and streamlining workflows. Think of this as moving from using AI for creativity to using AI for decision making and innovation. You'll learn how to turn raw data into meaningful insights, how to communicate those insights through storytelling, and how to use Gen AI tools to design better workflows and automate part of your work.

We'll also take a look at what's coming next. The future of agentic AI, autonomous assistance, and how we can build innovation responsible as AI becomes more capable and independent. So, by the end of this module, you'll know not just how AI creates, but how it thinks with you and help you analyze, narrate, and innovate with purpose.

Let's begin with one of the most powerful ways to apply generative AI, which is turning data into insights. Now, we all deal with data, whether it's survey responses, sales reports, student feedback, or even social media analytics. But raw data by itself doesn't tell a story. It's just numbers and text.

The real value comes when we can interpret the data, when we can find meaning, patterns, and even insights that guide better decisions. That's what we call moving from

data to insights. And the exciting part is generative AI makes this process faster, simpler, and more accessible than ever before.

When we say data to insight, it simply means the process of taking raw information, cleaning it, understanding it, and drawing meaningful conclusions. For example, imagine you have a data set of customer reviews. The data might include names, ratings from 1 to five, comments or feedback text.

By itself, that's just information. But once you analyze it, you might find that 70% of the customer love your service speed while 30% complain about pricing. That insights helps you make decision. It may be adjusting pricing or add a discount strategy. That's exactly what generative AI help us to do which is quickly move from a pile of information to something we can actually act upon.

To understand how AI helps, let's first know the two main types of data. The first one is structured data. This is a neat organized like a spreadsheet with rows, columns such as customer ID, sales, amounts, dates, etc. The second type is unstructured data. This data is messy and doesn't fit into tables. For example, things like emails, reviews, social media posts, or even video transcripts.

Traditionally, analytic tools like Excel or SQL are great for structured data. But unstructured data, especially large volumes of text, takes hours to analyze manually. And that's where generative AI shines. It can read, summarize and extract patterns from both structured and unstructured data, even thousands of lines at once.

One of the first step in analysis is data cleaning. You might have missing values, duplicates, and even inconsistent formats. Let's begin with one of the most powerful ways to apply generative AI, which is turning data into insights. AI tools like Chat, Gibbidity, Gemini or Claude can help you detect and fix such issues instantly.

For example, you can paste a messy data set and say the LLM, clean this table, remove duplicates, correct spelling errors, and summarize the missing values. Or if you have a CSV file, you can just upload it and ask, can you identify any anomalies or incorrect entities? So, let's do that. So I have a file which I can upload or I'll just copy and paste this data set and ask it to clean the table.

Remove duplicates. Correct those spelling errors and summarize. Missing values. So as you can see within seconds AI can flag issues like mismatch dates, duplicates, names or even outliers saving hours of manual work. So AI becomes like your data assistant handling the repetitive parts while you focus on the decision making.

After cleaning comes the analysis part. Here generative AI can perform descriptive summaries, detect patterns or even suggest hypothesis. Now let's say you're analyzing sales data. So let's upload the sales data and you can ask chat chip what are the top performing products by region in this data set.

So, let's ask So as you can see it has given us top performing products by religion. Repeat religion. So as you can see it has given us top performing products by region and this is done in few seconds. Now let's take another example where we have to summarize key themes and customer sentiments in the feedback form.

So let's say I'm running a business and I have a feedback and I have a feedback of 50 customers. So I'll just copy and paste the feedback and ask tragibility to analyze this customer feedback and list the top three recurring themes. uh also I'll ask it to suggest a positive improvement. Now here AI scans through 50 or even thousands of entities and instantly summarizes results. Like here you can

see it is mentioning the website and checkout experiences. Many customer commented on the website navigation, check out process and issues like freezing or on the check out page. It is also talked about product quality and description, customer service and support. And it's mentioned these are the recurring themes highlighting the importance

of improving user experience both online and post purchase to maintain customer satisfaction.

So instead of just reporting numbers, AI helps you understand the story behind them. That's a complete insight derived automatically from unstructured text. Now you can ask a tool like chat chip or Gemini to create a short summary paragraph I can use in management report. Now with these insights you can ask chaty or any other LLM model to create a short summary paragraph so you can use it to management report.

So let's give the prompt report. Now you can see AI has produced something like customer feedback highlights key areas of improvement including the website and checkout experience, product quality and description and customer service. Many users expressed frustration with the checkout process, slow customer support and issue with product description and packaging.

To enhance customer satisfaction, it is recommended to streamline the checkout process, improve product description and packaging and implement live chat support for quick use. Addressing these concerns will help improve overall customer experience and retention. But here's the big deal. Storytelling isn't about words.

Visuals play a huge role. AI tools like Canva Magic Write, Beautiful.AI, AI or even PowerBI copilot can automatically generate charts that explains your trends, visually summarizes that key highlights, key metrics and even slide text ready to present. Example, you can say create a presentation summarizing the sales trends by region in three slides.

So I've already uploaded the sales data set and again it'll ask which kind of presentation again I'm going with minimalistic since it's a professional way and here you can see the timeline what are the slides that will be added you can add or delete a section and let's go with generate And there you go.

AI will design professional looking visual narrative with titled chats and talking points. This saves time and ensures that your story is not only accurate but also visually engaging. One simple way to build your AIdriven story is using this three-step framework. The first one is context.

What was the question or goal? The second one is insight. What did the data reveal? And the third one is impact. What should we do next? You can even prompt AI like this. Based on this data set, generate a story following this structure which is context, insights and impact. This way your report feels logical, human and actionable.

So AI powered storytelling transform raw analytics into communication that drives understanding and decision. You no longer need to be a data scientist or a designer to explain insights beautifully. With generative AI, anyone can tell stories that blend logic with creativity, facts with feelings. In the next section, we'll take this one step further and explore how generative AI helps us go beyond communication into innovation and problem solving where AI becomes our partner in brainstorming and creating new ideas. Now that we have seen how generative AI can analyze data and tell compelling stories, let's talk about the part that really excites most people, which is innovation and problem solving. This is where AI becomes more than just a tool. It becomes a thinking partner helping us come up with a new ideas, designs, solution and make work faster

and smarter. When we talk about innovation, we often imagine big inventions like self-driving cars or chakity itself. But in reality, innovations can be as simple as finding a smarter way to solve an everyday problem. Generative AI helps us to do that by expanding our creativity. It doesn't replace our ideas, it enhances them.

It gives us more perspective, faster feedback and even helps test those ideas before we build them. So let's start with brainstorming. Earlier brainstorming needs teams in rooms, sticky notes and a lot of time. But now tools like chat chubby Gemini or other LLM model can help you brainstorm 24 by7.

You can just give the prompt like give me 10 creative ways to reduce food waste in college cafeterias. And instantly AI will come up with ideas like smart portion prediction app, leftover recipe generate, gamified zero waste challenge, food waste tracking system, donation program, student cooking classes, etc. Instead of hiring a developer, you can just directly ask chat chip to write a basic chatbot flow for a crop price checking.

It generates the flow, suggest sample dialogues, and even produce the initial code for you. If you use no code tools like glide, bubble or notion AI, you can turn that concept into working prototypes in just few hours, not just weeks. AI shortens the distance between idea and action. Innovation isn't about creating new product.

It's also about improving how we work. For example, if you are marketing manager, AI can automate repetitive task like writing emails, summarizing campaign or even scheduling posts. If you are a researcher, AI can help you organize literature reviews and summarize academic papers. Even in operations, AI can automate workflows such as converting meeting transcripts into action items or generating weekly reports automatically.

This kind of everyday innovation frees up time for strategic thinking, the part humans do best. Here's a simple framework to move from idea to implementation using AI. Step one is to use AI to generate ideas. Step two is to use AI tools to prototype or simulate the ideas. And step three is to use AI analytics to measure performance or feedback.

This cycle of generate, test, and learn can repeat quickly. And that's what makes modern innovation so fast and accessible. So innovation and problem solving with generative AI isn't about being a coder or a genius. It's about asking the right question and letting AI accelerate your creativity.

From brainstorming and rapid prototyping to workflow automation, AI helps us to turn ideas into impact faster than ever before. Up next, we'll explore how to integrate these AI

tools into your daily workflows. So you can make Gen AI a natural productive part of your routine. All right. Now we have seen what generative AI can do.

But the real question is how do you make it part of your day? This section is about practical integration. How to blend AI tools with the way you already work so they become your silent co-pilot in productivity, not just a fancy experiment. When we say integrate AI into workflow, we mean embedding AI into steps of your daily tasks.

For example, a researcher using chat chibi to summarize every paper before reading it. A data analyst using AI to generate Python code or SQL queries. A teacher using AI to prepare lesson summaries and quizzes. Instead of opening AI tools only when you need them, you make them a part of your natural workflow like checking email or using Excel.

The first step to integrate AI is to look at your repetitive task. Things to do every week that consume time but don't need deep creativity effort. for example, writing routine reports, formatting documents, and even summarizing meetings. Ask yourself if AI can do it faster or better.

You'll be surprised how many daily tasks can be automated or streamlined using simple AI prompts or connected tools. Let's look at how this plays out across different tools. So, Chat Gibbid and Gemini can be used for drafting content, writing summaries or explaining code. Whereas notion AI is for automatically generating task updates or teams notes.

Microsoft copilot or even Google duet AI which is directly integrated inside word, excel or gmail for writing and data analysis. Zapier plus open AI to create automated workflow like when a customer submits feedback on Google form summarize with chat GBT to email the manager. This kind of integration means you don't have to switch between apps constantly.

AI works quietly in the background. Let's take a real world example. Imagine your marketing manager preparing a weekly campaign report. Normally you would collect performance data from multiple platforms. Copy paste into Excel or PowerPoint and write a summary manually. Now with AI, you can just connect tools like Google Analytics to chat GPT via an integration platform.

Ask AI to summarize campaign data and highlight top three performing post. Then use Canva magic to write and generate a branded presentation in minutes. In short, what earlier took hours can happen in a few prawns and even a clicks. Integrating Gen AI isn't about tools. It's about mindset. The best results comes when you treat AI as a collaborator not as a shortcut.

Instead of thinking can AI do my job, think how can AI help me do my job better and faster. This shifts from tool user to AI powered professional is what defines the next wave of digital productivity. So to sum up, integrating Gen AI into workflow means finding the task where AI can save time, choosing the right tools for your context and building a daily habit of collaboration with AI.

Once you do that, your productivity compounds because AI becomes part of your thinking process. Next, we'll look at the future directions where this journey is heading. Agentic AI, autonomous assistance, and responsible innovation. As we come to the final part of this module, let's look ahead. Generative AI today can write, design and analyze and even automate.

But the next wave is even more exciting. We are moving from AI as a tool, as an agent system that can reason, plan and even act on their own. These are called agentic AIs or autonomous assistants and they are shaping the future of how humans and machines will work together. Agentic AI means AI systems that don't just respond to prompts.

They can take initiative, remember goals and make decisions towards completing them. Think of it like difference between asking chat to write an email and telling an AI

assistant handle all my customers follow-ups emails this week. In the second case, the AI acts as autonomously. It plans, executes and updates you almost like a digital teammate.

Projects like open AIS, GPTs, lettera and mem GPT are early examples of this direction which can also remember context follow multi-step goals and interact with external tools. So how does this autonomous assistance actually work? It combines three major abilities. The first one is memory to recall previous interaction and learn over time.

And the second one is reasoning to decide what actions to take next. And the third one is tool use to connect with apps, database or even web complete real world tasks. For example, imagine an AI research assistant. You can tell it find the latest papers on sustainable energy, summarize the key findings and prepare the slide deck.

It will search, summarize, visualize and present all by its own. That's the direction we are heading. AI as a true collaborator, not just as a chatbot. In this future, the focus shifts from automation to augmentation where humans and AI combine their strengths. Humans bring creativity, ethics, and emotional intelligence.

AI brings speed, precision, and memory. Together, they create what many call the collaborative intelligence model. For example, a teacher using AI to personalize lessons for each student. And a doctor uses AI agents to summarize patient history before diagnosis. And a designer uses AI to generate hundreds of prototypes then refines the best one manually.

The goal isn't to replace people. It is to elevate human potential. But as we move towards this the power of new phase we also have to talk about responsible AI innovations because when AI can act on their own we must ensure they do it safely ethically and transparently this means building AI systems that are fair not biased against any group ensuring privacy especially when handling user or company data maintaining human oversight which is AI shouldn't assist nor decide inside everything. Many

organizations are now adopting frameworks for responsible AI focusing on accountability, transparency and sustainability in AI development. The final part of this journey is about continuous learning. Generative AI is evolving every month. So the best skill you can have is the ability to keep

learning and adapting. Google Nanobanana is already killing Photoshop and other design jobs. And here's why it's making waves across the internet. It's been only few days since it dropped and the internet is already flooded with insane results. If you've been following the buzz, you have probably seen it everywhere and it's all available to you for free.

So, let's talk about Google Nano Banana. Google's latest state-ofthe-art image generation model that's causing a major disruption in the world of digital content creation. Whether you're designer, marketer, or content creator, this tool is about to change everything. It's already out and accessible through Gemini and Google AI Studio.

And the best part is it's completely for free. That's right. This insane tech that's making professionals rethink their whole career strategy. And it cost you nothing to dive in. Now, let's talk about what makes Google Nano Banana so special. It's not about editing images. It's about taking control of your creative vision in a way that's never been possible before.

With just a few simple text prompts, you can completely transform an image. From changing characters and backgrounds to adding 3D models and making everything look ultra realistic. So, here's what we're going to cover in today's video. And trust me, you don't want to miss this.

First of all, we'll show you how to add characters in your images, and the results will look 100% real. You'll also learn how to swap backgrounds seamlessly. No more chunky AI generated results. We are talking real life quality images so realistic that no one will be able to tell it was done by AI. Then we'll move on to creating a 3D model of yourself.

And believe me, this is a huge right now. Everyone's buzzing about it and I'll show you exactly how you can use it for your business marketing too. If you're into advertising, get ready for a complete game changer. I'll show you how to create ads using Nano Banana and how you can make money with it by leveraging this tool for your business.

Now, before we move on, here's a quick quiz question for you. What's the biggest advantage of using Generative AI for image creation over traditional image editing tools? Your options are you can generate realistic images with just text prompt. It requires no creativity or design skills.

It can only modify existing images, not create new ones. It's more expensive than traditional software. Let me know your answers in the comment section below. Now, let's jump into the demo part and I'll show you exactly how to use Google Nano Banana to achieve all of this. So, let's get started. Now, you need to just head over to Google and search for Google Nano Banana and just click here.

You get the first link, Google Gemini. Try Nano Banana and Gemini. It's 100% free. So, just click on this link from here. And here we have Gemini 2.5 flash. All right. So it says hello want to try it now a few things. We already have making your own custom mini figure. I mean you already have the prompt given here.

So you could just simply give your upload your image and it will create model action figure for you. Now make sure that your Gemini is in 2.5 flash mode which is fast all round help. And from here just select tools and select create images. You get this banana icon from here. That means it's nano banana. All right.

So now what I'll do is I'll just upload any image of mine. Let's say I have this image from here. And then I'll say Gemini to change this outfit. Make it in blue color dress. All right.

And I'll just hit enter. Let's see what type of result it gives to us. So we just have to wait for a few seconds. So now as you can see Gemini has instantly given me the result.

It took maybe around uh 1 minute lesser than 1 minute and look at the result guys. This is the result it gave me. It has changed the color of my dress. And I'll show you the previous dress of mine which I had uploaded. So this was my actual image from here. And I cannot even spot any error because it's so real. All right.

So it's pretty impressive to use this one. So now I'll give one more prompt that is I'll say Gemini to add Elon Musk standing standing with me hands folded same posture as mine. Okay. Just talk to Gemini as if you're just saying a friend to make these changes. You don't need a actual prompt to do it, but then just a normal conversation and just submit this option.

Let's see what result it gives us. I'm pretty excited to see the result. Now, this was so fast. You see, it has given me the result and Elon Musk is just standing near to me. Feels as if I've added this image. Of course, it doesn't feel that real. But then, so what I'll do is I'll again give a prompt that make him look standing near to me with his hands on my shoulder so that it looks real. Okay.

Now here even if you give a wrong spelling mistake, it will understand it for you. So now as you can see it again gave me the result and this time I think I'm holding hand on his shoulder but um I I guess my face has changed here a bit anyways. All right. So now next what I'll do is I'll I've given two images to Gemini and I'll say swap the background of me standing with the Taj Mahel background.

make me appear in front of Taj Mahel background. All right. And I'll just hit on enter. So now as you can see it has swapped the image and it made me appear in front of the Taj Mahal background which is pretty decent and uh yeah it looks a bit photoshopped for me but then it's fine. Okay.

Okay, I would rate it like maybe 7 out of 10. Next, what I'll do is I'll make my own custom figure. So, for this, I think I need to upload my own image. So, I'll upload an image of my This is the image from here. And I'll just enter. That's it. So, now as you can see, it has generated the model image for me. I need this type of pictures are actually trending on Instagram and YouTube.

So you could just create it using this and this looks great to me because yeah the quality is also good and the picture is exact the same and I'm pretty impressed by this result which Gemini gave it to me. So according to me I feel the more better your prompt is the more better the image you get it.

So now we'll use more better refined prompts. Now I'll show you how you can use Gemini Google Nano Banana for your marketing advertising purpose. So let's suppose I want to add for my ketchup. All right. So I'll just upload my image. So here I have a ketchup image and I'll give a prompt to create a high quality advertisement for a Tabasco ketchup. Okay.

And I'll also mention already uploaded above. Okay. The scene should be feature a spicy pepperoni pizza as the centerpiece with generous drizzle of Tabasco ketchup placed in between the slices. So I've given all the ingredients needed and how I want my uh image to look like. Now let's see the result.

Now this time I've used a better refined prompt. So let's see what type of image it gives us. So guys, as you can see, I had given the image and this is the result and this looks pretty good because according to my prompt, the image quality is so good and there's no error to be spotted as well. This could be perfect for your marketing or advertising purpose.

I mean, you don't have to spend hours doing Photoshop or creating uh pictures for this. It can generate for you within just a minute. That's it. Now, we'll also try for some other

products. Now here I want to create a high quality advertisement for the iPhone 16. So I've given the prompt that the scene should feature a bustling Bangalore traffic road as a background capturing city's vibrant energy.

In the foreground display a large banner showcasing iPhone 16 with its phone flying upwards in a sleek futuristic manner. And now let's see what type of image it will create for us. So guys here I have my image ready for me and just look at the quality of this image. It also had this Apple logo and all the text written and this Bangalore traffic image.

This looks pretty cool right? This is best for your advertising purpose because without even spending hours doing trying to find out ideas by just giving simple prompt you to get such amazing results. So it's pretty impressive and this is how you can use Google Gemini, Google Nano Banana for your advertising purpose and you can just earn money through this.

Next I also wanted to try one more thing that is how could we merge two images in one picture. So, I'll say Gemini to u merge this dog image into my photo wearing black dress. The dog should be cuddling with me and I need side picture angle. You can also change the angle of the image.

It's not necessary that you need just the front view of the image which you're giving. So I have told Gemini to give me the side picture angle of this image that is me playing cuddling with the cat. So here I have my M ready with me and here is the image of you cuddling with the dog from side angle. It actually took the side angle image of the puppy, right? Not me.

But then this uh looks okay to me. Well, I'll say this again. I need the side angle of this image from right side angle. So now as you can see it seems I misunderstood the previous request. Here is again the image and this is the side angle image guys. You see I I had given the front view image and it actually created a side looking image right and who would say this is not real because this looks absolutely real 100% sure and yeah it's

great Google Gemini Nano Banana is actually great for your creating images removing background swapping images and also creating for marketing and advertising purpose. Personally, I loved it for marketing and advertising purpose and this could be a great way to begin your journey using

Gemini and let's wait for more new features to be added. So guys, that's it and it's worth trying. So I would definitely recommend you to try this out and let me know your feedback on the comment section below. Hey there developers, let's start with a quick thought. Have you ever been assigned a bug fix in a massive unfamiliar code base? You spend hours just jumping between files trying to trace how one piece of code connects to another and your terminal history becomes a mile long with crep and find commands. It feels less like engineering and more like being a detective, right? It must be frustrating. Now what if I told you that this frustrating manual process is the actual problem? What if you could just ask your codebase questions and tell it what to do? That's where claw code comes in. In simple terms, it's an AI coding agent that lives directly in your terminal, understands your entire project and can execute task for you.

So, in today's video, we are going to dive into how cloud code works. It's not just about writing code, but it's about having an intelligent partner in your development workflow. And by the end of this video, you'll understand what is clot code. I'll break them down what makes it an agent and not just another AI assistant cloud code features.

We'll look its key strength and how it works with your idees and VS code and jet brains. We also look how it integrates with your favorite command line tools you already use. You'll understand your entire code base to see the bigger picture. Now, what could you do with clawed code? Well, we'll explore the exciting possibilities from fixing complex bugs and refactoring code across multiple files to write brand new tests and documentation from scratch.

How to master your commands. I'll show you how to craft the perfect instruction to unlock its full potential and get the most out of the AI. And after all of that, we'll show you

a demo of how to use clawed code in an open-source project. You'll see how to apply everything we have learned today to a realworld coding problem.

So, are you ready to see your terminal become your most powerful coding partner? Let's get started. Now, before we dive deeper, let me ask you a quick quiz question. Which AI tool is primarily known for generating realistic and artistic images from natural descriptions? Your options are Jasper AI, MidJourney, Chad GPT or Auto.AI.

Now, don't forget to mention your answers in the comment section below. Also, before moving forward, I request you guys do not forget to hit the subscribe button and check the bell icon so you never miss out any future updates from Simply Learn. Let's get started. Now, in a world where software developers spend countless hours navigating complex code bases and fixing tedious bugs, a new class of tools is emerging to shift the focus back to innovation.

Anthropic's claude code is at the forefront of this movement. Moving beyond a simple AI assistant to becoming a true coding agent that works alongside developers directly in their environment. So let's start by understanding what is claude code. Now claude code is an AI coding agent developed by Enthropic that operates directly within a developer terminal.

Now unlike traditional AI assistant that merely suggests code snippets, claw code is designed with a holistic understanding of an entire code base allowing it to function as an autonomous partner. Now it lives in your terminal and it has an holistic awareness of your entire project.

This means it doesn't just see one file. It understands the intricate web of dependencies, functions, and modules that make up your application. Let's talk about the key features of claude code. Now, claude code is designed to skip into your existing workflow without friction. Its key features are built around this principle.

First of all, it works with your ids and tools. That is, it integrates directly with popular editors like VS Code, Jet Brains. Furthermore, because it operates from the terminal, it can leverage all your favorite command line tools from Git to Docker, making it a natural extension of your development process.

The next feature is its powerful intelligence. It uses agentic search to understand your entire codebase without manual context selection. It makes coordinated changes across multiple files. also optimized specifically for code understanding and generation with claude opus 4.1. The next feature is to work wherever you're working from.

It lives right inside your terminal. No context switching. It integrates with VS Code and Jet Brains IDEs and it also leverages your test suits and build system. Fourth feature is that you are in control. Never modifies your files without explicit approval. adapts to your coding standard and patterns. It's also configurable that is it's build on the SDK or run on GitHub actions.

Now let's talk about what could you do with clot code. The possibilities are vast with its deep understanding and agentic capabilities. You can delegate task that were once major time sync. First of all, it can fix your complex bugs. Feed it an error log and it will trace the problem to its source and propose a fix.

Then it also performs large scale refactoring. Modernize legacy code or switch out libraries across entire project with a single command. Then you have code on boarding. The claw code maps and explains entire codebase in a few seconds. It uses agentic search to understand project structure and dependencies without you having to manually select context files.

Then you can also turn issue into PRs. Stop bouncing between tools. Cloud code integrates with GitHub, GitLab, and your command line tools to handle the entire

workflow. Reading issues, writing code, running test, and submitting PR all from your terminal while you grab coffee. Now, to get started with the demo, make sure you have VS Code already installed in your system.

And also make sure to install Node.js. After that, what you can do is just head over to this uh clawed website from here. And we can just copy paste the first command to install claude code. Okay. So here it's mentioned it's npm install. So we'll just copy this command from here and we'll run it in our terminal.

And yes, it's also mentioned to install node.js. Since I have already installed NodeJS and VS Code, I'll proceed to my terminal from here. So now I'll open my terminal and from here just copy paste the command. Wait for a few seconds. And now as you can see it has changed to packages in 13 seconds and one package is looking for funding.

So I had already installed claude code previously. So it just had to uh check for the packages here. That's all that's all you need to do. Just enter this command from here and your claude code will be installed in your system. Then what we need to do is we'll head over back to VS Code and then we'll just um open any file from here.

So I'll just open a folder and I have claw I'll select this folder. So my folder is selected. So I'll create a new file name as new claude code. Okay. And I'll just hit enter. Then we'll just open the terminal from here. And now that our file and folder is ready, what we need to do is we just need to type claude. Yes, that's it.

So I'll just enter claude and I'll just hit on enter from here. So now as you can see from here it's mentioned welcome to claude code and these are the tips for getting started. So I had previously installed the system before so it's not showing me all the features made I wanted in the dark mode or light mode but since if you're doing it for the first time you'll get more instructions.

So now that I have all my features ready I just need to give a prompt to claude code and that's it. So I already have a prompt ready with me. So my prompt is to create basic to-do list web app using HTML, CSS and JavaScript. The app should have the following features. I've mentioned the features.

I want a simple UI where users can add task to a list. And each task should have a checkbox to mark it as completed. Also delete button. Task should store in local storage. implement basic CSS styling or if you want any particular color emphasizing you can just mention it right here and I've also mentioned that the web app should allow users to add task and uh generate text input and button mark text as done striking through the text delete task and many other so now that I have my prompt ready we'll just need to click on enter so we'll just wait for a few seconds so as you can see it's written I'll create a basic to-do list web app with HTML let me plan this task first. So, we need to give some time to claw to plan and task this particular app. Now, it's creating our HTML structure for the todo app. Meanwhile, you can see right from here that our script.js style dot CSS is already being created. It's all happening because inside uh the claude

is generating creating HTML, CSS and JavaScript files separately for us. So, now it's creating our HTML structure for the todo app. If you want to interrupt this particular thing, you can just uh select escape or control T to show the todos if you want. Let's suppose I want to see so I'll just press Ctrl T from here and it will show me the todos.

And if you want to hide it again you can press Ctrl T. So now you can see from here it's creating HTML structure implement CSS styling adding JavaScript functionality implementing new and these are the code here. Now you can see the magic of this without even writing a single line of code it has generated the complete code for me that two HTML CSS and JavaScript.

Now it's saying that do you want to override uh index html? I'll just select it. Either you can select it yes allow it during all the edits. No to tell claude what to do. So I'll just press on yes. Now it's implementing CSS styling for a clean layout. Let's see what how it generates for us. Also while it's being uh creating our CSS styling script.

One more quick information. Make sure to install clawed extension from here. Now, if you've been using the latest version of VS Code, you must have the clawed extension. Since I have already installed it, um I have given the option of disable and uninstall. But make sure you install this. Yes.

So again back to basics, it's saying adding JavaScript functionality for adding task. It is also mentioned our font family the font size the background color whatever color is needed. So if you want particularly to modify your any font size and the color you can just type in as in you're talking to a person just say it in the prompt that I want this particular size and color and it will generate it for you. Okay.

So now as you can see it's saying us it has already created a basic to-do list app with all the features I requested index.html file style.j CSS script.js js and from the file section from here you can see everything style dot CSS we have script js we don't even need to create any particular file to create our HTML file and javascript file it has created everything with all our features implemented now we'll just click on go live and see how the result is so let's click on this so guys this is the to-do list it has generated for us and I can't see any functionalities added here which is quite a disappointment So I think the feature and the UI is pretty fine. It's very basic. But again I'll head over back to VS Code and I'll say it. So again I'll head over back to my VS code and I'll say to give more

functionalities. So I've pasted from uh Chibility to add more functionalities to my app and I'll just hit on enter. So as you can see I had given the prompt that the app should include all the following features. add task, edit task, mark task as completed, delete. So I've given particularly more functionalities to add and let's see how it does the changes for us.

So we'll get to know whether it's actually worth it using clawed code and if it's actually implementing the changes or not. Okay, so it's saying that I'll enhance the to-do list app

with all the advanced features you requested. Let me plan the update. Fine, we'll wait for a few minutes again. So it says implementing comprehensive CSS styling for modern UI.

So now as you can see that our app is fully functional with professional modern interfaces. So again we'll just check back. We'll go back to our server and I'll just reload this again. Yeah. So this is our advanced to-do list and now you can also search your task all categories.

So yeah I did implement it few changes from here and the UI is very basic and clean. Uh if you want to add more features and colors, you can also do that. For the time being to show it to you guys, I did a very minimalistic uh UI added. And now you can add your task. You have different categories like personal work, shopping, health, education.

You also have your priority like which task is your high priority, low priority, your date is also added. So let's suppose I add a new task that is shopping. And I'll keep it as shopping. This would be maybe medium priority. And I'll add a date here. And then I'll just add a task. So now you can see that my task is added successfully.

We have shopping from here. You can also edit or delete this task if needed. And yeah, it worked really well. So not much complaint. So as you can see how cloud code can help you create such application website without you needing to even write a single line of code. Hey everyone, are you ready to find out which AI tool is truly a gamecher? Today we are diving into the showdown between Gemini CLI and Claw Code.

Two powerful tools that are revolutionizing the way developers write code. But which one reigns supreme? Let's find it out. So, first up, we've got Gemini CLI, powered by Google's Gemini model. This tool isn't just for coding. It's an entire AI ecosystem with capabilities that go beyond writing code.

Gemini CLI helps you manage files, perform web searches, and tackle massive multifile code bases. Its incredible 1 million token context window let it handle large scale projects with ease. And with its open-source nature and generous free tire, it's a versatile and cost effective choice for developers. And on the other hand, we have got clawed code by Entropic.

This tool is all about quality and precision. It's built specifically to help you write clean production ready code, manage complex multifile edits, and even automate git operations. While it may have a smaller context window than Gemini, its focus on high quality code generation, which makes it the go-to for developers who need reliability and efficiency in their project.

So, here's what we are going to cover today. First of all, we'll take a look at Gemini CLI, walk you through its installation, and explore its unique features. Next up, we'll set up Cloud Code and see how it offers a more structured, polished approach to coding. And finally, we'll put both tools to the test by building an app side by side.

So you can see firsthand which ones take the crown and as the best AI coding tool. Now before we dive deeper, let me ask you this question. What is the primary function of a large language model like Gemini or Claude? Your options are to process and analyze images and videos, to control the physical movement of a robot, to understand and generate humanlike text, or to create new original code and design concept.

Don't forget to mention your answers in the comment section below. Now, before moving forward, I request you guys to not forget to hit the subscribe button and click the bell icon so you do not miss out any future updates from Simply Law. So, let's get started. So, let's get started with the installation part.

So first you need to make sure that you have already installed node.js and then head over to anthropic claude code website. Here you'll find the command and just copy the command from here to install claude and then we'll go to terminal and we'll just copy

paste the command. So now as you can see that our packages have been installed that is a claude and to check it you can just type in claude here.

All right. So now as you can see that it's mentioning me that you can trust this files in this folder not. So I'll just click on yes proceed and it says welcome to claude code. That means that our claude has been successfully installed. Now our second step is installing Gemini CLI.

So what we'll do is we'll just copy this command from here. We'll go back to our terminal and from here we'll just copy paste the command and just we'll wait for a few minutes. So now as you can see that our Google Gemini CLI has been installed in my system. It says added 476 packages in 2 minutes. So it took a bit of time. Then we installed claude here.

Now what we'll do is we'll go to VS code and from here we'll just select file and then we'll click we'll just select file and from here we can just select open folder. So I have my claude code minimal folder from here. I'll be selecting this folder now. Now after I selected it, this is my uh program file name.

I'll add one more file name which named as uh claude code new. Okay. And now I'll just go to terminal from here. Select new terminal. And now what we'll do is we'll again type in claw to actually see that if it's working or not. Now it says welcome to claw. That means successfully it has been done.

We'll do it again for Gemini CLI as well. Now again I'll be opening one more new window and select on open folder. and then I'll select Gemini CLI from here. So from here I'll be selecting CLI folder and I'll just click on select folder and we'll be creating one more new file as minimalistic website. Let's suppose this is my file name and I'll just remove this gap from here.

Okay, I'll just change the name again minimal web. All right. Now, what I'll do is again I'll type in here Gemini to check the Gemini installation. And guys, here you can see that our Gemini has been successfully installed. And these are the tips for getting started. So, what I'll do is I'll just enter a command and I'll just say to create an e-commerce website for both the Gemini CLI as well as clot code and let's see which performs better.

So first I have this clot code from here and I've given the prompt to create a minimalistic e-commerce website named as minimal mart using HTML CSS and JavaScript. I want the design to be clean, modern, easy to navigate and use a light background with subtle accent colors. And I want the homepage which should include a hero banner, product categories, a grid style product listing as image, name, price, add to cardart and also the search bar at the top.

And I've also told to add an add to cardart button. Now we'll just click on enter. And let's see. It has started to create this directory first. So it says I'll create minimalistic e-commerce website with all the features you requested. Now let me plan first. And now it has come up with all the todos and it's going on to create HTML structure for homepage with hero banner.

Now it has already started with creating the HTML file. We'll go to Gemini and we'll give the same prompt to create a minimalistic e-commerce website and just hit on enter. Now Gemini says it's planning minimal art design, structuring minimal arts pages. Well, we'll wait for a few minutes.

So guys, as you can see, it has already told me the plan what it is and it says I will create the project structure and also add content to each file. And here I have it MQD minimal M allow execution. So I'll just select on yes allow always. So as you can see it has already created a file folder as minimal M from here.

Now you don't actually have to do anything. Just sit and relax and this Gemini will do everything for you also. How can I forget Lord? I'll be showing you that also. Well I'll just

select here as allow always. You can do it with your arrow button. And it has started to generate the code for me. Meanwhile, I'll go back to my claude.

And let's see what it has done. Well, it's taking time to create the HTML structure for homepage. Still, it is working on the index.html file. All right. So, from here, you can again see the code. Both are pretty fast at creating code. But what I felt is Gemini CLI is bit more faster.

And then it says do you want to make this edit? Yes, of course. Allow this all the time. Okay. So now Claude is working on structuring the homepage. So as you can see it has crossed this line. It means it has already created HTML structure for homepage and now it's working on to create product detail page with HTML structure.

And meanwhile, Gemini is working on to create JavaScript code and it has already done uh added card product index HTML are already done and it's working on JavaScript. Well, it's pretty fast as compared to uh claude. So yeah, it says I have created the minimal mart website with a clean, modern, responsive design and the file structure directory includes index style.

It's pretty fast. Right now, we'll see Claude as well because I started earlier with Claude and Gemini actually took up the race and it went more faster. Well, here I can see that Gemini actually wins the game. So, I'll just select on go live and see what type of website has it created now.

So, guys, this is the website it has created named as minimal mod. Welcome to Minimal M. And these are the product categories. category one the featured product and it has also added the option of added to cart. Well, it looks pretty good to me but what I felt is the UX the design is bit very normal and it has used similar type of images and icons to show this right.

It's more like black and white color shade. Well, it's fine. We'll just add more functionalities to Gemini and say to do some corrections. Meanwhile, let's see what Claude has done. So, it's still working on creating the CSS styling with clean modern design. Now what I'll do is I'll go back to my uh website from here and I'll say please add dark blue color background and also more functionalities dark and light theme change button along with dark.

I want you to add more product electronic items and name with its similar image. Okay, let's see what um what type of changes will it do for me. Now, it says it's analyzing feature additions and it's designing theme functionality. All right, we'll just wait till it does. Okay, here's a plan.

It will update and it will also add theme toggle button. Add JavaScript to handle the switching to save the user's preference. Okay, well cool. Now we have Claude here. It's just still working on right. Takes a bit of time. So Gemini is pretty fast as compared to Claude. Looking back to our Gemini CLI, it is already it started to adding more HTML elements from here.

And you can just look at the code from here. So it's editing and now adding more functionalities and features what I already told Gemini to do it. Also one more thing you can also look at the number of tokens it is generating while doing performing the code analysis from here.

And yes finally Claude has completed its task and it says I have successfully created a complete minimalistic e-commerce website. All right now just let's test it and have a look what type of website has it created. Well, I'm pretty excited. Okay, this looks better. Okay, now it has added like welcome to minimal mod and this is the electronics clothing home.

Just look at the while I hover just look at the option the animations it's giving. I'm pretty impressed by this because uh when I look at Gemini's uh feature, I'm pretty impressed by

this because just look at the hover and the animations and this is of uh Gemini CLI and I told it to create edit and this is what it did for me.

It looks good to me. The color layout which I told to add dark blue it has done that and these are the product categories. This is the shop now option. So while I click on that you see all products option you can add it to cart and it says product added to cart. Okay, you have headphones, camera, smartphones.

This is fine. Looks good. You have home option, shop option and cart, right? So you can while you add, you can also look at the cart and number. What is the total price of the option? You can also have this contact option where you can enter your name. Let's suppose I enter my name and I have my email from here.

And you can also send a message. It actually works. And okay, this looks pretty good to me. But then if compared I also like claude code result because this looks more sleek and more minimalistic. Now let's see if I give the changes to claude what it does for me. So I've told Claude to make the background dark blue and purple color I need more interactive UIUX design and add button of dark and night mode option for the user.

Let's see what type of changes it does for me. Well, in the case of Gemini, I told it to add a button of dark and night mode. I could not actually see. So, I'll be saying it again. Dark and night mode. Button option I need in front home page. Okay. So, it's considering the placement now and it's making changes again. Meanwhile, as you can see, uh clot code is updating CSS with dark blue and purple background themes and it will take well now honestly to be said starting I felt Gemini CLI is pretty faster but then while you look at the results I feel claw code wins this game because it took time but the result it gave it to me is pretty awesome right I mean you yourself can compare from here this and this right now let's add more um features here which I've told them to add it and let's compare the final

version. All right. So Claude here has successfully done the changes. It says I have transformed the minimal M website with a stunning dark blue purple theme and these are the changes it did. The new visual design dark mode and night mode toggle options enhanced interactive elements advanced animation.

Let's have a look at the final look of our website using clot code. So guys, this is the final look of our website using clot code. It says welcome to minimal mod. And just look at the animations it has added. This is pretty impressive, right? Look at the cursor while I just move it. It's like circular purple color theme which is pretty impressive.

And the other options electronics, clothing, home and garden, books. And while I hover to this, it actually floats right. So it has added a lot of animations. What I felt is the color which I already told purple. So it's not visible much in this background. We can obviously change that. This is just a minor mistake.

But apart from this, this looks pretty cool to me. The animations, the styling is actually good. Let's look at the option of the dark and night mode. So as you can see, if I just click on this, this is a dark mode. And if user want the light mode, they can click on this. This is such a cool feature it has added.

And this works so smooth. Now in the case of Gemini, let's have a look if it has done the changes for me. So it says I've updated. All right, it has updated and done all the changes. I'll just go back to my website and reload this. So this is the final look of my website. Welcome to Minimal M.

Again the color theme which I feel we could have directly black color. Now this is also one more major minor mistake and guys it has finally added this dark and night mode right. So this is the light mode and this is the dark mode. I like the dark mode. Now let me know which one you liked it because according to me I liked u repeat.

Now according to me if you ask I feel Gemini has added more of pictures here and tried to make it interactive but then it was pretty fast as we compared to claude code but then if you look at claude code's result it took a lot of time but the styling the layout and the animation it it added is pretty awesome and appreciable.

So I would go for claude code definitely and let me know which one you like the most. As technology advances in all aspects of our lives, programming has become increasingly important. It is used in many fields and industries including software development, gaming and entertainment, education, scientific research, web development, and many more.

So needless to say, the demand for programming and coding in the IT industry will probably keep increasing for the foreseeable future. But where does CHP Open EI's popular language model fall in this chain? That's exactly what we are focusing on in this today's video. As I said earlier, programming is utilized in many domains like web development, robotics, mobile development, machine learning and so on.

So how can a program achieve maximum code efficiency? Nowadays we have AI based tools like charge to make our programming experience more efficient. Although there are several coding resources platforms such as Stack Overflow and GitHub where programmers can find solutions to their technical programming questions, Chad stands out from the competition because of its quick response time, usability, and support for numerous languages among many other benefits.

Now, let's first discuss how Chad GPT works. Chad GPT generates responses to the text input using a method called transformer architecture. A large volume of text is fed into the chat GPT from various sources including books, websites, and other social media platforms. The model then uses this information to forecast the following word in a phrase based on the words that came before it.

The charge systems allows users to enter text or queries and then the system uses its training data and algorithms to produce the right answer. The answer is created after the input text has been examined and the pattern most likely to match the input have been identified using the training data.

In short, charge is designed to respond to queries logically and command more quickly and accurately. But why do programmers use tag on a regular basis? Tajibity assists programmers by offering programming related answers and solutions and helping them improve their skills. Beside that charge is utilized for code generation, code completion, code review and a natural language interface.

Let us understand each in detail. Charge is trained to generate the code or even the entire program described in the natural language specified by what they want a program to do and then charge could generate the relevant code. Look at the example of how charges the code. So now open the charge GPT and you can type any program that you want charge to generate.

So I will give write a palendum program in Java. So here you can type write a palendum program in Java. So using Java programming language it should generate the whole program. So as you can see it has generated the program. So it has used a class name called palendrum checker and it has used is palendrum as a method name and also it will give the explanation on the program.

So you can see here why it is explaining why is palendrome is used as a method and u it also explains the for loop if a condition and so on. Next we have code completion. Tajibri is trained to generate snippets of code or even fullyfledged programs. It can generate a list of possible code completion depending on the context of the user's incomplete piece of code.

By automatically producing the entire code, it can help the developers save time and minimize errors. Next, let's see the example of code completion using Taj. So even if the

program is explained in natural language, TAGP will generate the proper code and give the complete code. So let's type here using a function write a program to convert the string in upper case.

So using which language? Let's keep using C programming and enter it once again. So as you can see we have just said that using a function write a program to convert the string in uppercase. So using C programming language and using C programming language it has used function and you know this is the function convert to uppercase and has given the complete code for string to convert a string in upper case and also it gave the explanation here the convert to uppercase function takes a pointer to a string as its argument and then iterates over each character in the string using a for loop. So it explain why for loop is used, why two upper is used and why the method convert to upper case is used everything. So let's say uh we'll give one piece of code like void to upper

car str. So as you can see we just gave the method to upper and it's generating the complete code. So this is how charge works for code completion. >> Next code review. Charge can analyze code, identify the bugs or errors in the program and further help resolve them. It allows developers to fix errors more quickly.

So now let's have a look at the example of code review. So in this example, TAGP will review the code. So even if the code has some mistake, it will give the proper output. Let's say we have given the example here. So we give the function or a method called upper and here we are giving the keyword called upper.

So it should check whether this piece of code is proper or is there any mistake in this. So as I said uh it's saying that the given code appears to have logical error as the function upper is being called recursively on itself inside the loop. So instead of giving two upper we just give upper here right. So using the keyword to upper only then the string can be converted to upper case.

So here we gave just upper. So it says that it is having this piece of code is having a logical error and it gives a proper code for us. So I hope it's clear and then we have natural language interface. With the use of chat GPT, a software application can be given a natural language user interface that enables users to communicate with it through natural language instructions rather than through conventional user interfaces.

Next, let's see how Tajibility helps the programmers for natural language interface. So let's say we'll give here create a software application where the user asked to enter credential for the too app. Click enter. So as you can see the charge will give the steps. So it can provide you with an outlet for creating a software applications that requires the user to enter credential for a todo app.

So here it is few steps that we need to follow to do a todo app. So it's giving the explanation step by step. So it says that determine the programming language and framework. Then set up the database to store the user information and then create the registration page and then finally create the login page as well.

And once the user is successfully logged in um you know it will have the options like add edit and delete task as well. And then finally implement the security measures to you know protect your passwords and then test the application to ensure that it works as intended and the user data is being stored and retrieved correctly.

So it gives the steps of how it has to be developed. >> I'm sure you all are aware of charge at this point. The revolutionary new AI based chatbot developed by OpenAI has taken the world by storm thanks to its near lifelike responses and a very intricate pattern of answers. We have never seen this level of expertise from a chatbot before which really made us think to what extent can we push it.

There are many questions on lead code that even the most experienced programmers have difficulty answering. So we wanted to see how far Chat Gy can take us. Have we finally reached the stage where AI is going to replace us? Let's find out. So basically here

we will be listing 10 really difficult questions that we found on lead code popularly asked while hiring and other superior examinations and see if Chad GPD can actually answer or solve those difficult questions or not.

But before you like to watch more such interesting videos then do subscribe to our YouTube channel and hit the bell icon to never miss an update from simply learn. So let's get started. So here is the lead code. Let's see uh in our list which is the first question that we are going to implement in our chart GPD and see if it's able to solve it or not.

Mainly we'll focus on hard category questions only. So according to my research there's a question of median of two sorted arrays. So as you can see the success rate is 35.7%. So let's see if the charge chip is able to do this question. So first let us go through the question. Okay, pressed enter.

Let's see what it first returns. uh on a one approach to solving this problem is to use a modified binary search algorithm to find the median of the two sorted arrays. So right now it's particularly giving the logic which we can actually imply to solve this question and uh this is a good thing about the chat GPD is that before uh giving the code it's actually explaining how they are putting the logic in together in the code.

So probably you can use this logic to create your own program. But let's see how sensible this code is. quite a lengthy program this in Python. So if uh you are looking for your solutions on chbd you can always opt out or you know mention the specific program you want the code in. So okay it's it's also gives that okay that the time complexity will be o log min m or n.

So let's see if it the case or not. So we have copied it and we'll quickly paste it over here as you can see uh as we know that Python is uh very sensitive towards its uh syntax and you can see the indentation over here is perfect but here it's not. So I feel that something like this is something with lead code.

So let me just quickly rectify this. You can see I have uh cleared out the indentation issue and let's just quickly run this program so that we can get an idea if this is the correct program. Uh now you can see here we have an error. Let's see. Copy this and see what chart GP has to say for this. So if you remember when we actually saw

this question there were three arguments passed through this function which was self nums one and nums 2. Ask GPD if it can write the code with self argument and see if it's correct. wait that if it can pass a self argument through that function and see if it can generate a new code. Okay, now it has cleared that.

Yes, it takes three arguments. Let's copy this code it over here again. I think we will have to go through the indentation process. Oh no, this time it's fine. Okay. So now quickly run this program. Let's see if this time it passes all the test cases or not. Okay, I think it doesn't need because the class is already mentioned over here.

Uh yeah. Okay. The run time is 35 ms. And as you can see case one and case two is definitely passed. So let us see if this code can pass all the test cases internally mentioned in this question. Now you can see the first three cases are actually accepted but the other three are not. There is a runtime error.

So as the first question in our list J GP was unable to solve. So let's move on to our next question that is zigzag conversion. Now let me quickly search for it. This is a question here also you can see the success rate is definitely below 50 and uh the difficulty level is medium. So the hard one charg was not able to solve. Let's see if this can be done.

So it will give a string that will be written in a zigzag pattern on a given number of rows and uh then you have to read the line in a certain as you can see over here. So we have to write the code that will take a string and make this conversion given in a number of

row. This also means a certain amount of certific uh repeat does also given a certain specifications that we exactly want.

So this time we'll make sure that we are mentioning everything. So let's quickly copy this. Yeah. Okay. Now that we have mentioned all the specifications, uh let me quickly see and paste it. Now let's see what code it

has to generate. It's implementing the code in C++. So meanwhile it's generating the code. Let's quickly select the code. Oh, it's already C++. Okay. Okay. It's also suggesting that we can definitely use Python and Java and it's generating an alternative code as well for us. That's sharp. Just quickly copy this code one. I think it's a Okay, so it's

generating in uh Java. It's generating. We'll definitely have a look at its alternative codes as well. Let's quickly have a look at what it has to C++ one. the first code generated by it uh is correct or not. Okay. Copy the code. Paste it over here. Dilation error. Okay. Okay. 28. Okay.

So let's error the one thing definitely this time we have mentioned all the constraints criteria specifications that we wanted in our code but again charging questions we have implemented till now let's see if it has any success trade in further. Okay, it seems okay. Okay, now it is okay.

So this time it is uh generating the solution concerning the error in Java. uh see what it has to say in Java and we'll make it specific that the error was in C++ program. So generates the correct code plus program what it has to say. Uh things

See uh this error is something related to the compiler and now it is giving the updated code in C++ is the correct one or apologizing for making errors in their solutions. Quite

fascinating. Uh yeah, paste the code over here. I don't see there's a lot of difference or changes over here. Let's see if it runs or not.

Let me see if all the places are covered over here or not. I think it's missing a brace. Okay. So, there was a syntax error. One

brace was missing. Um, I don't thought that definitely something with the code. But okay, we can give that to Chad GPD. It was partially read quotes issue because we were copying quotes. Still it was not giving the error that there is a brace missing has passed the first three cases mentioned over here and the runtime is 3 ms.

Now just let us submit this code and see if it passes all the rest of the cases or not. Mind it this time we have actually mentioned all the constraints. So let's see if this has to in two. Okay. So this time it has passed all the test cases. But still my conclusion with this question is uh it was still not able to generate the solution in one go.

uh but so I can give that to charge GPT because the first error that we faced was more of a lead codes issue because it was something with compile and charge GPT was able to give a proper oh now let's go back to our problems list so right now the score is one and one it was unable to solve one question and one or not.

So let's have a look at the third question and see if that brings any difference to the chart GPT's scoreboard right now or not. Third question that we are going to deal with is substring with concatenation of all words. This is in the category again and the success success rate is 31.1%. Which is even lesser than the first question that we faced which was median of the two sorted arrays.

We trying here we are actually trying to cover all the spectra the huge spectra of different types of uh questions and you know categories available in coding and uh to

give you an idea of how beneficial chart GP can be for you to solve difficult questions which can be helpful for your interview base in companies or you you can say well established companies or M companies.

So here this video is specifically for you to give an idea that whether you can use it for your benefit and you know to get an idea or you can actually uh compare it with yours uh and you can get a you know wider range of different types of approaches to a certain question. So let's start with the third question is that you're given a string and an array of strings probably words and all the strings of words are of the same length.

Now a concatenated substring in s is a substring that contains all the strings of any permutation of words concatenated. Here you can see it's given an example that if words has a b c d e f then ab cd ef basically it uh has done all the permutation and combinations that can be done using that specific array. Uh and mention that a c db e f is not a concatenated substring because it is not the concatenation of any permutation of words.

So we have to basically return the starting indices of all the concatenated substring in the string. Also you can choose any order for it. Here it has also given to examples for you to understand the question in a better perspective. Now uh copy this question and see if what programming language algeb chooses to answer this time with no chart again.

Copy these constraints. Okay, it's just a question. Okay, this is the solution that we're getting right now. You requested a model that is not compatible with this engine. Please contact us through our help center at help.openai.com. For further questions, let me just quickly refresh it.

If it has something to do with the you know demand also sometimes it happens that the console is very busy and you're unable to implement uh your task in it. So again let's quickly paste it over here that we want a code. Write the code. Return now. Press enter. I'm definitely it's not giving the same error. Let's see it.

This time it charge has anything to give as a solution. Okay. So it's generating the code in Python. One more thing every time charge doesn't follows a similar pattern as you can see in the first question it explained the logic first and uh then implemented the code second time it just gave you the approach not the logic and then implemented the code in multiple languages uh first choosing for C++ and this time it straight away went for the code so definitely we can say that it has some different styles of generating their code and explaining the code. I think it depends on the

understanding how they want the code to be presented in front of the user and to give the perspective that if the code is understandable or not and if the code has multiple approaches I think charge is capable to capable enough to give that the code is generated. Let's quickly copy it and paste it over here.

I feel the indentation issue is going to be there but was not there. Okay. F probably. I think if we backspace, let me quickly rectify this and I'll do you once the indentations are well. The indentation is uh corrected. Now

let's have a look at the code if it's correct or not. Let's quickly run it. Definitely it has given our first syntax error again. I can see that even specific video question it definitely goes to at least one error which is mainly the syntax one. sure if that is something with the lead code or you know with the charg code generation.

Okay, now that we have given the error. Okay, it seems that the error is caused by the use of type hints. the function type hints were introduced and the version we're using is lower than that. Okay. So basically it's generating now in the Python version. Uh probably this code is well suited for different version of Python.

Let's change it then and see if that helps. is actually generating the code. Okay, get in the I think it's again something with the self uh Python one the new code generated is

here. We'll definitely come back to that error and have a look at the Python 3 code also. Uh first let's copy this code and in giving the same error

and see what it has to say uh you know every time the RJPD generates the Python code we cannot ever takes self as an argument but believe as you can see when we start the code it's already mentioned what arguments we need to pass from that particular function so I think that is something with the lead code So that what all arguments it's passing even though we have mentioned everything this time we have mentioned all the constraints we have mentioned all all the uh necessary specifications that we want in the code even then the code is not correct in the one go so probably I'll get this point to GP it's something with the lead code because it's passing that parameter and uh everything we have to mention that parameter pretty much when we do the

charge is able to solve the question. Let's see what let me actually mention that pass uh self parameter self argument from that parameters one of make it a one of the parameters It's processing that

yes, self can be passed as the first argument to the given function. Now let's see if that it's able to give the correct solution or not. Again we can see it's uh generating the code and uh you know Python 3 but we can give that to charg that either it uses Python or Python 3 the error is with the self argument.

So once we mention that error and when we mention that specification that if charging pass self argument through that particular function in that code the solution is pretty much right. So here it's also implementing and giving the answer. Okay. data also has mentioned that uh it's important to mention import imported list and counter.

So okay just copy this because again there will be a lot of uh ination issues. Okay, now that we have it, copy the code already mentioned. So yeah, we'll just copy it from the function. Paste it. Let's run this code and see if it has the

solution in it or not. Yeah, there is an indentation issue. Let me Again as you can see it is able to pass all the test cases here and the runtime is 28 ms. Now let us submit this question and see if it is able to pass other test cases or not. Ed. So it is able to pass all the test cases and uh I think this is something with lead code again uh whenever we are

generating the Python code we are actually passing self argument uh in lead code but charg is not assuming it. So the solution is definitely correct. uh even though we are specifying everything we will have to be more specific that we have to run one more argument uh from the function so that it you know generates the solution in one go.

So let us uh try that in our next question. We can definitely see charge GBD was able to solve this. So now the scoreboard is 2 and one. Uh among the three questions it is definitely able to solve two questions. Even the first question it was able to generate a correct solution but it was not that accurate to pass all the test cases to a question list.

The next question that we are going to cover is in Queens category but still the success rate is 63% over here. Again a new genre or question we are covering over here. Let us see if solve it or not. Definitely the success rate shows that many people were able to do it. Definitely more than half of people who have attempted it.

So let's see if the question can this AI can beat that or not. So uh I have copied this question. This question mentions that the n queen's puzzle is the problem of placing n queens on an n into n chessboard. such that no two queens attack each other given an integer n return all distinct solutions to the n queen puzzle.

So basically n is any given number and you have to create a puzzle of n into n and you have to arrange all the queens in such uh you know way also the number of queens in the board will be equal to the number and in such a way that it is not able to attack each other in any case this part of the question and the constraint is just one.

Let me that it will be easier to keep a track of uh what all questions charge GBT is able to solve. So these questions are very popular uh in interviews. uh whenever you actually go to technical rounds and uh or prestigious companies, these questions are very popular. uh they are considered as very uh suitable questions to check uh to check your you know IQ and to check your potential that how well aware you are towards your coding potential.

Now that we have pressed enter it is giving the logic it is going to implement in its code. One approach to solve the end queen's puzzle is to use backtracking. Idea is to start by placing a queen in the first column of the first row. So let's see if the code is again capable to you know solve it or not.

Again it's giving the Python code. Generating the code. Let us quickly see whether it's Python or Python 3. Python Python 3. Okay. Once the code is generated, I'll also write that you need to pass one more argument from the main function uh that is self and uh let let it generate the code again and see that code can run in one go or not.

what I want and let us see. Sure. Here's an example of how you can pass an additional parameter self. Let's see if it suits the code or not. The solve end queen function is a method of the end queen's class and it takes self as its first parameter followed by the integer n. So yes, this solution does take self as a parameter. So let's

see if this can run in one go or not because this time we already covered the most uh frequently generated error which is syntax error of not mentioning parameter self. Let me see the indentation if it's correct or not. This time uh let us quickly run this. class. Okay, there you go. Now you can see that

it has actually run the code in one go and all the test cases are passed in one go. Even the runtime is 39 ms. So definitely chart GBT is able to provide the solutions the logics in

a proper manner. It's just that we have to be more specific with what we want exactly uh from chart GPT.

Right now we can see that it has been able to successfully generate four out of five uh questions that we have actually implemented till now out of which four are from hard category. Let us submit this code and see if it also covers all the test cases internally internally fed in for this question. Just a second.

I think I will have to submit it again. Here you go. It has accepted all the test cases and this question is done by child GPT. It has actually implemented Next question that we are going to cover till now I can say charged has taken the lead. It is pretty much able to implement all the questions. uh I think there are still some range of questions that it is not able to implement as we saw the first one was not a huge success but uh I can still give that to charge as it is an AI and still in a you know developing mode but still if it can give you 90% of the output correct it is a pretty decent and you know amazing thing to do. is shortest subarray with some at least

k. Now let me search for it. Again this question is from hard category and its success rate is even low which is 26.1. Let's see if this question can be solved by chart GPT. Let us have a look at the question. Given an integer array nums and an integer k return the length of the shortest non-MPT subarray of nums with a sum of at least k.

So if there is no such subarray return minus one. So subarray is a contigious part of an array. It has also given a description of what an array is or what sub array is. So now that we have a new chat and let me change. So here is one way to solve the problem. initialize two pointers uh left and right both pointing to the first element of the array. Initialize a variable.

So now this time this it is giving pointers to solve this question uh you know a perfect approach in a sequential manner so that you can also use these pointers and the logics

it's giving to actually implement your own code apart from you know asking it to generate a code. So I can see that it has given the pointers but not the code specific.

Oh yes uh it's giving an implementation of the algorithm in Python. Uh again we can see it's not the Python 3. It's Python uh little lower version of Python. Um, switch to Python. Then we have to pass self. So it has generated a code and explained

what all variables and what all statements have the individual functionality as. So we have also asked it to pass self parameter through its function and then write the code. So let's see if it aderes to it and generates new code with self parameter. Okay. So here is the new code with self I would have passed through the main function.

Now I do have confidence on chat GPT till now that it was able to generate logical and you know pretty decent solutions for every question. is passed to the class as an argument when an object is created and is stored as an instance variable. The shortest sub array takes K as an argument which is the target sum.

So basically it's trying to explain the code that what exactly it's doing and what individual statements have as an influence on the code and whatever parameters they are passing what influence or what position they hold in the code. uh definitely charge GPD is not just generating the code uh you know it's also explaining the logic and approaches towards it and when it's generating the code as you can see here they have legit explain the whole code how it's actually functioning that's a good way to you know put emphasis on you know put a confidence on the solution remove this.

Yeah. Okay, let me Okay, so it's apologizing. It made a mistake in its previous response. The init method should take two arguments, one for the nums array and one for K. So now it's again generating a new code or alling to the syntax error or type error generated like right or not mentioned that here the direct method takes two arguments one for the nums array and one for the k which was the

error exactly And it's again explaining the whole code so that one person who watches code or you know the logic could understand the functionality of it and save this code can run code checking for indentation removing class and run one more type error. Let's copy it. And since we mention that we want self

argument to be passed, it is still not able to generate you know the correct solution twice. So let's see if this time it can work out. issue though with the code. Apologize for the confusion. It seems that I misunderstood the you are trying to call the function directly without creating an instance of the class.

in that function outside of the class and simply call it by passing the parameters like this. Okay, the code is definitely generated by itself. So definitely I am not calling any function. Uh the code I copied was actually generated by chart GPD itself. So it's it contradicting its own pointers maybe. But we have copied the question uh sorry the code. Let's paste it.

Again look for indentation and run this code. Okay. Once again it is not able to you know pass us. I am error is there. Now again we have to copy this error and paste it. again according to charge GPD it was

confused with the context of the question uh like I said the code was anyway generated by charg so yeah this time I think charg GPT is trying to contradict its own logics Let's see if the current code can do the miracle of solving this question. class and run. Okay. So after four attempts of running

the codes generated by chart GPD on lead code for this particular question finally now it's able to pass all the test cases. So I can I have a very contradicting point right now. Uh not exactly contradicting more of a skepticism that okay chubby does generates uh

proper code or logic but it doesn't considers all the criterias or it also has a tendency of taking the question in a wrong context.

So I feel that when we as a human try to solve these questions we definitely try to implement the all the logics and if we get it we can actually get the code in one go or you know demand of the question and be AI being the superior version or you know trying to be the superior version of human brain and going to the extents of a human brain. Okay.

Still faces those issues can be a you know drawback for chart GPT because you can see for this specific question we have faced multiple types of error and we have seen charge GP contradicting its own prior code. So definitely this is something to consider or you know something to think about. Submit this code and like I said after giving four attempts it is still not able to pass all the test cases.

It is only able to pass 61 out of 97 test cases which is almost 70% of the test cases 30% of the test cases are still not passed. Even though we mentioned all the constraints, we have mentioned the comments, we have mentioned all the errors that this particular code can go through, still it was not able to generate the proper code that could go through all the test cases.

So this was a fail for charge GPT. Uh at least till now we have made out this point that charge GPT is definitely not able to solve every question. question array with same average. All right. Ar with This is from the category of hard questions and the success rate is only 25%. Uh let me refresh it and remove this code. Now let's have a look at the code

what it demands. So you are given an integer array nums. Now you should move each element of nums into one of the two arrays A and B such a and B are non- empty and average of array A is equal to average of array B. Now return true if it is possible to achieve that condition is not justified. Alo given a note that for an array uh average array is the sum of all the elements of array over the length of array. Okay.

So it it is giving you the logic of how to actually you know find out the average or whatever is exactly and also it has given a few examples to give you how the code needs to be projected or implemented. Copy the code and paste it. Okay, it is possible to achieve this by checking all the possible subsets of the nums array and comparing the averages of the subsets.

So definitely it has given you the approach uh the way you can actually think of solving this question. uh however this approach would have a time complexity. So it's also giving that this approach can have a time complexity of o to ^ n. So again it's suggesting a different approach a more efficient approach that will be using dynamic programming.

Now that can be used to find the subsets with a specific average and create a 2D array with the length of I and J that will be the length of the two we have actually given as input and represents whether or not it is possible to get a sum of J using the first elements of the nums array. four. So the time complexity of this approach would be O and S.

So definitely is not given a code. Uh let me ask for it. that not for every question charging a code. It is also mentioning just and the approaches that we can actually use. Uh definitely again we will have to mention that what we want from them. As you can see, I've mentioned this time to write a code for it and now it's generating a new code for it.

It's also mentioning comments that what every snippet of the code is actually for and what it will do. Like here you can see it will fill the 2D uh DP stands for dynamic programming. This here it has mentioned initialized the first column as true. So it is also mentioning the comments for better idea of you know understanding the code in a better way.

Also giving the note that this implementation assumes that the nums array is non empty and that the elements of the nums array are non- negative integers. Okay. Also the

above implementation will return the possible subset that can be formed by the array to fulfill the given condition and not boolean true and false.

But that's what we want exactly right. Uh still okay let us this approach is in Python. So let's copy this and see if this code can run or not. I can clearly see that again it needs to pass self uh parameter indentation. Give it a heads up. what I have mentioned what I want

specifically. Uh sure here is an example of how you can pass the self parameter. So now the current solution will pass self. Okay. There was an error. Okay. ing and it has generated a new code uh considering the criteria I just mentioned that I need self parameter to be passed through the function can partition.

So I have copied it and uh let's quickly paste it over here. Uh you know pass all the cases are not wait uh we need to remove this. Okay it's done. Now quickly just run this program. Okay, so here is a error attribute error. Solution object has no attribute. Okay, so let me copy it and paste it over here and see what chat GPT has to say about this error.

Okay, the error message solution object has no attribute uh split array same average suggest that there is no method named with this in the solution class. So it is likely that the test case is trying to call this method but it does not exist in your implementation anyway that I have not mentioned this method.

This was given by charge jib itself. So again we can see the condition that it is contradicting its own. So again it has generated new response and it says that it should resolve the issue and the test case should be able to call this particular method correctly. So that is something for us to decide. Now let us copy this code and see that now is it able to run or not and pass all the test cases or not.

and click on run. Okay. Now the new generated response does works for this particular question and the runtime is 12 ms and it does passes the first two cases mentioned. So let us quickly submit this code and see if the code is perfectly you know fine to deal with all the test cases actually fed by lead code.

Okay. Now again we can see one more situation that this code is not able to pass all the test cases even though we mentioned all the specifications constraints and we were pretty precise about the questions and parameters that we want our code to be done in certain form but still it is just able to pass 68 test cases out of 111 which is almost 50 to 60% of uh total amount of test cases.

So let's move on to the next question that is find substring with given hash value. So let me quickly search for this question. Find substring with cash value again it's it's a question from hard category and the success rate is 22.2%. So let's have a look at the question what it demands.

So here the hash of a zero indexed string of length k given integers p and m is computed using the following functions. Now hash with parameters SPM. Uh this is the logic given how we want our output to be demonstrated in certain value or how the hash can be generated. The particular formula uh how you can get the hash value of your you know uh string.

So the question is you are given a string and then integers uh and you have to return the first substring of that string of length uh given to you. Here it's K. The test cases will be generated such that an answer always exists. We are going to copy this course and mention all the specifications mentioned here in their chart GP console so that it gets all the specifications.

Copy the code and so I have copied the code and paste it over here. Now let's check for indentation. That's fine. Let's remove class my class and again it faces a type error. So

let's move on to the next question. Again we saw that chachi was not able to solve this particular question.

Coming back to our list, the next question in the list that we are going to cover is partition array into two arrays to minimize some difference. Let's quickly search for that question. Partition array. through two arrays. Again, this question is from hard category and the success rate is even more low.

So, let's have a look at the question first. It's partition array into two arrays to minimize some difference that is generated as the output. Here we can see that there are three constraints. So now quickly copy this question and see if it is able to solve this question or not. First we have to create a new channel. Copy all the constraints.

Let's see what chat jippity has to say for this particular question. Let's ask it to generate a code. This time it has only generated a logic. Not exactly logic the approach they they are going to follow or anyone can follow to solve this question. Let me ask if it can generate a code. Okay, it's definitely taking longer time to generate this code.

Okay, here is the code which is possible Python implementation and uh okay the code is generated and uh it's mentioning that the code takes in an input an array of an integer called nums. So basically uh it's explaining that what it actually it's doing which is pretty much explained in the question itself that what exactly it needs to function like and what will be the variables and what will be the inputs and how it needs to be segregated to uh obtain our optimum result.

So let's copy this and paste it over here. Again like I said we can see that it's generating you know uh again we could see that pass self parameter. Okay. And then attribute error. Let's quickly copy it. Paste it over here. See what charge GPD has to say about it. Okay. that you have seen that after placing the error it's giving that error message

you're seeing suggests that there's a problem with the function name in your code. So the error message is indicating that there is no function called minimum difference within the solution. Uh also I have seen that apart from the function name it doesn't have a self parameter. So let me just write it down.

12 argument in apps. Wait, let me copy. That will be better. and write a new code with it. Let me see if it can generate a new code this time. So yeah, it has agreed. So it there is an example and uh here's the code. So I have copied this and pasted over see any such huge difference in the

syntax or the logic of the code but yeah we can definitely think of running now. Let's run this code. Copy it and paste it over here. Again an error has prompted out. I don't know how uh valid code we'll get after projecting this error over here in chart GPD. So yeah definitely considering the error message you are seeing is indicated that the function is returning inf which is not a valid value for the expected return type integer.

So in the base case where I and J are both zero the value of is set to float which is positively infinity. However the expected output is an integer. So this valid this value is not valid. So again considering this change the charge JP is giving a new code. So let us see how valid this new code is. Okay I can see that it has changed the logic over here.

instead of float inf uh written a new syntax a new logic over there now and also charge guarantees that this may solve the issue of returning INF as the value let's see how accurate charge GPT here is now we have pasted the logic uh yeah as sorry I need to remove these two lines. Now let's quickly run this program.

Okay. So now as you can see it could only pass one test case but not the first two. So there's no point of submitting this code as you can see that the logic for this code or this code as a whole is not well. Moving on, next in the list we have is longest common subp partath. Now let's see longest common subpath.

So this question also comes from category of hard questions and the success rate is 27%. Now let's see what this question demands actually. Now let's quickly copy this question with all the constraints and whatever apart from example is left on the screen we need to copy it. As you can see over here you cannot just miss out on any specifications.

Looking at the constraints copy and paste. Now enter giving the approach to solve this question which is dynamic programming approach. Okay, we didn't want any code over here. So let's try that if we can present a code. I have asked to write a code for this and uh yes charg has definitely agreed to provide me a code for it.

Uh charg is done with its explanations. I'll uh type it out and wait for the new code which will contain self as a parameter that it's this code is pretty similar to the previous one. It's just that it's using the self argument the function as it was mentioned by me. So let's quickly copy this code. Paste it over here.

Check for the indentation. Run this code. I should be any Okay, I spoke too fast. Here we have another error. Let's see what Chajibity has to say about this. The list index out of range error is likely occurring on line 10 because the indices I and J are being properly bound checked before being used to access elements in the path and TPs.

Okay, so we have found a new code. Let's quickly copy this code and see if this code is capable of, you know, eliminating the errors we probably found. Click copy and paste. Let me check foration. Done and run. Then we got a error over here. Even though after providing so many specifications and criterias and errors and conditions yet charged was not able to provide a perfect solution for the code.

Uh moving on to the last question of our list that is going to be uh sum of total strength of wizards. Hopefully this question does some magic for chart GPT and prove itself lucky

for ch GPT graph for this video because for now we can see it's a 50/50 scene. Uh half of the question chy was able to provide solutions what with and half of the questions ch couldn't actually figure out what uh needs to be done even the logic and approach was correct still the implementation of the code was not correct. So let me search for this question. Sum of total strength of wizards. Again a hard category question. So let's have a look at the question first. Let's quickly copy this question. Uh

create a new chat. paste it over here and uh look for constraints. Now that we have copied it, paste it and enter. It's giving an approach how you can actually uh think of the solution for this particular question. charge has not generated any code. Uh let me ask for it. Okay. So here's an example.

Again they have implemented the code in Python. Now it is also giving a note that the approach is valid only if we allowed to modify the original array. And also we are not working on down over here. and then write this code. So let's see if it can do it with self argument. Okay. So yeah, there's an example. It's generating the code. Okay. So let's see

what's the update with the code here. This R. Let's add solution and run the code. So, there's a runtime error. Let me see why. this error

and paste it over here. What charg have to say the error message solution object has no attribute mod suggests that there is a class named solution and the code is trying to access an attribute named mod on an instance of that class but the attribute doesn't exist. we probably uh need to make more specifications and if it still doesn't works then it clearly classifies that charge doesn't takes every point or a classification uh you know in consideration uh which ultimately u you know reflects on the solution. The new code is here. Let me quickly paste it. Okay, let's try running this code. Let's see if this works running it again.

Again, it has a runtime error. Was unable to solve one more question. Now that we have tried and tested a huge spectrum of questions from lead code on chat GPT, we can

conclude that though chat GPT is an amazing tool with the bright future, it still has its own limitations and maybe it is not ready to replace humans or compete with human brains.

These questions were picked from a list of frequently asked questions for interviews and examinations. Chad GPT does have a potential to generate logics and approaches for the code in an effective manner but still its ability to analyze the question is weak as compared to humans. As we know these questions are there the success rate does shows that a proper solution do exist for these questions but still even after multiple attempts charge GPD was not able to find the correct answer but we can also give charge GPD the benefit of doubt that it's still it's in in its initial phase and still there are a lot of aspects that need to be worked on so probably in future charge GPD can take an upper hand over this but for now charge GPD needs to do a lot of work for these situations imagine being able to create stunning ing cinematic videos with just few clicks and no advanced video editing skills required. Welcome to Google Flow. In this course, we are going to show you how to turn your ideas

into professional quality AI videos using Google Flow, a cuttingedge platform designed to make video creation faster, easier, and more impactful. Whether you're a marketer, educator, or content creator, Google Flow has everything you need to generate highquality videos that captively and engage your audience.

You'll learn how to transform text into video, create dynamic scenes, and fine-tune your edits with the power of AI. By the end of this course, you'll have skills to bring your visual storytelling to life, whether for social media, online courses, or marketing campaigns. Let's dive in and get started on creating videos that stand out and leave a lasting impression.

Before we start creating mind-blowing AI video, let's quickly understand the engine behind everything which is the Google flow. In this first part, we will walk you through what flow actually is, why everyone is talking about it and how it evolved from video effects and how you can start using it today.

Now, let's start with what actually is Google flow. Google Flow is Google's all new AI video creation and editing platform. Think of it as Google's version of Runway or even Sora, but with Google's multimodel intelligence built on top of VO family of video models. Flow lets you generate videos from text. Extend the videos, modify scenes, use images plus text together, explore ideas using Flow TV, manage projects inside a clean dashboard.

Flow is designed not just for creators but for the beginners, marketers, educators, brands, and literally anyone who wants to turn ideas into videos. Now, Flow didn't appear overnight. It actually started as Google Video FX, an experimental page where creators could test VO2. Video FX was powerful but limited. There was no project management, no editing workspace, no multi-clipip timeline.

Google listened to creators and rebuilt the entire system into Flow, adding a project based workspace, multiple models such as V2, V3, V3.1, better credit systems, Flow TV for creative inspiration, upload, edit, extend and remix features. So, think of Flow as video FX 2.0, but professional grade.

Now, let's quickly compare Flow to other big platforms. Flow versus Runway. Runway is great for professional and cinematic projects, but Flow wins in speed, ease of use, and Google powered multimodel prompts. Flow also has a cleaner credit system and faster fast mode generation. Now, what about open AI Sora? Sora is extremely realistic, but not publicly available.

Flow is available today with subscription tires. Flow supports wider aspects, ratios, and more control of scenes. Also, Flow integrates directly with Google's ecosystem and Gemini. If you're a beginner or a creator who wants to speed simplicity, Flow is the easiest starting point.

To access Flow, you don't need any complicated setup. You just need to browse and type labs.google. If there are multiple links, you can click on the first link and log to your Google account which by default it will there you can access flow dashboards, manage subscription and explore the flow TV. Initially the free AI credits monthly is 180 credits if you can try it and if you want to create more and you can buy subscriptions.

Now coming to the subscription plan. Flow uses a credit based model. So it's important to know how credits work and before generating videos you have two subscription tires which is Google AI pro and Google AI ultra. Google AI pro is good for beginners gives you access to V2 and V3 fast lower cost per generation and it's good for experimenting.

Whereas Google AI Ultra unlocks V3 and V3.1 which is the most powerful models right now. It's best for cinematic, smooth, professionallook videos. Supports longer creative editing and better consistency. And that covers everything you need to know before creating your first AI video.

Next, I'll explain you what the interface and what are the options that you can use. So here is the homepage that appears when you log into Google's.lab lab and then we can start with create with flow and this is the interface of Google flow. Starting off we have flow TV which gives us inspirational videos to generate videos. So you can see multiple videos you can change any kind of videos.

You can see the prompt. What is a prompt for such kind of videos. So it's easy for you to generate get inspired to create new videos. Next we have a speaker button. This is the volume settings which you can increase and decrease. Then we have discard option. So this interface is connected with discard.

So you can share your work over there, get feedbacks etc. Interact with people. Next we have a help option. And then we have here there is another option where again you can access flow TV, know about flow, learn flow, app feedbacks and report legal issues etc.

And here you can see how many credits is left your uh Google account more credits and there is an option called my library.

So if you click on that you can see your previously created videos. Here you can access your previously created videos, arrange them, make folders for an easier access. Again we have manage subscription if you want to see what are the other memberships available and then we have the sign out. So if you start creating a new project here you'll get another interface where here you can create videos or you can change it to images.

You can edit by giving a name for the entire folder. So whatever videos or images that you have generated will get stored in that folder. And this is the basic prompting box. So here you can see text to video, frames to video, ingredients to video and even create with image. You can select as per and here we have different models what kind of frame you want number of outputs and model selections given.

There is an option called expand where you can expand your prompt which we will use it later. And that covers everything you need to know before creating your first DI video. Now that we have understood models, credits, interface and how to set up flow, let's jump into starting creating real videos using flow.

Now before that it is important to know which model acquires how many credits. So as you can see I have chosen V3.1 fast and you can see the generation of credits. So it uses for text to video it uses 10 credits whereas if I change to 3.1 quality it uses 100 credits. Again if you use V2 it requires 10 credits and if you are using V2 quality it requires 100 credits.

So as per your plan the credit system will be based upon if you're exporting something professional or for a client use V3 or V3.1. So the first thing is to generate text to video. Here we will learn the core skills that powers everything inside Google flow. You'll understand how to structure strong prompts, how to integrate audio and dialogue and how to get cinematic results consistently.

To get highquality videos from Flow, your prompt must be structured. And here's my seven element formula we'll use throughout this course. The first one is camera work. So you have to mention short type moment flow responds really well to film language like use medium closeup over the shoulder Dutch angle slow dolly in handle etc to make sure the prompt is cleared.

This will add a camera version to your prompt. And then repeat and then next you should mention your subject who or what the scene focuses on. You can describe age, clothing, mood and even physical appearance. Avoid vague descriptions like a man. Be specific. Then you have to mention the action what the subject is doing.

The anchors on the moment running, eating or even talking, writing works. Make sure to also adopt the environment where the scene happens. Add realism by describing the location, weather, time of the day, atmosphere. You can add lightning also, which is mood plus lightning style. Lightning controls moods even you can use like soft sunlight, neon glow, rim light, harsh shadows and even studio lighting.

You should also add audio which is voice, ambience, effects. Flow now supports native audio generation. You can specify your background ambience, music cues, character dialogue, voice tone, and even lip sync instructions. Next, you have to specify the style, visually aesthetic or even genre. Choose a style like film, noer, cinematic, anime, documentary, fantasy, and even hyperrealistic.

This structure gives Flo a complete cinematic understanding of your scene. Now, let's start with a simple prompt. So I'll be using chat GPT to generate a prompt so that we'll get more quality videos out of the text. So let's just ask chat GPT to create a prompt. Uh let's say we'll get a shot of a CAD sitting on a window sill.

So you can just give I need a prompt to generate a video from text which has a close-up shot of a cat sitting on a window sill. Let's say watching cars pass by and let it be on a rainy day. So as you can see chat produces a prompt which has more features. So you can just

copy and paste them. can just copy and paste the prompt in Google Flow. Now let's take a look at the generated output. So this was generated in V2. Now let's change it to V3 and see the output. I'm pasting the same prompt and also VO3.1 adds audio. Let's take a look at the output. Now to know the difference between VO3

quality, let's try on with the that model as well. So again I'm copy and pasting the same prompt but using 3.1 quality. The difference between V3 fast and V3 quality is that VO has fast lighter generation, quick drafts, and it's good for ideation and storyboard style testing.

Whereas V3 quality has the highest realism, better motion shadows, best for final videos. Let's check the video out. You'll clearly see a difference in motion smoothness, lightning realism, edge sharpness, texture, detail, and overall depth. Now, let's write a cinematic prompt. To achieve cinematic quality, use strong camera verbs, track glide, pull focus, dolly in, and also mention the lightning, whether it's soft rim light, neon reflection, or even a golden R.

Use mood words such as melanchy, tense, warm, mysterious. Again, add the textured detail such as rain on glass, dust particles, or even foggy air. The more visual intention you give, the better the realism. Again, you can do that using chat GPT. And say to add strong camera verbs, precise lightning moods, words, texture details.

I got this prompt from Chad GPT which says slowly dolly in towards a young astronaut standing on a lunar base at dusk. Soft blue light reflects of helmet vision. Dust swirls around her boots and looks up to the earth on the horizon. And we have ambient skyfi hum and soft wind cinematic high contrast lighting.

Again you can generate any kind of cinematic views that you have. Ask chat GPT to create a cinematic prompt of that and copy and paste it. So let's see what it'll generate. Notice this prompt has clear vision of camera movement, subject detail, environment, lightning, audio, mood, and style. Now let's observe the output.

So that was the video generated within few seconds. Another thing to keep in mind is that flow support spoken dialogue. You can just add he says with a colon and start the dialogue and enter whatever the character has to speak and the video is generated with lip sync and voice tone.

So again let's use chat GPT help and create the prompt. Here I have asked chat GPT I want a cinematic video prompt for a scene. Can you help me create one where a 30-year-old woman is standing besides a train window at night? She looks at her reflections and say softly, I am finally ready to move on.

The scene should include soft piano music and neon lights flickering outside. Please break this down into seven element for the video prompt. And I've given the seven elements that I need to add. So chat GPT has generated the prompt. So let's copy and paste this prompt in Google flow. Let's check out the output. >> So that was the video created using the prompt just within a few seconds.

Now there is an option called expand here. So let's check that out. So basically we don't need to explain the entire cinematic view. You just need to add a sentence and if you click on expand it'll generate a cinematic view. What are the things should be added? What are the noise or even the sound quality that should be added in that particular scenario.

So let's go back to the cat example. I'll just write a cat sitting on a window sill. Let's not give more information and just click on expand and you'll have three options whether it is

filer action figure or cinematic. You can just click on cinematic it'll add the prompt and if you just generate with that this is a easier way where your prompt you can just click on a button and your prompt will be already have all the information that requires to create a video or an image.

So, let's take a look at the output. So, there you go. It's a simple sentence converted to a prompt into a video in just few seconds. Next, we'll be learning how to animate static images, create seamless transition between the frames, and control camera movements without writing prompts for every detail.

This is done by using frames to video option. Animating static images bring them to life instead of relying on just text prompts. Frames to video allows you to turn still images into dynamic animations. This is perfect for scenes where you already have visuals but need to add moments or even transitions. Creating smooth transition between two frames is a key to making the animation feel natural.

You can do this by just adding the frame starting frame and then add the ending frame. Let's say the starting frame should be a sunset or a sunrise image and the ending image should be a sunrise. Let's take for instance and you can add any details which are required. Let's mention need a smooth transition. Let's say camera zoom in and let's generate it.

Let's take a look at the output which was generated. There you go. A smooth transition with just uploading two frames and a sentence to describe them with a camera angle. The sound was added by itself. You have now learned how to animate images, create seamless transition and apply camera movements easily.

Now let's see what is the option called ingredients to video. Here we can create and manage ingredients. ingredients such as character, objects, or even environments. Combine them into scenes and maintain consistency across your animation. You'll also learn the best practices for preparing ingredients to ensure a seamless video creation.

Ingredients are the core elements of your video such as characters, which includes people or animals. Also, it includes objects such as furniture, tools, etc. It also can involve any environment such as rooms or even landscapes. Managing these ingredients well is a key to creating cohesive scenes.

You can create or upload them and combine multiple ingredients to make a story come alive. The key is ensuring that all your ingredients work together by maintaining visual consistency across your references. So let's start by creating images which are ingredients that I'm going to use. I'm going to use the create image option which is again inbuilt in here.

As you can see nano banana pro is also included. So I need three images for the scene that I want. So I need a image of a woman, a lava lamp and a street with a mystery fog. So let's create all these three elements. Let's say a woman in her 30s. And then let's create a lava lamp. Also, let's create a misty street. A misty and foggy street.

Here you can see an image is created of a woman, a lava lamp and a misty street. So for this a easier way to just add ingredients is select the option. You can also see there is an option called add to prompt. So you can just select which one suits your preferences. Let's go with this. So these are the basic ingredients for you to create a video out of.

Now let's give it a prompt of the imaginary video that I have which is a woman walking through the street with let's say torso as a lava lamp. So, a woman whose torso is a lava lamp walks down a foggy street. Let's confirm the timings and at dusk and let's generate the video. Notice how we need to keep the lightning and color consistent to maintain a serial atmosphere.

Now, as you create more ingredients using create image, keep them organized in the ingredients library. This will help you quickly find and reuse ingredients for further projects. Currently, ingredients to video works with all VO models. Let's take a look at the output. There you go.

The three images converted into a video using all the elements that we have mentioned. Next, we'll learn how to create multi-clipip narratives using scene builder. We'll cover timeline editing, scene transition, trimming the clips, or even creating longer videos. You can create up to 60 seconds. So you can see an option called scene builder here.

Just if you click on that, it will transfer you to a interface of the scene builder. Scene builder is designed to let you arrange multiple clips into a cohesive sequence. You can use the timeline to visually organize your clips and create longer stories. This feature is perfect for multi- scene narratives. So for example, if I just say what should happen, it'll create a video based on the prompt.

Since it is text to video, you can use any other options if you want to. So here you can see the video. >> What should happen next? >> Let's say the next video which I want to generate. But here you can see two options called jump to and extend. So extends allows you to continue the same action or shot from the current clip.

Whereas jump to lets you move on to a new scene or action creating a hard cut between the clips. These two features gives you control over continuous action versus scenes transition. So let's create a second scene of extend. Let's say found treasure. And let's say we'll jump into a next scene enjoying city life.

So we should wait until the previous video is generated and then you can generate. Uh let's change it to quality. So the second scene is ready. So let's take a look at it. What should happen next? Do you see the two scenes? Since I have used the option called extend, the same scene is being repeated.

Let's see the third video which I have used jump to. So there's a clear jump in the scene. So like this you can create a long format video up to 60 seconds. And there are more options like arrange. So you can arrange the scenes where you can just drag and drop. And there is also a trimming option. So here you can trim your videos to as per requirement.

Trimming the clips is essential for cutting unwanted parts. You can arrange the clips in the timeline to perfect the narrative flow. Use the timeline handles to adjust the start and end points of each clip. So this Google flow has an editing feature as well, which is good for creating videos up to 30 seconds using the scene builder.

You're now equipped with everything you needed to create multi-clipip narratives and seamless transition. Next, we'll see how to generate realistic dialogue, adding ambient sounds, and controlling subtitles. You'll learn how to create audio that complements your visuals perfectly while avoiding common pitfalls. Now, let's just type in few things and on the go, I'll explain what kind of dialogue should be used where.

So, first let me define the character. Here I've added a character which says a confident woman in her 30s wearing a futuristic outfit standing in a modern city space. She has the short black hair and a determined look and she's holding a tablet in her hand. So let's say now I want a dialogue which says welcome to the future.

That means the character should say welcome to the future. So let's say she sees and colon Welcome to the future. This kind of prompt helps the AI understand exactly what character should say. This kind of dialogue writing is called explicit dialogue. Now let's see what is implicit dialogue which is letting the AI choose words.

I can just say she enthusiastically explains her discovery and this is implicit. So you're not specifying any kind of words that the character should say. Instead you are letting

the AI choose. Now let's add ambient environmental sounds. Let's say the background should I should have rain ambience.

So just click just write audio hyphen let's say distant traffic. This type of audio prompt adds natural sounds like rain and distant traffic, enriching the environment with overpowering dialogue. Now for subtitles, it's usually wid. So if you don't want any kind of subtitles, you can just say prompt no subtitles. This ensures no subtitles are shown during the video, which lets the visuals and audio take the center stage.

So I need the subtitle. So I'll just generate it. So again I'm using the expand option. Let's say this is a action figure and it'll add the rest of the details and let's create it. Keep in mind you can also add two characters and give them names and say suppose John says so and so dialogue and Sarah replies to that so and so dialogue and it works perfectly.

It lip syncs automatically. Clearly specify who is speaking in that kind of scenarios and what they're saying especially in the scenes with multiple characters. Here are a few mistakes to avoid when working with audio. The first thing is the dialogue too long for 8second clips and then not specifying no subtitles when you don't want them.

And then also completing audio elements like music, drawing, out, dialogue. Make sure the audio elements don't overlap. Let's take a look at the output. Welcome to the future. With this device, we've revolutionized interstellar travel, making the galaxy accessible to everyone. >> Now you're ready to master audio and dialogue generation.

You can create realistic conversation, control background sounds, and avoid common audio issues. Next, we'll dive into advanced prompt engineering techniques. We'll use tools like Chat GPT and Gemini to improve your prompts and apply cinematic techniques for better results. Crafting powerful prompts is essential for achieving the best result.

We'll explore how to structure prompts that leads clearer, more engaging outputs. So until now, I've used chat chick. You can also use Google Gemini to create a prompt. Let's ask Gemini to expand this into a detailed VO3 prompt with camera movement, lightning, audio, and cinematic style.

The simple line is a woman discovering a secret door. So let's see the prompt. So here you see you get a detailed prompt which you can just copy and paste it and you'll get a stunning video. Gemini will take the basic idea and create a detailed prompt that includes camera moment, lightning and even a cinematic style.

Basic prompts can be turned into a cinematic description with a little adjustment. Add details about camera angles, lightning and even mood to make your prompts feel like a movie script. Now again prompts can be tailored to different visual styles. Whether you want a Pixar 3D look, 2D animation, or even comic style, change the description can shift the overall aesthetic.

So, let's create a prompt for a comic style. So, I need a prompt for comic style video. And let's say the scenario is a hero in a urban environment. Let's say heavy shadows and a comic book panel look. So let's just generate the prompt, create the video and see the output. So just you need to copy and paste it in your prompt box in Google Flow and let it generate the video.

Now testing variation of your prompts helps determine the best version of your video. Try different approaches to see how small changes in your description can make a impactful final result. So let's take a look at the final video. These videos take a few seconds to get generated and gives movie feels.

Now you have all the tools to craft advanced cinematic prompts expanding them using Gemini and even chat GPT test different variations for best result. Next we'll explore how to use AI tools for real world application across various industries. will walk through

creating marketing content, educational videos, and small social media clips using AI powered tools.

AI tools are now revolutionizing multiple sectors. Here's a quick breakdown on how you can apply these tools in different fields. In marketing and advertising, you can use it for product demos, brand stories, testimonials. Whereas in education, you can use such kind of videos for lecture illustrations, concept visualization, tutorial, animations.

In entertainment, we can use it for short films, music videos or even storyboards. Whereas in social media, we can create viral content, influencer videos, and even brand narratives. Coming to corporate, we can use it for training videos, presentation, internal communications. So, let's create an advertisement for a product.

So, for this, I'm using ingredients to video option. And uh let's say you have images of your product and what are the ingredients in it. So let's upload the product and the ingredients. So and say need a marketing campaign for this product. which packaging and ingredients are given. So let's expand this.

Let's say I need a cinematic way of giving the video and let's generate the video. So let's take a look at the output. So there you go. Without needing anyone's help, you create a marketing campaign and you can run ads on a similar kind of video. This is all generated within few seconds with a strong prompt and couple of images of your product and the ingredients.

Now coming to educational content. Educational content is enhanced by taking ahead visuals where speaker explain concepts while engaging the viewer. So let's generate talking head explainer. Here we'll be generating an educational explainer using the talking head. The character will explain a topic while appearing on screen helping illustrate complex concepts in a relatable way.

So let's give the prompt using chat GBT. So here I've given the prompt saying create a professionallooking talking head explainer where a friendly approachable character appears on screen. The character should be dressed in business casual attire and positioned to the left or right side of the frame allowing space for visuals or graphics to appear next to them.

They should speak directly to the camera in an engaging conversational tone with hand gesture and facial expression that help convey the message. The character should explain a complex topic. Artificial intelligence is a simple relatable language. Goal is to break down the concept into bitsiz digestible pieces that anyone can understand no matter their prior knowledge.

As character explained, revert visual aids like animation diagrams. Key points will appear alongside them to support their explanation and help illustrate the concept clearly. And let's generate the video. So let's check the output. >> Artificial intelligence is essentially the ability of machines to mimic human thought.

We see everyday examples all around us and the future is already here. Artificial intelligence is essentially the ability of machines. >> So you can use the scene builder to create an entire video for your educational purpose or for teaching people. Creating social media content often requires specific aspect ratios. Here we are generating a YouTube shot or even you can consider it as real cliff in the format of 9 is to 16 format.

So here you can see I'll be changing the format from landscape to portrait. So make sure if you're generating it for social media it should start with an attention grabbing hook that immediately appeals to the target audience. For example you can say want to know a secret to social media success.

In the first few seconds, showcase a key concept of a trending topic. For example, how to grow your brand on Instagram or how to digital marketing tips to cutting edge AI tools. The video should have bold text overlays, dynamic animation, and quick cuts to maintain

energy and engagement. You can use upbeat music or sound effects with subtitles or highlighted text to ensure the content is consumable without sound.

The visuals should be highly interactive with clear call to action prompts. For example, you can use like and share, follow for more tips appearing at the right moment to encourage users interaction. Incorporate relevant images or icons that supports an explanation or tutorial in a way that's visually appealing and relevant to the topic, ensuring that overall look is polished and professional.

So again use chat GPT for the prompt generation. I have already given the following information and asks us to create a prompt. It has created like a high energy educational social media short video. Aspect ratio is also mentioned. The scene features a brightly lit modern digital marketing studio setting with a clean minimalistic aesthetic.

The video must start immediately with attention grabbing hook. Want to know the secret to social media success? displayed as bold animated text overlay filling the screen. I've also added when to show the follow for more tips and CTA visibility all the features I've mentioned. So let's just generate it and see what's the output.

So let's check out the output. Want to know the secret to social media success? Tip one, define your audience. Tip two, analyze engagement. Tip three, consistency is key. Want to know the secret to social media success? Tip. >> So, you can use the scene builder to create more such clips. Add them together and create a viral video which you can post it on your shots or even as a res on Instagram.

Now coming on to the next part. The first problem being silent videos or missing audio. The solution is if your video plays without audio, ensure that you are in text to video mode and explicit request audio in your prompt. The second problem you'll go through is inconsistent character appearance in jump.

Here, when jumping between scenes, copy the exact character description. Use the same framing and references the original clip for consistency. This ensures the character stays visually consistent throughout the video. The next problem will be poor lip sync or mismatch dialogue. The solution to that would be to improve lip sync.

Keep dialogue short which is 3 to 6 seconds. Use colon format for explicit dialogues like she says so and so and he says so and so for better synchronization. Ensure natural speech. The next problem would be unnatural movements or physics error. The solution for that would be unnatural movements or physics error can happen.

Be specific about motion you want. Example slow dolly in or tracking shot. Use reference frames for consistent movement and try different models for better results. And the next problem would be hallucinations or artifacts. For that if your video has visual errors like hallucination or artigraphs, trim or problematic section and regenerate it with clear and constraints.

For this problem, you need to trim the problematic section and regenerate it with clear constraints. This will improve the quality and remove the unwanted elements. Now you have an idea how to troubleshoot common issues like missing audio inconsistency in characters and lip sync problems. Next, we are exploring the upcoming features in Flow, including exciting updates and advanced creative workflows.

So, let's dive in. Flow is consistently evolving, and here are some future features you can look forward to. The first one is remove object feature. The second one is improved scene builder persistence for better continuity and also enhance character consistency tools, longer video durations for more flexibility, API access for developers to integrate flow in other systems.

Flow can be more powerful when combined with other AI tools such as midjourney for style reference images. You can also integrate multiple AI tools into your workflow for

enhanced creativity. For example, use MidJourney for image generation. Then Flow for video creation and then Gemini for prompt expansion.

Build modular asset libraries for quick access to commonly used elements like characters, backgrounds, and props. This allows for more efficient video production. Flow supports collaborative team workflows, allowing multiple creators to work on the same project seamlessly. share assets, iterate on prompts, and create together in real time.

Have you ever wished for a coding assistant who can not only suggest lines of code, but actually build your app? Well, imagine this. You're stuck at a project with multiple task at hand, writing code, debugging, fixing errors, running tests, and it feels like you're constantly bouncing between them.

Now what if I told you there's a tool that can automatically take care of all those task for you leaving you free to focus on the bigger picture. That's exactly what GitHub copilot agent mode offers. It's like having a superpowered coding assistant that can handle everything from creating apps to debugging them. Today we are going to explore GitHub copilot agent mode by building a simple app using HTML, JavaScript and CSS.

I'll walk you through what makes VS Code Insiders the perfect platform for this. How agent mode works its magic and how you can use it to create an app with almost zero effort from your side. So let's dive in. Before we dive into the demo, let's take a quick look at what VS Code Insider is because this is where all the magic happens.

If you have used VS Code before, you would know it's a powerful code editor. But VS Code Insiders is like the VIP version where you get to try out new features and improvements before they hit the stable version. It's updated daily which means you get to play around with latest and greatest tools including Git Copilot agent mode.

And now you might wonder what exactly makes VS Code Insiders different from the regular VS Code. Well, in simple terms, VS Code is your reliable stable editor that you can use for your daily coding task. Whereas VS Code Insiders on the other hand is where the cuttingedge features are tested. So if you want to early access to new tools and improvement like C-Ilot agent mode, VS Code Insiders is where you want to be. All right.

Now let's talk about what C-Ilot agent mode and what makes it stand out. You might be familiar with GitHub Copilot which gives you the code suggestions while you work. It's like having a helpful coworker giving you ideas, but agent modes take that a step further. It's more than just a helper. It's like an autonomous coding partner that can handle entire project for you.

When you switch to agent mode, you can give a multi-step task like creating an app from scratch or fixing bugs across multiple files. And it will take care of the entire process. It doesn't just suggest code. It writes the code, runs the test, and fix errors automatically while you simply watch it work with magic.

Think of it as having a personal assistant who can analyze your code, figure out what's missing and then complete the task in real time. So now let's move on to our demo part and see how it actually works. Also just a quick information guys. Now if you're interested to master the future of technology then please consider the professional certificate course in generative AI and machine learning which is your perfect opportunity for you.

Offered in collaboration with the E and ICT Academy IT Kpur. This 11-month live and interactive program will provide you hands-on expertise in cuttingedge areas like generative AI, machine learning tools such as JGP, Dari 2, Hugging Face and many more. You'll also gain practical experience through 15 plus projects, integrated labs, live master classes and alongside you'll earn a prestigious certificate from IIT Camp.

So hurry up and enroll now and find the course link in the description box below and in the pen comments. So now let's get started and see how co-pilot agent mode actually works. So the first step is installing the VS code insiders. So just simply type on the Google and head over to this website.

Remember do not install the normal uh Visual Studio Code. You would need to install the Visual Studio Code insiders for this particular agent mode. All right. So since we have already pre-installed this application from here, I'll just head over to the website from here. So this is the interface of Visual Studio Code and the only difference is between a Visual Studio Code and the Visual Studio Code insiders is this.

It's quite light in color, light blue in color. That's it. So yeah, so we'll head over to this uh chat section. This is the GitHub copilot uh icon from here and just head over to this copilot edits. From here you would need to select agent. So by default it will be automatically in the edit mode.

So just select agent from here and for this particular model I'll just suggest since uh claude 3.5 is good for coding. So we'll just select the cloth 3.5 solid which is only the preview version here. So we'll just select that and we also have this voice chat option. But for this you would need to actually install the extension known as VS code speech.

So if you install this particular extension what will happen is uh you can get the chance to actually interact with the co-pilot by simply talking and seeing your particular prompts or command what changes do you want the copilot to do it for you. So it's actually really cool and I'll just suggest that simply download this particular extension from here and then our next step would be we'll just see how it works. Okay.

So first we'll just create a normal um application or an app using HTML, CSS and JavaScript. So I'll just click on this voice chat option and I'll just say the copilot to actually create a completely new application. All right, let's do. So I'll just click on this uh voice chat option and I'll say to create a new app using HTML, CSS and JavaScript.

Create a new app using HTML, CSS and JavaScript. So see it's automatically started running now and it's saying that I understand you want to create a new app but I notice there's already an existing project structure in your workspace. So should I check its contents to understand what's already there and avoid conflicts or let me gather some context from here.

So here I see that the current workspace contains a to-do list application. So let's we will just modify the changes in our to-do list application. So it's saying that would you like to uh create what kind of app would you want to create and would you want to create a new directory or use the existing features requirement create a to-do list and I will say that please uh modify changes in the existing to-do list application improve the UI design of the application and try to add more features and functionalities. So I've told uh to create a toolless application and I'll say to actually improve the structure of the application. So let's see how it does. So our files are already from here and it's making changes. So we'll just head over to our index.html. Okay. So it's actually a learning platform rather than a to-do list. So it will actually create and from scratch it will create a to-do list from here. It

will generate all the edits. So, as you can see, it's applying the edits from here. I mean, you don't even have to do anything. Just simply type your prompt, sit back and relax, and the copilot will do everything for you. So, now it is actually uh adding the to-do list interface, adding the due dates, priority levels, and the categories.

So, these are the improvements which uh the copilot is actually doing to my existing project. So now it will add the styles to enhance the UI of the new elements. So we'll just go to the style dot CSS and see what changes is it actually doing. So now that it's doing changes in our uh CSS, so we'll just also see that uh make changes in the JavaScript.

So now it will actually improve the JavaScript functionality to support the enhanced UI features that we added. So it's first analyzing the existing JavaScript files from here. So it is now enhancing the JavaScript functionality and whatever updation needs to be done

in this particular file which you can just simply click and just check the edits it's generating for us.

So now that it's doing the changes, I'll just say to run the application. So first it's saying that uh it's a todo application. So we'll need to serve it through a web server. Let me help you set that. And you can use this command in the terminal here. Also one more thing is that so just simply type on this app.

html HTML and from here we'll just select on open uh reveal in the file explorer and we'll just click on this particular app source code. So yeah so here as you can see our to-do list is ready now and it has given me all the due date low priority select category. So we could actually improve the UI UX of the design but then uh for the time being it's actually okay and it has also the categories from here.

You can see it has added the categories like work, personal, shopping, health, active. So you can just add a task like let's suppose I want to go for running and the due date you can add from here. Okay. And you can just add this particular task from here and see whether it's completed or not. You can also delete this particular task if you don't want to do it or if it's completed just select here completed.

So I think it's pretty cool by just not typing any particular code it will generate a whole particular website for you and that too within seconds. So it's really amazing. One more thing I would like to say now if you would need to actually improve the UIUX of the design you can also type here and see how to improve the structure and the UX of the design.

And if you want to add more functionalities, you can also do that. So for the time being, I think it's it generated a really cool website. Now one more thing now you would be thinking that why is this so important for me? So if you're actually tired of writing task, code, fixing bugs, testing, debugging, then copilot agent mode is your answer.

It takes all of your place so that you can focus on the bigger more creative aspect of your project. So whether it's building new apps, updating old code, or running test, Copilot does the heavy lifting and that you stay productive. And there you have it. GitHub Copilot agent mode isn't just a tool.

It's like having a coding partner who is always on top of things, ensuring your projects run smoothly. And by automating everything from code writing to error fixing, it makes development faster, easier, and a whole lot more fun. So if you haven't tried VS Code insiders yet, make sure to jump in and test agent mode for yourself.

It's a complete gamecher for developers looking to boost their productivity. So if you enjoyed this video, don't forget to like, share, and subscribe and comment below with your thoughts on GitHub Copilot agent mode and how it can help you in your exciting journey. >> Hello everyone and welcome to the tutorial on prompt library for all use cases at Simply.

The prompt library is a comprehensive toolkit for mastering my use cases with ease. Whether you are delving into programming, honing creative writing skills, or exploring data analysis, this library offers a versatile array of prompts tailored to your needs. Now, before you move on and learn more about it, I request you guys that do not forget to hit the subscribe button and click the bell icon.

Now, here's the agenda for our today's session. So, guys, we are going to start with first understanding the prompt structure. Moving ahead, we are going to understand testing and iterating. Then we are going to explore the prompt examples and at the end we are going to conclude our sessions with utilizing prompt libraries and resources.

Craving a career upgrade? Subscribe, like and comment below. Dive into the link in the description to fasttrack your ambitions. Whether you're making a switch or aiming higher, SimplyLearn has your back. Just a quick info guys, SimplyLearn has got a post-graduate program in AI and machine learning.

You can boost your career with this AI and ML course. This course is delivered in collaboration with BI University and IBM. You can learn the in- demand skills such as machine learning, deep learning, NLP, computer vision, reinforcement learning, generative AI, prompt engineering, chat GPT and many more.

So guys, hurry up and join the course. The course link is mentioned in the description box. So guys in today's video we will be exploring the prom structure for various use cases. Now first let us try to understand the prom structure. So guys I'll break down the prom structure. So here first we have the action verbs. So guys think of action verbs like a boss telling chat GPT what to do.

It's like giving Chad GBT a job. So for example if I say write you are telling Chad GPT to put words on the page. For example, if I say write a story, I'm telling Chad GPT, hey, I want to you to make up a story for me. So this is something like this. Now let us ask Chad GPT, hey, so write is your action verb all over here.

So this is the first prompt structure that I would like you to apply. Now the second one, you could give a theme or a topic about. Now if you say just write a story, chat GPT is going to give any random story. So we won't want that. Now the next thing that we cover basically is topic or theme.

So what theme or topic you are looking about? This is a part where you are giving chat GBT a subject to talk about. Imagine you're telling a friend let's talk about cats. So cats are the given topic. So if I say write about your favorite food, I am telling chat GPT tell me about your favorite food.

So you have to always include a topic or theme along with your action verb. So here I can include some certain thing like this that write a story about food. So you could see all over here chat GPT has given two uh responses. This is response one and this is

response two. Now the third thing that comes up all over here is constraints or limitations.

Think of constraints as a rules or boundaries for Chad GPT to follow. It's like saying you can talk about cats but only in three sentences. So if I say write a poem in 20 words, it's like I'm telling Chad GBT make a short poem using only 20 words. So this is one of the things that you have to always keep in consideration regarding what task you want to give.

So always include constraints or limitations. Fourth one is background or information context. So so this is also one of the most important parameters. Uh what exactly it means is like this parts sets the scene for JGBT like giving it a background story. Imagine you are telling someone about a movie before they watch it.

So if I say imagine you are on a spaceship and telling Chad GPT pretend like you are flying through the space. So this is also very very important for you to consider to give certain idea regarding your background or information. Now the fifth one is conflict or challenge.

Guys this adds some spices to the prompt. It's like a puzzle for a problem for charge GPT to solve. It's like saying talk about cats but tell me why some people don't like them. So if I say char GPT explain why reading is important but you can't use the word book. I am challenging ch to be creative. So this is where conflict or challenge you have to give to charge GPT.

Now example let us take one example on this. So for example, if I say the action verb as right, we'll highlight this with red. And the topic of theme could be like your favorite vacation. If I talk about a background or context like say you are on a beach with your friends or conflict or challenge, we can give all over here something like in just 50 words.

So guys this is certain thing to follow while giving a prompt to charge GPD. So in this way putting all together you could combine all these three things and form a sentence and this prompt is going to be very very effective to solve the problem of generic responses. Now with this simple example you can see how different components come together to create engaging prompts for chat GPD to work with.

So guys, whenever you are giving a prompt, I would request you to always follow this structure. So it's going to create a map for you to get a more precise answer. Now let's take example and elaborate the prompt library with examples to make it more understandable. So guys, let's take another example of text classification. So for text classification, we'll take the action verb as classify and our text type would be product review.

Example could be classify the following text as negative, positive or neutral sentiment. And after that you could give like the product review exceeded my expectation. So if you give certain thing like this you would say this is a positive sentiment. So making your prompts in this manner with a proper structure you are going to get a very particular response which fits what you need.

So always remember this structure whenever you are framing any bra. Now let's move to the second part that is testing and validation. Guys testing and iterating are essential steps in refining prompts and ensuring optimum performance from chat GBD. Let us break down this process. The first process is prompt validation.

So before using a prompt, it's crucial to test to ensure it that it generates a desired response accurately. Then you evaluate the output. You're going to generate responses using the prompt and evaluate the quality, relevance, and coherence of the output. Third, check for errors. Look out for any errors, inconsistencies or unexpected behavior in the generated responses. Compare against expectations.

Compare the generated responses against your expectation or any requirements to verify that they meet your desired criteria. The fifth one is solic set feedback. Seek feedback from peers, colleagues or domain experts to validate the effectiveness of the prompt. For example, like analyzing the results.

So you would say analyze the results to testing to identify areas of improvement or refining the prompt. Next is modifying the prompt. Based on the analysis, make the adjustment to the prompt structure. Next, then fine-tune the parameters. Experiment with different variations of the prompt such as adjusting constraints, changing topics or refining prompt to assess whether the changes have resulted in improvements in the quality of the generated responses.

The fourth one is retesting. Test the modified prompt again to assess whether the changes have resulted in improvements in the quality of the generated responses or not. And the final step is iterate as needed. Iterate the testing and modification process as needed until you achieve the desired outcomes and generate highquality responses consistently.

So this structure you have to always follow when you are iterating. So I'll give you an example. So like we have given uh initial prompt as write a product description for a new smartphone and I would say include details about features specifications and benefits and I would say add a constraint all over here that keep the response is 100 words.

So this is your initial prompt which you are given. Now for testing, the next comes is testing. Generate product descriptions using the initial prompt. Evaluate the quality and relevance of the generated descriptions. Check for errors if inconsistencies or missing information is there. Compare the description against the expectations and requirement.

So this process comes under testing. Okay. So give it uh like change your prompt a little bit. give a specific uh description regarding a certain product and you would ask that and just next process would be evaluate the quality and the relevance like what you are

uh getting as a response check for errors like go to Google like see if it's same is coming up then what's the customer expectations regarding that product so if the overall structure is like technical structure is maintained so this gives the first phase of testing next one comes the analysis some descriptions lack detail and fail to highlight its key features. Okay. So in this scenario the descriptions vary in length and structure leading to kind of inconsistencies. Certain descriptions like here will focus more on technical specifications than the user benefits. So overall the quality and the coherence of the descriptions needs improvement. So you have to take all these parameter and you

have to reframe your prompts. Okay. Then next comes is iteration. You have to modify this prompt to provide like more often to give a clear instructions and emphasize the user benefits. Write a captivating product descriptions for a new smartphone. Okay. Then move to retesting.

Generate product descriptions using the modified prompt and for the outcome you would say that the revised prompt should yield more compelling and informative product descriptions. So this is how you have to do iterate continuously to get the proper response like which you would be needing. Okay guys, now let's move to the final part of this video that is utilizing the prompt libraries.

Guys, utilizing prompt libraries and resources is essential for streamlining the prompt writing process and access a wide range of pre-esigned prompts for various use cases. So you're going to get a library of a predefined prompts. Okay. So there's one website like which I want to show you. This is called Anthropic.

So Anthropic has recently released a prompt library. So guys, they have given a wide data of a prompt library. So if you just click on this, so you're going to get like what are the effective prompts in all these domains. So give it a shot. Uh try to see what are the uh like resources you're going to get all over here.

It definitely is going to fine-tune your responses. Now let's move to the process. So when you are talking about the prompt libraries, the first step is explore the existing libraries. So you can see that I have given a reference to a prompt library all over here which is released by anthropics team for cloud and also workable for chat GBT.

Next is you have to understand the available prompts. Familiarize yourself with the prompts available in this library and including their structures, topics and constraints. You have to also analyze how prompts are categorized and organized within the library to quickly locate relevant prompts for your needs.

Third is adapt to prompts to your needs. Customize existing prompts to suit your specific objectives, audience and use cases. You can modify prompts by adjusting the action verbs, topics, constraints or background information which aligns with your requirement. Create your own prompts like combine different components such as action verbs, topics, constraints to craft prompts that addresses specific tasks or challenges.

Next process you have to do is sharing and collaborating. You will share your prompts with the community to contribute to the collective pool and resources. So this is one way of learning that I really really want you to follow. Now you have to keep experimenting and iterating at the same time. And finally you have to see the documents and organize all your prompts for the scene.

So what you can do best is see all the existing prompt libraries like I'll show you one more. So prompt library for chat GPD GitHub for all use cases. So you could see explore various repositories on GitHub like what are the uh kind of like prompts available like this repo specifically focuses for the academic writing.

So just visit this uh repository and uh you could see they have given a lot of thing like for brainstorming they say so you can see the action verbs all over here try to like uh uh try this prompt and see how you are getting a response. Then for article sections like what's

it's there. So you're going to get a lot of like things and uh more of the experiment and more uh you are exploring the more idea you are going to get regarding this.

So my advice would be just explore as much as libraries you can and depending upon your use cases you have to make an organized prompt structure. So following this format which I've told you follows the action verb the topic or the background information. Then what are the constraints you have to give? Okay, it's any particular theme is there.

You have to include all those things and use the existing prompt library also. So you can refine your uh prompt and always to get a good response. It's my personal experience that you have to keep fine-tuning, keep testing, iterating, analyzing so that your result comes very fine. >> Hello guys, welcome to this reasoning with 01 course.

What if we told you that there's an AI that does not just give answers but actually thinks? So introducing 01, the latest and the most advanced reasoning model from Open AI. So unlike other models that rely on patterns, 01 is designed to reason through problems, self-correct and explain its thinking step by step just like a human would.

So today we will explore what makes O1 so special. Then we will come across how it reasons. Then we will see where it's useful. And finally we will see why learning about this model is essential in the age of intelligent machines. So let's just dive in. So what is 01? So 01 is open AAI's new AI model.

But it's not just a smarter chatbot. It's a reasoning engine. While models like GP4 are brilliant at language and creativity, O1 gives a step further. It can think logically. It can solve problems with multi-step reasoning and analyze the complex data. In short, O1 is the closest we have come to building an AI that can think through our task, not just respond.

So why does this matter? Because the real world problems in maths, science, code and life require structured thinking, not just quick answers. So let's now move on to what is reasoning and why needs it. So before we understand how O1 works, let's ask what is reasoning. So reasoning is a process of analyzing information, finding connections and drawing conclusions like solving a puzzle.

For example, you see dark clouds, people are carrying umbrellas. So even if it's not raining, you reason it as it's about to rain. So AI needs this same skill. So without reasoning, an AI is like a parrot. It repeats. With reasoning, it becomes a problem solver. And that's exactly what O1 brings to the table.

So let's now understand how 01 reasons. So let's break down how O1 actually reasons step by step. The first one is chain of thought reasoning. So O1 solves problem step by step just like we do on a paper. For example, the question is what is 12 into 15? O1 does not just say 180. It thinks it as 10 into 15 is equal to 115.

Then it calculates 2 into 15 is equal to 30. Then it add final two results that is 150 + 30 is equal to 180. So these chain of thought method reduces errors and shows the logic behind every answer. So let's now move on to the next that is reinforcement learning. So Oman gets better over time by learning from success and failure just like a person.

If one method doesn't work, it tries another, sees the result, and improves next time. It's like a chess player who gets sharper with every game. Then we'll come to reasoning tokens. So when solves a complex task, it uses invisible tokens to keep track of its thoughts like a mental sticky notes.

So imagine solving a math problem. You scribble your steps on a paper. 01 does the same internally helping it to stay organized and accurate. So let's now dive into the real life examples. So let's see 01 in action. The first one is solving a math problem. So the question is what is 25 + 37? So 01 thinks like 20 + 30 is equal to 50 and 5 + 7 is equal to 12.

50 + 12 is equal to 62. And the final answer it gives is 62. So it's not magic, it's methodical reasoning. That's what makes 01 more trustworthy and transparent. Then we are going to the next example that is debugging a code. So now let's shift to a coding scenario. If you ask why is my Python code giving me an error. So here's what 01 does.

It reads the code line by line, spots where the error happens, then explains why it occurred, and then it suggests a fix step by step. It's like having a calm, logical coding partner who always explains their thinking. So let's now move on to the next one. We have what makes 01 different. So here is a quick comparison. Say we have a capability.

So GPT4 predicts the next word and same O1 also does. So GPT4 thinks through steps. It doesn't think through steps but 01 does. So GPD4 selfcorrection is a basic thing but the selfcorrection in 01 is a very advanced thing. Then it solves a complex problems. GPT4 sometimes does it but 01 is consistently doing it.

Then transparent reasoning. GPT4 has no transparent reasoning but 01 has a transparent reasoning. So let's move on to where can 01 be used. So because of its structured reasoning, 01 is useful in multiple domains. The first one is maths and science, solving equations, explaining experiments and so on.

And then we have coding. So writing, debugging and understanding programs. Then we have research that is analyzing data, summarizing papers and verifying facts. And then we have creative work. So planning content, brainstorming logically are all falling under creative work. Then we have image analysis.

This interpreting visuals step by step. So wherever thinking is needed, 01 delivers clarity. So let's now come to the common mistakes avoided by 01. So previously older models had a few common issues such as the jumping to conclusions directly then ignoring small details and then we have repeating errors in the new context.

But O1 avoids all this by reasoning in steps checking its own logic and learning from feedback. And that's why it's more dependable especially when stakes are high. Let's now move to why O1 is so important. What is the future of work and thinking? So because reasoning AI is the future and it's already here.

Learning OI works helps you use AI more effectively. Ask better questions obviously then understand how machines think and then stay ahead in the technical research and educational field and become a critical thinker yourself. So this is not just about AI. It's about preparing for a world where thinking with AI is a part of how we work, learn and create.

So let's now move on with the key takeaways. So let's recap what we have learned. Owen doesn't just answer, it reasons. Then it learns from mistakes and improves. It explains its logic step by step. Then it's accurate, explainable and helpful across industries. then it's future of intelligent systems and you can be a part of it.

So whether you are a student, developer, analyst or just curios is just a tool that helps you think better. So let's dive into deeper in the next module. Hello guys, welcome back to the second module. So in this module we will learn about prompting in 01 models. So prompting is a way we interact with the AI by giving instructions and receiving responses.

So with 01 models like GPT40 things have gotten simpler. So these models understand prompts with more accuracy which means you can be clear, concise and direct without overexlaining. So there are certain principles of prompting in 01 models. The first one is it is simple and direct. Then number two is no explicit chain of thought needed.

Then the third one is structure over description and the fourth one is show don't tell. So there is a summary for this like to get the best out of the O1 models remember the four

principles. You have to be simple and direct. No need for explicit reasoning. It's already built in. So give structure not long descriptions and show tone with examples.

Don't just tell. So these principles will help you get better faster results from 01 models while keeping things smooth and efficient. So let's then get into the demo part. So here we'll write our first code as. So the first thing import warnings warnings dot filter warning ignore.

So this is a basically a warning control. So this is Python code snippet does two main things. It suppresses warnings and retrieves the open AI API key for use in the later interaction. So I will just break it down step by step. So the first line import warnings imports Python's built-in warning module.

So this module is used to manage the warning messages in the program which are typically notifications about the for the potential problematic or unexpected events in your code. It helps developers keep track of the minor issues without even interrupting the program's execution. So next warnings dofilter ignore is used to tell Python to ignore any warning messages that might happen.

So normally warnings are shown to alert the developer about something. So that may be critical issue but could be important. So then the third line from helper import get open AI API key imports a function called get open AI API key from a module named helper. So this function is responsible for retrieving the open AI API key which is necessary for interacting with the open AI services such as generating text or using other machine learning models.

So finally the open AI API key is equal to get open AAI API key calls the imported function to get the API key and stores it in the variable open AI API key. So these variables know how the key needed to access the OpenAI's features and it can be used for even making the authenticated API request.

So this code is designed basically to suppress any unnecessary warnings while ensuring that the program has the power of the API key. So let's just run this. So then basically we have imported JSON. So the firstly import JSON allows working with the JSON data obviously commonly used in the API interactions from ipython dot display import display then markdown HTML imports functions to display etc.

So then from open AAI import open AAI imports the open AI class so which is used to interact with the open AI services. Then client equal to open AI API key equal to open AI API key creates an open AI client using the API key enabling communication with the open AI's models.

Then finally we have GPT model equal to GPT 40 mini and 01 models equal to 01 mini defines the name of the two different models that can be used in the API requests. So in short this code sets up the OpenAI client and defines the two models for use in natural language processing task. So we'll run this.

So here basically the code snippet defines a prompt for generating a function using the open AI's model then sends the prompt to the openi API to get a response. So let's now break it down step by step. So bad prompt equal to generate a function till insulin creates a string variable called the bad prompt because it is a statement basically.

So this prompt is a request for the model to generate a function that outputs the smiles ids a way to represent the molecular structure basically for all molecules involved in the insulin. It also guides the model to think step by step outlining task like identifying the models creating the function and looping through each molecule to return the smile ids.

So then we have response equal to client.chartcomp completions.create model. So this basically sends the prompt to the open AI API for processing. It uses the client.comp completions method to interact with the open AI's API specifying the model that is 01

model which was defined earlier. The messages parameter sends the bad prompt to the model as input under the user role.

So in short this code defines a prompt using the open AI's model to create a function for returning the smiles ids for insulin molecules and sends that prompt to the openi API to generate a response. So let's now move on run this and move on. So then basically we have display HTML division style equal to background color.

So basically displays the formatted HTML and markdown content in a Jupyter notebook. So then we have the first version, the first one, first line displays an HTML division with a light background color and a border also. So it also includes a heading that says markdown output beginning with a downward arrow emoji.

Basically the HR tags are used to insert the horizontal lines before and after the heading. Then we have display HTML function is used to render this HTML in the notebook. Then we have display markdown. This line takes the content of the response from open model stored in this and it displays it as markdown.

The markdown function is used to render the content in a more readable formatted way. So this content could include text, headers, list or any markdown elements generated by the model. So then we can run this and you can basically see the heading. Then the step one is identify all the molecules involved in insulin chain A, chain B.

Then step two, prepare the smiles and strings for each amino acid. Then we have create the function This is basically the output. So then we have this good prompt generate a function that outputs a smile ids. As we have given a bad prompt, it is necessary to give a good prompt as well. So yeah, we have given a good prompt.

So let's now display that good prompt. Mark out beginning understand insulin step by step. And you can see the difference clearly. So this is basically a great code. But don't get overwhelmed. I'll break it down step by step. So this example is an structured prompt basically and used to guide an AI assistant to behave like a customer service representative for a company named any corp.

So which sells storage solutions such as beads. The purpose of this prompt is to make sure that the AI behaves professionally, respectfully, and in line with the company policies while responding to the customer queries. The first part of the prompt labeled as instructions clearly tells the assistant what role it is playing a customer service assistant and emphasize the tone it should maintain kind and respectful at all times.

So this ensures assistant responds in a friendly and helpful manner just like a real support agent would. So next comes the policy section which outlines the six specific areas that the assistant must follow. So these include handling refunds, recording complaints, providing accurate product information, maintaining the professional conduct, complying with the policies and data privacy rules, and refusing to answer questions outside these topics.

For example, under refunds, the assistant is allowed to process them as long as it follows any corpse rules and documents everything properly. If a customer complains, the assistant should listen carefully, reassure them and escalate the issue if necessary. So if asked about topics unrelated to these areas, the assistant must politely refuse to answer and guide the user back to the allowed topics.

The final part is a user query which shows the customer's actual message like hey I would like to return the bin I bought from you as it was not finded as prescribed. So these falls under the refunds category in the policy. Since the assistant is allowed to handle the refund request, it will process this message accordingly possibly by confirming the issue, starting the return process and recording the complaint.

So in short, this structured prompt creates a safe and clear framework for the assistant to operate. It tells the AI what kind of support it can offer, how to behave, what topics are allowed, and how to respond when faced with customer issues. All while making sure the customer feels heard, respected, and healed. So let's now print the statement the structure prompt.

Now you can see that is printed. So now this code is from the Python program using the open AI API which allows developers to interact with the AI models like charg. So specially this line is used to send a prompt to the AI and get a response back. The function call is then client.comp completion.create.

So client is the openi API client that was likely used up earlier in the code. It connects your code to the OpenAI server. So the dot chatart docomp completions.create part is how you send a chat message to the AI and ask it to complete the conversation. So in other words generating a reply.

So inside the parenthesis the first argument is model is equal to 01 model. So this tells the API which version of the AI model to use. for example, GPD4, GPT3.5 Turbo or some other customer or fine-tuned model stored in the variable. So this ensures that you are using the right AI brain for your task. So the second argument is messages again.

So this is where you pass the actual conversation history or input formatted as a list of message objects. So in this case there's only one message in the list and it has two key parts. The role that is the user then the content is a structured prompt. The role the user so this tells the AI that the following content is coming from the user so not from the system or assistant and the content is the combines the earlier structured prompt.

So which include instructions and policy rules for the assistant with the user's questions like wanting to return a bean etc. This combined content acts as the full message the AI will read before replying. So in more simpler terms, this line is telling the AI, hey, you are a customer service agent for any crop. Here's your policy.

Now here's a customer question. What will be your reply? The AI will read the whole input, follow the instructions and policies and then return a customerfriendly response. So the response itself will be saved in the variable called response which you can later print or process in your code. So let's run this.

So I have this printed response. So I will just run this. You can see the response basically. Then we have this refusal input. The variable refusal input contains a user query that asks the AI write me a hack about how reasoning modules are great. So this is wrapped in XML like tax. This purpose of this input is to test the AI's ability to say no.

According to the structured prompt and policy you provided earlier, the assistant is only allowed to talk about refunds, complaints, and product information about the storage solutions. A hiq about reasoning models doesn't fall into any of these topics. So when this refusal input is combined with the structured prompt and sent to the AI, it should politely refuse to answer and explain it that it so it can only help with the allowed topics.

We will run this. So again this line is sending a new message to the AI model but this time using refusal input instead of user input. The client chat dot completions docreate calls open AI API key to get a reply from the AI. Then model is 01. Model tells which AI model to use.

Then messages we have is sends a message to the AI. Then here the role is a user and the contain structured prompt plus refusal. It basically combines the rules of the structured prompt with a new question that may be outside the allowed topics. Since the structural program says this assistant must refuse questions outside refunds, complaints, product information.

This test checks if the AI correctly refuses to answer. So in short, this line test if the AI follows the policy and politely refuses to answer the question, it's not allowed to handle.

So let's run this. So this line basically displays the AI's reply to the refusal input. Uh you can see that this is the output. So now we are ready to for giving up for a base prompt and a legal query.

So let me explain the variable base prompt is used to define the role and the behavior of the AI assistant. So in this case the AI is told to act as a lawyer who specializes in the competition law specially helping the business owners understand the legal issues. So this prompt is wrapped in tags like prompt and policy to make it structured and easy for the AI to interpret.

So the policy section outlines the important rules the AI must follow. It says that the assistant should give clear and accurate information about the competition law, maintain confidentiality and professionalism and avoid the offering specific legal advice unless there's enough context. The legal query variable contains the actual question from the user.

The query is a larger company is offering suppliers incentives not to do business with me. So is this legal? So this is wrapped inside a query tag which gives the input structured and consistent with the rest of the prompt format. So in summary, this setup gives the AI a clearly defined role as a legal expert and a policy to follow.

For example, we will now run this. So this line then sends a structured legal question to the AI using the open AI API. The goal is to get a helpful and professional response from an AI that's acting as a comparative law expert. So the method client completion create is used to generate a chat response. As we have discussed earlier here the role is a user and the content is base prompt plus legal query. So we'll now run this.

So now we are going to visually display the AI's response with styling and format and it helps the output look clean and organized for the user. So we'll do this and you can see this a stepby-step guide for the so this is also again a uh like big query. So here the

variable example prompt is a structured example prompt that shows how an AI assistant acting as a competition law explored should respond to a user's legal interest.

So it's formatted using special tags like prompt, policy, example, question and response to keep everything organized and machine readable. So inside the prompt section, the AI is structured to behave like a lawyer who specializes in the competition law, helping business owners with their legal questions. So this gives the AI its role and sets the stage for how it should respond.

The policy section defines the boundaries the AI must follow. Then we have the example section provides a sample legal scenario and how the AI should answer. So then we have the response that follows is a well structured professional reply. It begins by referencing the US antitrust laws especially the Sherman antitrust act and explains how some types of collaboration like price fixing or market division are automatically illegal.

So then we have it also once the information sharing which can be risky and potentially lead to antitrust violations. In summary, this example prompt acts as a guide or training example to show how the AI should behave in legal conversations. So then I'll run this. So here then we are going to again display the output in a formatted style and you can see this displayed.

So let's now move on to the next module and learn about planning with 01. So in the earlier lessons we explored how to structure prompts and guide the AI models like 01 and 01 mini to behave in specific roles. So whether a customer service agent or a legal advisor. So here in this lesson we will learn how to plan with 01.

Uh it will introduce a powerful use case where the AI is not just answering a single question but actually creating a full plan to solve a task. So this kind of problem solving is perfect for 01 models especially when you are working with a fixed set of tools and some constraints. So instead of asking the model to do every step directly which could be slow

and expensive, we let 01 mini generate the plan and then use a smaller faster model like GPD4 mini to execute its step.

So this two approach planning with 01 mini and executing with GP4 mini helps us work efficiently. So please don't worry if your results don't exactly match the ones in the video tutorial. Now that's totally normal since AI responses can vary slightly every time you run them. So in short this lesson builds on what you have already learned and now takes it a step further like teaching the AI how to think ahead and plan not just react. So let's dive in.

So we will start with the first part of the code. This is about suppressing the warning messages as we have discussed earlier in Python. So Python sometimes shows warnings when it deflects something that might not cause an error right way but could lead to issues in the future. So this is for this and the second part is focused on importing your OpenAI API key securely.

So instead of hard coding the key directly into the notebook which is not safe the code calls a helper function from the helper import get open AI API key brings in a function named as get open API key. So we will run this first two lines import the two standard Python modules copy and JSON.

The copy module is used when you want to duplicate the Python objects, especially when you need to make changes without affecting the original. The JSON module is used for working with the JSON data, which is common format for sending the receiving structured information, especially when dealing with the APIs. The next line imports the OpenAI class from the OpenAI package.

This is what allows your code to communicate with the open language models like GPT4 or 01 mini. Then we import the 01 tools from a separate life named utils. py. So this likely contains the some common utility functions or tools. So then next we set up the connection to the open API by creating a client instance.

So client equal to open AI API. Here the open AI uh API key variable which was retrieved earlier using the get open AI API key is passed to authenticate your session with the open AI. So this line sets up the client object which you will use to send prompts and receive responses from the models.

So finally two variables are defined to specify which models to use. So these are simply model name shortcuts. So instead of writing the full model name each time you can just refer to 01 model and GP model. So 01 mini is used for planning the task while GP4 mini is likely used for executing the individual steps in the plan.

So we will run this. So here the code first line initializes an empty list called message list. So this list will likely be used to store the sequence of messages exchanged with the AI model. Next, we define a dictionary name known as the context, which sets up the initial state of a business environment for a planning task.

Under the inventory section, we are told that 50 units of the smart home hub x200 are currently in stock. Then under the orders we see a single order that is 0 RD3001 requesting 200 units of this product for a customer that is having the ID cd 901 located in the loss Angeles. The available suppliers and the supplier section list two suppliers and the components they can provide.

Notably only SUP101 has a component of COM X200 required to make the X200 product and they have 500 units available. The product capacity then section defines how many units can be produced. 100 right away and another 150 next week. This helps the model decide how to meet the demand of 200 units.

The shipping options provide two choices for delivering to Los Angeles. A standard service that cost $1,000 and takes five days. An express service that cost dollar 1,500 and takes only two days. The model can consider this based on the urgency or budget.

The customer's dictionary holds customer details like the name and the address of the client placing the order.

The product section defines what components are needed to manufacture each product. So in this case, the X200 device requires one unit of com X200 to build each finished product. The code sets up a realistic business scenario basically involving the product, inventory, orders, manufacturing, suppliers, and shipping. All the key elements the AI needs to plan how to fulfill the order effectively. So we'll run this.

So this is basically a prompt for planning the task. So this block defines a large string called the 01 prompt which acts as an instructional prompt for an advanced language model like 01 maybe. The goal of this prompt is to guide the model to act as a supply chain management assistant and generate a detailed logical plan for handling a complex order fulfillment task.

It's essentially the brain that will break down a business challenge and outline the best steps to solve it. The first paragraph in the prompt sets the context. It tells the AI that it will be given a complex task that involves managing customers inventory, production supplies, and shipping.

The next section list all the functions available to the exeutor AI. The functions allow the exeutor to do things like check inventory levels, allocate stocks to the orders, place purchase orders with suppliers, schedule production runs, and so on. Each function is briefly described along with its required inputs and expected outcomes.

So this ensures the planning AI knows what tools are available, how they work, and what each one is for. It helps the model build a plan that is realistic and executable by the downstream agent. So in summary, this 01 prompt sets up a rich and highly structured instruction environment for the planning model.

It tells the AI to think carefully, reason step by step and create a clear, detailed and logical order plan. One that another AI can confidently execute using the provided tools. It ensures consistency, safety, and clarity across every step of the supply chain process. So let's run this. So this is basically the prompt for the execution process by the system.

So this block defines a variable called GPT4 system prompt which is a string used to guide the behavior of the execution model especially the GPT4 meaning in this case while the O1 mini model is responsible for planning. The prompt is for the model that carries out each step of the plan. The assistant is told us to act as a helpful executor whose job is to strictly follow a given policy related to handling incoming orders.

The prompt defines a step-by-step workflow for the model to follow. First, the model should read and understand the full policy. Then, it needs to identify which step in the policy is currently working on. So, before acting, it should give a brief explanation of the decision.

So, finally, it should perform the action which typically involves calling one of the available functions with the correct inputs. So in summary, the GPT41 system prompt sets up the mindset and the behavior of the GPT4 mini executor. It ensures the model works step by step, explains it reasoning and only takes action that match the policy creating a controlled and transparent execution flow for order handling in a supply chain scenario.

So let's now this this code block defines a list named tools which contains a collection of the functions that AI can use to interact with the supply chain system. So these functions act like a toolbox and each tool or function performs a specific task related to the inventory production supplers orders etc.

So each tool is defined as dictionary that includes its type that is a function and a function section that deals the function's name, description of what it does and the

input parameters it expects. Some key functions include get inventory status to check how much a product is in stock. Then update inventory to add or subtract stock.

Then fetch new orders to see which orders are pending. Then get product details and check available suppliers to explore what's needed to make products or find suppliers. Then place purchase order to order components. Allocate stock to reserve product for a customer order. Check production capacity and schedule production to plan and carry out manufacturing.

Then calculate shipping options and book shipment to handle delivery. Send order update to notify the customer. instructions complete to mark that all steps in the plan are finished. So each function also defines the parameter it requires like product ID, order ID and quantity etc. So this is important because AI must pass exactly the right data when calling these tools.

The use of required fields and the additional property. So in summary, the tools list defines a complete set of functions the AI agent can use to operate the supply chain system. So let's now move on to the functions then. So this block defines all the Python functions that simulate the supply chain operations described earlier.

The first few functions deal with the inventory management. Get inventory status. Then get product details. This features the product information. Then get inventory status returns how many units of a specific product are currently in stock. Then update inventory product ID quantity change. Adjust the inventory count based on a positive or negative change making sure inventory doesn't go below zero.

Then allocate stock tries to reserve the inventory for an order. If there's enough stock, it reduces inventory and confirms the allocation. If not, it allocates whatever is available and marks the result with an insufficient stock warning. The supplier related functions start with the check available suppliers which returns a list of the suppliers the company can order components from.

Then comes the production related logic. Check production capacity. Time frame checks how many units can be produced with the given time frame. Then schedule production run checks the capacity schedules program if possible and updates the inventory. For shipping, the function calculate shipping options returns a list of available shipping methods and their cost for the specified destination.

Lastly, the function send order update simulates sending a message to the customer to keep them updated on their order status. So, at the very end, there's a dictionary called function mapping which links each function name to the actual Python function. So, let's then run this. So guys, knit together the whole process and make the most out of it.

Let's now move on to the next module. Hello guys, welcome back to this new module of coding with one. So initially we will start with the first part that hides the warning messages using the warnings do filter warnings ignore as we were using to. So this keeps the output clean by preventing the non-critical warnings from appearing during execution.

Next, the code imports get open API key function from the helper file. Finally, the function is called and its result is stored in the open AI API key variable. So in short, this setup hides warnings and securely loads your API key for using OpenAI tools. So we'll run this. So here then the first line imports the three useful display tools from ipython dot display image and markdown.

These are commonly used in Jupyter notebooks to visually show images format output using markdown or display objects in a cleaner way. Next the code imports the openi class from the openi library. So this class is used to create a client that communicates with the OpenAI's API to generate responses, run models or perform other AI tasks.

So then the client client is equal to OpenAI API key. Open AAI API key initializes the OpenAI client using the API key we previously stored. This connects your code to the OpenAI services. So you can start making the API calls. So finally two variables GPD model and 01 model are defined with model names GPD4 mini and 01 mini.

These are likely short and labels are used in your code to specify which AI model you want to work with when making the API request. So we'll run this. So this code defines a function called get chat completion which is used to get a response from an openi chatbot model. It takes two inputs the model the name of the model you want to see like GPT4 mini and prompt the user's message or question.

So inside the function it calls client.art or completion.create. So this is the open AI API method used to send the prompt to the chosen model and generate a chatbased response. So the message is passed in the format open AI expects an array of message objects with the roles like user and assistant. Finally the function returns just the content of the model's reply using the response dot choices dot message.contain.

So this keeps the output clean and gives you only the AI's reply text, not the entire API response structure. So we'll run this. Now here the code defines a multi-line string variable called the React demo prompt. So it's a prompt written in the natural language that will be sent to an AI model like chart GBT to generate a react component.

The prompt is asking the AI to create a React component named feedback form for collecting structured interview feedback. It outlines four main requirements. Reading candidates with rubrics, giving specific evidence for each rating or to calculating a recommendation score and guiding the interviewer to provide the behavioral feedback.

It also includes additional instructions for the AI like using the intelligent UX features like thoughtful rubrics, smart validation and helpful prompts. Finally, the prompt includes clear coding instructions. The component must begin with the use client. In short, this

prompt is designed to generate a polished intelligent interview feedback from using React.

So, we'll run this. So this line calls the get chart completion function which was designed earlier to get response from the open AI model. It passes two arguments GPD model which specifies the AI model to use like GPT4 mini and react demo prompt which is the details instruction asking the model to generate a react component.

The function sends this prompt to the model and returns the AI's response which will be block of code as requested. The result is then stored in the variable called GPT code which now holds the React component code generated by the AI. So we'll run this. So let's now simply print the code and you can see the code is printed.

So this line uses the display function from the ipython.d display to show an image inside a Jupyter notebook or similar environment. So image GPT4 app you can see this image GP4 app image loads an image file named GPT4 app image dotpng from the current directory. together. This command displays the image in the notebook output which can be useful for showing the visual results, UI screenshots or diagrams alongside your code. So let's now move on.

So it now sends the same React demo prompt to a different model called the 01 model which is likely a lighter or experimental version like the 01 mini. It uses the get chat completion function to generate the code by calling the OpenAI API. Now 01 holds the output generated by the O1 model for the same prompt allowing you to compare the results across models and you can just print the one code and you can see now it is printed and you can now display this.

Now this is nice and with this the function process orders text in a list of orders and optional parameters settings debug and notify customer. So it processes each order by validating its fields calculating totals applying discounts and queuing the customer

notification. It collects the results errors and notifications in a separate list and returns them in a dictionary.

So however these are the few issues in the code. First using a mutable default argument settings is risky because it can lead to unexpected behavior. It's better to use none and assign an empty dictionary inside the function. Also comparisons like if debug is equal to true should be simplified to if debug.

Similarly if items valid is equal to true can be just if items valid. So ensure the function logic works for basic order processing but should be cleaned up for the Python best practices performance and maintainability. So let's now run this. So basically it creates a prompt string called prompt using the f string.

So it tells the AI to clean up and improve the Python code stored in the code snippet. So the prompt specially asks the AI to return only the updated code. Next the get chart completion sends this prompt to the selected GPT model using previously defined function. So this functions calls the openi API and gets a revised version of the code as a response.

Finally we are printing the GP code displays a new version of the code so you can review or use it. Lastly, this line sends the same prompt to a different model, the 01 model using the get chat completion function. So, this prompt asks the model to clean up and improve the code stored in the code snippet.

The function processes the request and stores the model's response in the 01 code variable. So, this response should contain the optimized version of the original code generated by the 01 model. The next line we are having print 01 code displays the improved code in the output allowing you to see the and compare the 01 models version with GPD models output.

This is useful for evaluating the differences in the code quality clarity and optimization between the models. Lastly, the result is stored in the result variable and then displayed using the display markdown result which renders the output in the nicely formatted markdown great for readability in Jupyter notebooks.

In short, this step helps you to use the AI not just for generating code but also for doing a quick sideby-side code review and analysis as well. So let's now move on to the last module. Hello guys, welcome back to this last module. So let's now get into reasoning with images.

We will start with importing the warnings and this is a built-in warnings module which is used to manage the warning messages during code execution. The second line uses the warnings or filter warnings to suppress all the warnings. This means any non-critical alerts will be hidden from the output. This is commonly done to keep the output clean especially in the notebooks or scripts where you don't want warning messages to clutter your results.

So we'll run this. So then here the first line imports the JSON module which is used to work with JSON data. This is helpful for paring formatting or storing structured information. The second line imports the OpenAI class from the OpenAI package. This allows you to connect to the open AAIS API and make requests to generate the text code or other outputs from the AI models.

Then the line from IPython display import display markdown image imports tools to visually display the markdown text images or formatted output commonly used in the Jupiter notebook. So after that the code imports the get open AAI API key function from a custom file named helper. This function retrieves your OpenAI API key securely.

So it doesn't need to be hardcoded. The function get open AAI API key is called and the return key is stored in the OpenAI API key variable. This key is needed to authenticate

with the OpenAI API. The lines defining the GPT model, GPT 40 mini and the 01 model that is 01 specify the names of the AI models you want to use for generating responses.

These can later be passed when making the API calls. Finally, client is equal to OpenAI creates an OpenAI client instance which you will use to send prompts and receive completions. So this should ideally be open AI API key. include the authentication unless your environment handles the key automatically.

So we'll now run this. So this means the image is expected to be located inside a folder named data in your project directory. The second line uses display image image file path to load and show the image in Jupyter notebook or similar environment. It visually renders the organization chart image.

So you can see directly in your notebook output. So this is commonly used when working with the visual data or presenting diagrams, charts or anything like that. So let's now move on to the next one. The code starts by importing the B 64 module which is used to encode the binary data like image files into a B 64 string format.

This is necessary because APIs like OpenAI's vision models accept images as base 64 encoded strings. The function encoded image that is image path takes an image file path reads it in binary mode that is RP encodes it in using B 64 and returns it as UTF8 string. This prepares the image so it can be safely passed to an AI model in the required format.

The main function 01 vision is designed to send both an image and a text prompt to an open AI vision and a built model like GPT4. It first encodes the image using the encode image function. So inside the function, it checks whether JSON model is true or false. If JSON mode is true, the response will be requested in the structured JSON format using response format type and JSON object which is useful for extracting the structure data from the model's reply.

So if JSON mode is false, it simply sends the prompt and image together and expects a regular text response from the model. Both the versions use client.chart.comp completions.create method to make the API call. Finally, the function returns the draw response object from the API allowing the caller to further process or display the model's output.

In short, this functions lets you send an image along with the custom prompt to a vision capable open AI model with an option to get the result either as plain text or structure test. So this line calls the 01 vision function which sends both an image and an prompt to an open AI vision model for analysis.

So it passes three arguments file path that is image file path to the image file prompt equal to what is this? This is asking the model to describe the image and the model is 01 model which specifies the vision capable model to use like 01. The function processes the image and the prompt, sends them to the model and then stores the model's response in the variable response.

So this response will contain the AI's interpretation of the image which you can later extract and display. So this line is used to display the AI models response in a clean and readable format inside a Jupyter notebook. So this response dot choices zero dot message.contained contained accesses the actual text contained as generated by the model in the first response choice.

So wrapping it with a markdown allows the output to be rendered with the proper formatting such as bold text, bullet points or headings if the response includes markdown syntax. Finally, display renders the the format markdown content in the notebook output area, making the model's reply easy to read and visually structured.

So let's now start with understanding the image. The code defines a variable called the structured prompt which is a string that will be sent as a prompt to the AI model, most

likely a vision enabled one to extract the structured data from an image. The prompt includes special instructions tags to clearly tell the model what to do.

It says that the model is acting as a consulting assistant where job is to process organization chart data from the image. If then specifies the desired output format, a JSON structure that includes an arbitrary ID for each person, the name, role, and array of ids they report to and an array of ids that report to them.

So this is a clean and structured way to actually represent an organization hierarchy. So finally we are printing the structured prompt says the full prompt string. So you can review or verify its contents before using it as a function like the open version. So this code calls the O1 vision function to send both an image and a structured prompt to the open AI vision model.

This is O1 asking it to extract the organizational hierarchy data. The arguments pass include file path equal to image file path which points to the organization chart image model is equal to 01 specifying the model to use. Then we have prompt is equal to structured prompt which contains a detailed instructions on the output format and the JSON mode is equal to true.

Setting JSON mode equal to true tells the model to return the result in a structured JSON format. So instead of plain text making it easier to purse and use programmatically. The final line actually prints the model's JSON formatted response from the first message choice which should include the organization structure with ids, names, roles and reporting relationships.

You can see the output. So this line creates a new variable called the clean JSON that stores a cleaned up version of the model's response. Sometimes when the models return the JSON code, they wrap it in markdown formatting using the triple back text like JSON and to indicate a code block.

So this line removes those wrappers using dotreplace and dotreplace with inverted commas leaving behind only the raw JSON content. The result is a plain JSON string that can now be pursed or processed further without formatting the issues. So this line attempts to purse the model's response directly using the JSON loads which converts a JSON formatted string into a Python dictionary order list.

So it accesses the 01 response. Choices dossage.contain which is expected to contain a JSON output describing the organization chart structure. The result is stored in the variable that is the organization data making it possible to work with organization hierarchy as a structured Python object.

However, the JSON file still contains the markdown wrappers like JSON. You should use previously clean version instead for space person. So the code defines a new prompt string called the analysis prompt which is meant to guide an AI model to answer questions based on a previously extracted organization chart. It starts with the special instructions like you are an organization chart expert assistant.

Your role is to answer any organization chart questions with your organization data and we have given the instruction. So this tells the AI what role it plays and how to use the data. then embase the structured organization chart data inside organization data tags by using an F string. This helps the model clearly understand that this is the data it should reference when generating answers.

So let's now run this here in the code. The first line is importing the get open API API key. This function is typically used to securely retrieve your open API key without hard coding it in the script. The second line calls this function and assigns the result that is your API key to the variable open API API key so it can be reused throughout your code for authentication.

Finally, the client is equal to OpenAI creates an instance of the OpenAI client using that key. This allows you to send the prompts and interact with the OpenAI's models through authorized API calls. So as discussed before, these are the messages and the role user content analysis report plus who has the highest ranking reports and which manager has the most reports and we will just run this.

So then we are finally displaying the markdown allowing my markdown formatting like bold text, bullet points or headers to be properly displayed rendered and the first line sets the variable image file path to the path of an image file named as year relation order.png which is located inside a folder called data.

So this image likely contains an entity relationship diagram. The second line uses the display image to load and visually display the image with the Jupyter notebook. So we'll just run this and basically you can now see the output the product ID this is the order product this is the ID the client C uh it also has a ID C ID product name client address product price and client credit card.

So basically this was all about the course that's it. So congratulations on completing your journey with reasoning with 01. You have gone from understanding the core foundations what reasoning with 01 is why it's crucial in today's AIdriven world and where it can be applied to actually putting it into practice.

We've began by setting the stage with the fundamentals, then moved on to hands-on exploration, crafting effective prompts, practicing logical and structured reasoning, then writing and refining code with O1, and finally understanding how each layer builds towards the AI interaction. You didn't just learn the theory, you experienced it through demos, challenges, and the real world examples.

So whether it was decoding logic, guiding AI through structured thought or debugging your own process, you have now got a strong grip on how to think with AI, not just about

it. This is just the beginning, guys. Keep experimenting, keep questioning. The more you reason, the smarter your results become. Thank you for watching.

Welcome to our comprehensive tutorial on creating a fully functional e-commerce application using React Tailwind CSS and Redux with the help of Chad GPT. In this tutorial, we are going to leverage the powerful capabilities of Chad GPT to assist us in building modern, responsive and featurerich e-commerce application.

You will learn how to set up your development environment. We will start by setting up the essential tools and libraries required for our project including React, Tailwind CSS, Redux and React Router. We are also going to design the application layout using Tailwind CSS. We will create a visually appealing userfriendly layout that includes the header, navigation bar, product listings, shopping cart and checkout pages.

We are also going to implement the state management with Redux. We will use Redux for managing global state, ensuring our application is scalable and maintainable. You will learn how to set up slices for handling user authentication, product data, and the shopping cart. We will also create reusable components.

And throughout this tutorial, we will build a variety of reusable components such as product cards, form inputs, buttons, and models to enhance the modularity and reusability of our code. for fetching and managing data. We are going to use React query so that we are going to fetch data from a backend API and manage the server state efficiently.

We will also learn to handle loading states, caching and synchronization with React query. With that said guys, watch this video till the end if you want to learn how to make an e-commerce application using chat GPD. So guys, this is a website which we are going to create with the help of Chad GPD.

So you can see all over here we have the navigation bar which includes home about products cart. Okay. So suppose I want to add this product. Okay. This is a lamp and say I want to add in this quantity say three. And if I click it up you can see there's a notification which is coming up and it's saying item has been added to the cart.

Now you can also go and check the cart. But before uh proceeding for checkout you need to also login. So suppose I have not created any account. So let us create an account. So say this is a mail id and say let us take some demo mail id and here is some password and you can say email is already taken. So let us give some random mail suppose name at the rate 1 2 3 and say the password is 1 2 3 4. Okay.

Now let's login. So it's saying email must be a valid email. So I have also put like we can say checkpoints where if the proper validation of the input field is also given. So name add the rate name 1 2 3 at the rateg gmail.com. Now I hope so it is going to work. So email or username has already been taken. So let us change our username.

So username name 1 2 3 and let us try to register. So you can see all the input validation has been done. So let us give some big. Okay. Now you can see it has completely logged in. Okay. So you can save the password. Now let us go to our repeat. So you can say I'm trying to login all over here and it's been logged in.

So now you can see if you click on our products you are going to see the products part like there's a chair there is a glass table king bed size and all these are the cards of the product. So basically these images have been taken from a link. Okay, I will mention you the link and you can do it and inside this you can also search the given product.

Suppose say I want I want a grade lamp or say chick chair. Okay, if I type this and uh if I try to search it you can see this is coming up. You can also select the company suppose if I just see all over here. So the company name is Luxora. Okay. So you can type all over there in the check box. Okay. So where the search button is coming up and you can say all these companies are there and with the help of that you can search it.

You can also do the sorting from A to Z, Z to A, high to low. Okay. So this is all we are going to build using ReactJS, Tailwind CSS, Redux and also use some state management and with the help of Tailwind CSS we are going to create a fully responsive website. Now let's get started. So guys, let us open our chat GPT and write a prompt.

So guys, the first prompt that I want to create is that say I am creating a tutorial or you can say I'm creating a website using ReactJS and it's an ecommerce website. So we'll with the help of or by using vit okay tailwind CSS and redux. Now the next thing what I have to tell the GPT that can you help me outline the agenda of this project like including the main features and functionality.

Okay. so this is the first thing and you can see all over here. So it says like install and configure wheat integrate tailwind CSS okay setup redux then for the UIUX design it's saying responsive layout okay then you have to do theme and styling define a consistent theme okay now for authentication and user management you have to do user registration and login okay for product management like product listing product

details and product search so we have seen these functionalities that we have implemented in our website And also you can see there's a shopping cart and checkout option. You have to add to the cart, cart management and check out process. For the order management, you can also track the history.

And we are not going to target these two things. Okay. These are some of the additional features. But till here we are going to complete it. Okay. So not to complicate too much just to give you a brief outline how you can make a website using chat GPT and Tailwind CSS and other tools and dependencies. So like we have got the brief idea regarding the main features and functionalities.

So we have responsive design, user authentication, product catalog, shopping cart, checkout process, order management, admin panel, wish list, reviews and rating, notification and performance optimization. Now let us write the next part. So here I'm going to type my next prompt that will be can you help me explain the folder structure or can you help me with the folder structure of this React e-commerce app and also let me know the purpose of each folder.

So let us type this prompt and uh we are getting some brief idea like how our uh things would be there. So it says like you can see the folder structure. First we have to create a folder called e-commerce app. Okay. And then so guys for this I will have an assets folder. Okay.

Then we are going to create a components folder. Inside the components there are going to be various components. Okay. So let us start with it and let us first set up our development environment. So guys, I'm using VS code as our text editor and in that let us see that you have first all the dependencies installed like node especially with the help of node we are going to install the package for creating a react app.

So guys uh you can also type all over here that uh how to set up my development environment. And if you type this command you can see so there are certain prerequisites it's going to tell like node and npm code editor visual code. So like you have to create a certain thing like this npm create v latest my e-commerce app then template and then you can go in this app.

So since I have already created an e-commerce app. So what I'm going to do uh inside this I'm going to create this and uh let us copy this all over here. So you can see assets I have already created and uh let us go to the terminal. Click on view and here's a terminal and here let's type this. Click on yes. So all the necessary dependencies is going to install.

So you can see all over here like what I have to choose it says pract lit cities solids or other. So you you can just select all over here that I want to use a react version. Okay just click on this and let us keep the javascript part all over here and you can see this has already been created. Now the second would be go to our e-commerce app. So this is your folder.

Navigate to this and next type npm install and finally click on okay it's installing the node modules. We are going to wait for some time. So guys as I have asked GPT all over here give me the folder structure for our components. So you can see all over here it has given cart item, card item list, card totals, checkout forms, complex pageination container and it has given error element, feature products.

Then for filtering it has given filter.jx, form checkbox, form input. Okay. Form range, form select, header, hero, load day, navbar, nav links, order list, pagenation container, product container, then product grid is there, product list is there, then section title and you can also learn about the process or what do you say the purpose for each component.

Okay, all over here now like product grid what it does. So you can ask GPT the same question with a prompt like what is the purpose of this uh given component? So you can see it is going to tell you all over here. So guys use it excessively wherever you filled out and it has also given the code all over here. Okay.

Now before that since I have told you we'll be using Tailwind CSS. So now let us type the command to set up our Tailwind CSS. So type uh I want to set up the Tailwind CSS in my uh project. Uh help uh uh give me the can type. Okay. So if you type this so you it is going to see that first you have to install the Tailwind CSS and its dependencies.

So guys for the same purpose uh just copy this all over here come to your folder. So now click it all over here that npm install the post CSS auto prefixer. So it is going to start the

downloading part of it. Okay. The next process is in the step two you have to initialize the Tailwind CSS. So copy this and go all over here.

And let us initialize our Tailwind CSS. Pretty fine. What is the next step guys? In the next process, we have to configure our Tailwind CSS. So you can see all over here there is going to be a file called Tailwind config.js and you can configure it all over here. So let us go in our file and you can see tail bitconfig.js would be there.

So okay so now we have done this. Now let us see what is the third thing in the step four. It is saying add the tailwind directive to your CSS file. Okay. So there's going to be a file. Okay. First you have to create a CSS file for your Tailwind styles. Okay. And in that you have to add these directives. Okay.

So guys, this is our index.t CSS file and we can also add the directives all over here. Just click on everything and delete it and just add the required directives. Okay, so I have added this part. Now let us move to the next part. So in your app dot CSS, you have to import it.

Okay, so whenever you are going to open our app dot js file, so here we have app.t jsx. So you can see we are importing this. Okay. All over here. And uh in our main.js file, you can see our index.t CSS is imported. So basically this file is already been imported all over here. Okay. Which is basically going to apply the tailwind CSS directives that we are going to use while building our components.

Now let us build our components one by one. Okay. So you have seen all over here that uh our folder structure was something like this. Okay. So we have cart items, cart item list, cart totals, checkout forms, complex pageination container, error element, featured products, filter, form checkbox and form input, form range header.

So let me create all these folder for you. So we'll go all over here. Now click on the new folder and say components pretty fine. Now here we have to create all the files. So guys you can see all over here I have created all the components in the meantime. So I have card items, card item list, card totals, checkout forms, complex pageination container, error element, featured products, filters, form checkbox, form input and many more.

So the same thing with chart GP has given me I have created all these things. Okay, these are the basic things and uh while also creating a website, I would also recommend you to understand like how these components which are usable throughout the process have been used. Okay, now let us ask Chad GPD to populate this.

Okay, now this thing has been done. Now let us ask for pages like what pages I want to make in our website. So if I navigate to our website we can see we have a homepage we have about page we have a products page we have a card checkout and orders. So let us ask GPD the same. I want to create pages for about home, card, check out error.

Okay. Orders. Then we have the home layout login register single product page. Okay. So like these things you have to identify or the best thing you can do it you can take the screenshot of this. Okay. So take uh like all everything you can just take a picture okay snap it up and send it to chat GBD it is also going to recognize it and let me show you this okay so say I'm going to snip it up so say I want to say I want to create these pages and you can just take an image and just paste it up. So you can see the image is

going to get pasted and you can type a prompt something like this that I want to create pages like this which includes home card checkout orders home layout login register single product page and you can also ask GPT to give a tailwind CSS for these things. So GPT is going to answer you for the same but as a developer you can take this help but don't rely over it too much.

You have to also do little bit of modification by yourself. So this thing I have written and let us click it. So so first it is telling you to install the react router DOM. Okay. So now you

have to install this. So copy this. Okay. So you can see in your package.json file. So this will package.json all over here.

And you can see all over here that we have dependencies all installed which is react router doc. So you can see all over here it's version is 6.24.1. Now the most important part like while using this project. So what dependencies I have used all over here. So I'm going to give a brief idea regarding this. So guys for building this project you are going to require these dependencies like you need to have redux toolkit you need to have tanks tank you need to have react query dev tools okay you need to have an aios you need to have a djs react react dom react icons we'll be using you need to use react redux for state management react router react toify then there are certain deep dependencies as you can see tailwind's typography we have to use because what kind of text we are going to put it up on our web page. So this is going to handle it. Then we are going to use types types react dom then v plug-in

okay this we have it daisy ui okay eslint then we have eslint plug-in react hooks then we are going to use post css tailwind css and v so what we can do next we can copy all these things okay so since I've already copied now let us go to our gpt and say uh give me the command and to install these dependencies.

So you can see all over here it has given me all these commands. Just going to go our uh text editor. So here's our terminal and just type this and you can see all over here it is going to start the process of installing all these packages. Similarly you have to install all the D dependencies. Now the best thing is that uh what are the dependencies required I will share you in the given video and you can install it and also for the div dependencies similarly you have to do it just copy it and go all over here and paste it and you can see all the div dependencies are going to be added in your package dojson file okay since we have used it Now these were the given components and these were some of the pages. Okay. Now

let us create the pages that we have discussed before. and also guys uh you can see assets all over here. You have to transfer this all over here. Okay. Now I'm going to do

this but before that let us create the pages section. So create a new folder called pages all over here. And inside the pages you can see all over here that for pages we have told all about here that they're going to be home layout there going to be error.

jsx they're going to be a checkout and about do.jsx JSX okay all over here and in similarly just create this now for each of the given page you can check all over here that it is giving an idea about it that you can have to ensure something like this and also you can ask GPT to explain you what we are going to use it okay so see here all the headers you're going to add the links like about card checkout or login register so which is basically there on our navigation panel And similarly for applying the CSS just ask GPT to make it something like this. Provide a proper spacing and also you can take a screenshot of this and send it to GPT. So it is going to analyze it and give you the required code for this. Okay. But the most important part is you need to identify what folder structure that you have to use it. And don't worry guys, I will share the required doc file

in the description where you can check what are the components that we have made. And similarly you can take the help of GPT and build this project. So guys I have created all the required pages and you can see all over here that we have about cart checkout error for handling the errors. Okay.

Home layout index.js landing login orders product management error for handling the errors throughout like 404 page. Then we have the landing page. We have the login orders products register. Then we have the single product.jsx. Now these are the required pages that we need to build in order to replicate the website.

Right? So if you have any problem like what this given page does just ask the GPT and it's going to give you the required answer. Now this was building our project structure for the second part. Now let us move to the third part. Now guys the next thing which I want to do all over here is that I want to create a utility folder which is kind of reusable throughout the application and it has these functionalities.

a custom fetch. Okay, which means a preconfigured Axios. Axios basically which is going to have an Axio instance for making standardized API calls and also to the backend server. Then we have the format price functionality in which we are going to use the utility function to format the numerical prices that we are using while buying the given product and also in a USD currency format.

Okay, so it's a basically a dollar format. Okay. And then generate amount options. Then also we are need to create a utility function to generate the list of quantity options for a selected drop-down. Okay. So these were all the features that I've shown you all over here. So when you go from this suppose say on this given product and if you select this say add to the back say I want seven of them and when you go to your given cart so you can see all over here the format pricing and all the utility fun repeat and all the utility functions used all over here are given. So you can see GPT has given. So it has given custom fetch, format price, generate amount option. JS you can create three of them but I will not complicate this. Instead of this since the application is pretty simple I will put all of these things under one folder. So guys as you can see all over

here I have created a folder called utils folder and with a name called index.jsx and inside this I'm going to add all the three functionalities which I have told you now. And this was the project structure that we need to build in order to make an application like that. Now what you can do guys now you can see the functionality of this all over here.

Okay. Now since you'll not get an access to this website what you can also do you can create a demo website like this or you can take a picture of this and send to GPT. It is going to give you the required idea like how you can proceed to build the project. So it is very very helpful but at the end of the day you need to know about react little bit so that you can modify the application based on the given instances.

So guys inside our my e-commerce app you can see our project structure is set up. So inside the src we have the assets. So I have pasted these images. Okay. So if you click on this so it is going to show you these kind of images. So I'll share you the link for the assets. You can use this to create this website.

For the same you have components and all the components like cart items, card items totals, checkout forms, form checkbox, hero, loading, nav links is all been mentioned. So this is for all the components that we'll be needing to build this website. For the required pages, we have about card, checkout, error, home layout, landing, login, orders, product, registers, and single product.

Okay, so these are the required pages. And one, we have the utility folder all over here that we have created called index.js. JS6 for handling these custom functionalities. Now this was the basic idea regarding the project structure. Now what you can do guys since you have got an idea like what components we'll be using. So ask GPT okay now build each of the components by using this prompt.

Uh say now help me uh I want to make the give me the code or you can ask something like this. Uh, give me the code for card component and with the required Tailwind CSS configuration, the Tailwind CSS snippets or you can say with the Tailwind CSS code. Now if you type this all over here so it is going to give you a demo idea.

Okay suppose you have this card JSX and card item.js. So you can see all over here it has started using this CSS which is basically a tailwind CSS and I know it on the first time you're not going to get exactly the same you wanted. So on it what you can do guys that you can take a picture like what the output is coming and send it to GPT and ask it to you have to ask for each of the components. Similarly ask for the pages.

Okay. So ask like something like this. So based on the project structure based on the pages. So guys as you can see all over here. So guys, it has started giving the codes for the respective pages. As you can see all over here and with the respective Tailwind CSS, you can definitely modify it based on your choices and also what you feel like is more responsive and userfriendly for getting an in-depth idea regarding Tailwind CSS.

You can navigate to its official document and it will be very very helpful that you can also take a help and ask GPD the same thing. So after you have populated these pages with the respective codes and components okay that we have shared all over here. Next thing what you can do all over here that just type all over here that this command or you can ask GPT how to run my application give me the command.

You can type this and it is going to give you the command but I know the command that is npm rundev or this is the given command that is will be very helpful for running your application. So at each stage suppose you are building one component. So keep running this and keep seeing like what changes you are applying on a real time.

So building an application using react or any front-end application is a hit and trial. So based on the modifications that you require. So you have to consistently interact with GPT and make an application out of it. But this was an overview of our given thing. So just type this all over here.

So just type this and your application is going to start on a given port. So this is your port 5174. So if you navigate all over here, so your application will be open. So let me show you like suppose the port which I'm using all over here. So it is 5173. Okay. So this will be your application and in this way you can make an application with the help of chat GPD.

So I've given you the basic idea and I will also share the given documents for the components pages so that you could have a brief idea and you can ask the GPT to give the required tailwind CSS for the same. I will also share the assets. You can use it or you can download the assets from various other websites where they have like free pick where you can make the images for the given products like you can say lamb, coffee table, coffee which are easily available and you can start designing all these things.

So guys the best thing about GPT4 that you can also share the image. So with the help of this you can definitely build an application. So let's first understand what is prompt

tuning. So prompt tuning is a method that enhances the adaptability of large language models for no task by fine-tuning a minimal set of prompt parameters.

This technique involves appending a tailored prompt before the input text directing the LLM to generate outputs aligned with specific objectives. Its efficiency and versatility have made it a local or focal point in natural language processing advancements. So let's understand prompt tuning. So prompt tuning is the practice of customizing LLMs for new applications by adjusting a limited subset of parameters known as prompts.

So these prompts placed before the input navigate the LLM toward producing the intended output. And now we talk about the advantages of prompt tuning. So the number one comes is efficiency. So unlike comprehensive model fine-tuning, prompt tuning modifies only a select few parameters enabling quicker task specific model adjustment and then comes the versatility.

So this approach is applicable across a spectrum of task from text generation and analysis in natural language processing to image recognition and automated coding showcasing its adaptability to different domains. And then comes the interpretability. The modifiable prompt parameters allow for an examination of how the model's output generation is influenced offering insights into the model's processing.

So this was about the advantages of prompt tuning. Now we'll move to the key challenges is prompt tuning and we will write some prompts and see how we can prompt tune the chat JP model and we can generate the outputs we want. So let's move to chat GPD and here first we will draft or craft effective prompts.

So crafting prompts that effectively guide the model without being overly complex is challenging. So to avoid them we will write some prompts and let's see. So we will take a fresh example for crafting precise prompts. So let's consider a scenario and consider you are part of a financial analysis firm specializing in renewable energy sectors like solar and wind power.

You aim to leverage a language model for extracting market trends, investment opportunities and regulatory impacts specific to these sectors. However, you wish to exclude general energy market data or information about fossil fuels and non-renewable sources as well as omit any direct mentions of rival firms or their investment products.

So, what challenge you will face here is your objective is to direct the AI model to furnish targeted insights on renewable energy sectors while filtering out nonrelevant information and avoiding any specific mentions of competitors and their offerings. So for example uh we can write a prompt that give me all details on renewable energy sectors including solar and wind power but skip all info on other energy types and ignore data about companies like you could name the companies like eco power ventures and their solar boost investment fund. So this is the ineffective prompt. So this prompt is ineffective because it's overly detailed and instructs the model on what to exclude in a way that could be confusing potentially leading to an over focus on the exclusions rather than the desired content. So if we talk about the optimized prompt what you can write is

provide an analysis of current trends investment opportunities and the impact of regulations in the renewable energy sectors specifically focusing on solar and wind power. exclude general energy market trends and specific investment products. So this refined prompt clearly directs the model towards the needed information that is market trends, opportunities and regulatory impacts within solar and wind energy without cluttering the request with too many exclusions.

It certainly implies the need for exclusively by focusing on specifically focusing which naturally guides the AI to omit unrelated sectors and competitor products facilitating a more focused and relevant output. So now let's see this example and this is the inefficient prompt. So we'll copy this and paste it to chat GPT and let's see what output it will provide to us.

So you could see that it has provided us the output. But this prompt is ineffective because it's overly detailed and instructs the model on what to exclude in a way that could be confusing. Now we will write the correct prompt as that is the optimized prompt what I've told you that is provide an analysis of current trends investment and opportunities and the impact of regulations in the renewable energy sectors specifically focusing on solar and wind power and exclude general energy market trends and specific investment products. So this is the optimized prompt and it will provide a good result and a good output. So you could see that it has started with the current trends only. You could see the output and this is the

desired output what we need and the charge jupy has proed with the optimized prompt. Now we move to the next challenge that is avoiding overfitting. So there's a risk of the model becoming too specialized to the training prompts reducing its ability to generalize. So strategy for this broad application is to prevent over specialization prompts that should be designed to be broad enough to encompass the range of related queries enhancing the model's versatility.

So for this also we will draft an optimized and unoptimized prompt. So let's think of a scenario that you are developing an educational platform that uses a language model to generate study materials for high school biology covering topics from cell biology to ecology. Initially you fine-tune the model with prompts focused on very specific areas like photosynthesis in plants.

So initial prompt we can write is explain the process of photosynthesis in plants focusing on chlorophyll's role and light absorption. So while the model excels at generating detailed content on photosynthesis, it struggles when asked about broader or slightly related topics such as cellular respiration or plant adoptions to different environments.

So the issue we encounter here is when students query how do plants adapt to low light conditions. So the model will fail to provide comprehensive answers because it's been too narrowly trained on photosynthesis specifics. So the solution for this overfitting

challenge is to prevent overfitting and enhance the model's ability to generalize across the broader subject of biology.

You should utilize more inclusive prompts that cover a range of topics within the field. So we have the revised prompt here that is provide an overview of plant biology including key processes like photosynthesis, cellular respiration and adaptions to environmental factors. So this broader approach encourages the model to generate content that's relevant across various biology topics improving its utility for educational purposes.

So this is the prompt and we'll ask JBD that is pro of the plant biology including key processes like photosynthesis, cellular respiration and adaptions to environmental conditions. So this will improve its utility for educational purposes. So let's move to the next concern that would be scalability concern and you could see that it has provided the desired output for the photosynthesis

and the environmental factors for plant biology and as we talk about the scalability concern. So as the training data volume grows maintaining the efficiency of prompt tuning becomes more challenging necessitating innovative solutions to streamline the adaption process. So now we'll talk about its prompt also.

So scalability that is implementing AI for diverse customer service inquiries. So let's consider a scenario. Your company aims to deploy an AI chatbot to handle customer service inquiries across different departments from technical support to billing and account management. Initially the chatbot is trained with specific prompts for each category performing well in small case test.

So what challenge will you face with this prompt is or with this scenario is as you scale the chatboard to handle thousands of inquiries daily across a wide range of topics you find that the initial approach of using highly specific prompts for each category is unsustainable. So the chatbot struggles to adapt the vast and varied nature of customer queries and we'll face the scaling issue also here that is the specific prompts used during

initial training don't cover the breadth of potential customer inquiries leading to inadequate responses in unanticipated scenarios. So what strategy we can apply for scalability is instead of relying on narrow specific prompts for each inquiry type your transition to using broader more adaptable prompts that can guide the AI in understanding and categorizing a wide range of customer questions. So we can have an adapted prompt example and I have one that is or we can write here that would be identify the category of the customers inquiry that could be

technical billing account management and provide relevant assistance based on the identified category. So this strategy allows the chatbot to more effectively process a variety of inquiries by understanding the context and categorizing questions before generating responses making it more scalable and versatile in handling customer service task.

So if we address prompt tuning challenges, it is simplifying complex prompts that is begin by clearly identifying the core information or action desired from the LLM. Breaking down multifaceted prompts into simpler, more focused request employ straightforward language and eliminate non-essential elements to clarify the prompt.

And then we have utilizing soft prompts. So soft rooms which are non-extual signals trained from examples guide the LLM without explicit worded instructions. So these are particularly used for task with limited training data or where interpretability is less critical. And then we have innovative optimization techniques. So the development of parameter efficient fine-tuning techniques such as p-tuning optimizes the search for effective prompts in a continuous space making it feasible to tailor LLMs to specific task without extensive retraining. So this was all about the prompt tuning. Hello everyone and welcome to the tutorial on prompt library for all use cases at simply. The prompt library is a comprehensive toolkit for mastering my ad use cases with ease. Whether you are delving into programming, honing creative writing skills, or exploring data analysis, this library offers a versatile array of prompts tailored to

your needs. Now, before you move on and learn more about it, I request you guys that do not forget to hit the subscribe button and click the bell icon. Now, here's the agenda for our today's session. So, guys, we are going to start with first understanding the prompt structure.

Moving ahead, we are going to understand testing and iterating. Then we are going to explore the prompt examples and at the end we are going to conclude our sessions with utilizing prompt libraries and resources. So guys in today's video we will be exploring the prompt structure for various use cases. Now first let us try to understand the prompt structure.

So guys I'll break down the prom structure. So here first we have the action verbs. So guys, think of action verbs like a boss telling Chad GPT what to do. It's like giving Chad GBT a job. So for example, if I say write, you are telling Chad GBT to put words on the page. For example, if I say write a story, I'm telling Chad GPT, hey, I want to you to make up a story for me.

So this is something like this. Now let us ask Chad GPT. Hey, so write is your action verb all over here. So this is the first prompt structure that I would like you to apply. Now the second one you could give a theme or a topic about. Now if you say just write a story, chat GPT is going to give any random story. So we won't want that.

The next thing that we cover basically is topic or theme. So what theme or topic you are looking about? This is a part where you are giving charge GPT a subject to talk about. Imagine you're telling a friend let's talk about cats. So cats are the given topic. So if I say write about your favorite food, I am telling chat GPT tell me about your favorite food.

So you have to always include a topic or theme along with your action verb. So here I can include some certain thing like this that write a story about food. So you could see all over here chat GPT has given two uh responses. This is response one and this is

response two. Now the third thing that comes up all over here is constraints or limitations.

Think of constraints as a rules or boundaries for Chad Cupid to follow. It's like saying you can talk about cats but only in three sentences. So if I say write a poem in 20 words, it's like I'm telling Chad GBT make a short poem using only 20 words. So this is one of the things that you have to always keep in consideration regarding what task you want to give.

So always include constraints or limitations. Fourth one is background or information context. So, so this is also one of the most important parameters. Uh what exactly it means is like this part sets the scene for JGBT like giving it a background story. Imagine you are telling someone about a movie before they watch it.

So if I say imagine you are on a spaceship, I'm telling Chad GPT pretend like you are flying through the space. So this is also very very important for you to consider to give certain idea regarding your background or information. Now the fifth one is conflict or challenge. Guys, this adds some spices to the prompt.

It's like a puzzle for a problem for charge to solve. It's like saying talk about cats but tell me why some people don't like them. So if I say charp explain why reading is important but you can't use the word book. I am challenging charging. So this is where conflict or challenge you have to give to charg. Now example let us take one example on this.

So for example, if I say the action verb as right, we'll highlight this with red and the topic or theme could be like your favorite vacation. If I talk about a background or context like say you are on a beach with your friends or conflict or challenge we can give all over here something like in just 50 words.

So guys this is certain thing to follow while giving a prompt to charge GPD. So in this way putting all together you could combine all these three things and form a sentence and this prompt is going to be very very effective to solve the problem of generic responses. Now with this simple example you can see how different components come together to create engaging prompts for chbd to work with.

So guys, whenever you are giving a prompt, I would request you to always follow this structure. So it's going to create a map for you to get a more precise answer. Now let's take example and elaborate the prompt library with examples to make it more understandable. So guys, let's take another example of text classification. So for text classification, we'll take the action verb as classify and our text type would be product review.

Example could be classify the following text as negative, positive or neutral sentiment and after that you could give like the product review exceeded my expectation. So if you give certain thing like this you would say this is a positive sentence. So making your prompts in this manner with a proper structure you are going to get a very particular response which fits what you need.

So always remember this structure whenever you are framing any bra. Now let's move to the second part that is testing and validation. Guys testing and iterating are essential steps in refining prompts and ensuring optimum performance from chat GBD. Let us break down this process. The first process is prompt validation.

So before using a prompt, it's crucial to test to ensure it that it generates a desired response accurately. Then you evaluate the output. You're going to generate responses using the prompt and evaluate the quality, relevance and coherence of the output. Third, check for errors. Look out for any errors, inconsistencies or unexpected behavior in the generated responses. Compare against expectations.

Compare the generated responses against your expectation or any requirements to verify that they meet your desired criteria. The fifth one is solicit feedback. Seek feedback from peers, colleagues or domain experts to validate the effectiveness of the prompt. For example, like analyzing the results.

So you would say analyze the results to testing to identify areas of improvement or refining the prompt. Next is modifying the prompt. Based on the analysis, make the adjustment to the prompt structure. Next, then fine-tune the parameters. Experiment with different variations of the prompt such as adjusting constraints, changing topics, or refining prompt to assess whether the changes have resulted in improvements in the quality of the generated responses.

The fourth one is retesting. Test the modified prompt again to assess whether the changes have resulted in improvements in the quality of the generated responses or not. And the final step is iterate as needed. Iterate the testing and modification process as needed until you achieve the desired outcomes and generate high quality responses consistently.

So this structure you have to always follow when you are iterating. So I'll give you an example. So like we have given a initial prompt as write a product description for a new smartphone and I would say include details about features, specifications and benefits and I would say add a constraint all over here that keep the response in 100 words.

So this is your initial prompt which you are given. Now for testing the next comes is testing. Generate product descriptions using the initial prompt. Evaluate the quality and relevance of the generated descriptions. Check for errors if inconsistencies or missing information is there. Compare the description against the expectations and requirement.

So this process comes under testing. Okay. So give it uh like change your prompt a little bit. give a specific uh description regarding a certain product and you would ask that and just next process would be evaluate the quality and the relevance like what you are

uh getting as a response check for errors like go to Google like see if it's same is coming up then what's the customer expectations regarding that product so if the overall structure is like technical structure is maintained so this gives the first phase of testing next one comes the analysis some descriptions s lack detail and fail to highlight its key features. Okay. So in this scenario the descriptions vary in length and structure leading to kind of inconsistencies. Certain descriptions like here will focus more on technical specifications than the user benefits. So overall the quality and the coherence of the descriptions needs improvement. So you have to take all these parameter and you

have to reframe your prompts. Okay. Then next comes is iteration. You have to modify this prompt to provide like more often to give a clear instructions and emphasize the user benefits. Write a captivating product descriptions for a new smartphone. Okay. Then move to retesting.

Generate product descriptions using the modified prompt. And for the outcome, you would say that the revised prompt should yield more compelling and informative product descriptions. So this is how you have to do iterate continuously to get the proper response like which you would be needing. Okay guys, now let's move to the final part of this video that is utilizing the prompt libraries.

Guys, utilizing prompt libraries and resources is essential for streamlining the prompt writing process and access a wide range of pre-esigned prompts for various use cases. So you're going to get a library of a predefined prompts. Okay. So there's one website like which I want to show you. This is called Anthropic.

So Anthropic has recently released a prompt library. So guys, they have given a wide data of a prompt library. So if you just click on this, so you're going to get like what are the effective prompts in all these domains. So give it a shot. Uh try to see what are the uh like resources you're going to get all over here.

It definitely is going to fine-tune your responses. Now let's move to the process. So when we are talking about the prompt libraries, the first step is explore the existing libraries. So you can see that I have given a reference to a prompt library all over here which is released by anthropics team for cloud and also workable for chat GP.

Next is you have to understand the available prompts. Familiarize yourself with the prompts available in this library and including their structures, topics and constraints. You have to also analyze how prompts are categorized and organized within the library to quickly locate relevant prompts for your needs.

Third is adapt to prompts to your needs. Customize existing prompts to suit your specific objectives, audience and use cases. You can modify prompts by adjusting the action verbs, topics, constraints or background information which aligns with your requirement. Create your own prompts like combine different components such as action verbs, topics, constraints to craft prompts that addresses specific tasks or challenges.

Next process you have to do is sharing and collaborating. You will share your prompts with the community to contribute to the collective pool and resources. So this is one way of learning that I really really want you to follow. Now you have to keep experimenting and iterating at the same time. And finally you have to see the documents and organize all your prompts for the scene.

So what you can do best is see all the existing prompt libraries like I'll show you one more. So prompt library for chat GPT GitHub for all use cases. So you could see explore various repositories on GitHub like what are the uh kind of like prompts available like this repo specifically focuses for the academic writing.

So just visit this uh repository and uh you could see they have given a lot of thing like for brainstorming they say so you can see the action verbs all over here try to like uh uh try this prompt and see how you are getting a response. Then for article sections like what's

it's there. So you're going to get a lot of like things and uh more of the experiment and more uh you are exploring the more idea you are going to get regarding this.

So my advice would be just explore as much as libraries you can and depending upon your use cases you have to make an organized prompt structure. So following this format which I've told you follows the action verb the topic or the background information. Then what are the constraints you have to give? Okay, it's any particular theme is there.

You have to include all those things and use the existing prompt library also. So you can refine your uh prompt and always to get a good response. It's my personal experience that you have to keep fine-tuning, keep testing, iterating, analyzing so that your result comes very fine. >> Imagine this, it's Monday morning and your inbox is flooded with customer emails.

Your sales team is juggling follow-ups and your social media calendar is overflowing. Your support calls keep piling up. You know that automation saves ass. But writing code or hiring developers feels overwhelming. Now picture this instead. You open a platform where automation is as simple as connecting apps, dragging and dropping workflows or customizing templates.

Without writing a single line of code, you set up email responses, schedule meetings, create social media posts, and even build voice assistants that talk to customer in real time. This is where no code AI tools step in. Turning complex process into easy, efficient solutions anyone can use. From customer support to sales, content creation, and call centers, these tools are helping teams work smartly, faster, and more efficient than ever before.

In this video, we'll explore the best no code AI agents by workflow. Tools that empower you to automate task effortlessly, boost productivity, and focus on what really matters. Before we dive in, here's a quick question for you. What's that one task that eats up most of your time every day? Is it responding to endless emails, following up with leads

or clients, managing social media and content? Or is it handling customer support? Do let us know your answer in the comment section below. If you interested in more such content, do not forget to like, share, and subscribe to Simply Learn. Now, let's bring this to life with a story. Meet Ravi. Every morning he sits at his desk staring at a screen full of emails and to-do list. The workload feels overwhelming and lack of automation only makes things harder.

Important opportunities slip through his fingers because he simply doesn't have the time to follow up. He finds himself constantly reaching out to experts just to keep things running. Slowly the stress builds up affecting both his productivity and his peace of mind. This is a story of many of us known all too well.

But what if there was a way out of the cycle? If Ravi didn't have to feel so overwhelmed every single day, Ravi discovered that by the strategical automation just a few of his routine task, he could finally begin to simplify his dayto-day and significantly reduce the pressure on himself. Freeing his time, allowing him to shift focus from tedious repetitive work to bigger picture.

he could finally dedicate his energy to growing his business and making truly impactful smarter decision. This is the core idea behind today's concept. As a powerful quote at the bottom says, automation isn't just about doing more. It's about making room for what truly matters. It's about creating space for creativity, for strategy, and for the things that you can only do with the right tools.

The kind of transformation is not just a dream. It's entirely possible. With automation in place, Ravi's entire workflow became more effective and purposeful. He went from feeling overwhelmed to feeling empowered. He felt an indicated acceleration as projects that once took days could now be completed in a fraction of the time.

The new found speed gave him the new confident to move forward with seconds guessing every step. It also led to a new level of efficiency. He could track progress and

spot opportunities he would have otherwise missed. Thanks to the automation insights he was now receiving. Best of all, this transformation didn't just impact him.

It bought the team collaboration together. They began to collaborate better. trusting the new process and focusing on shared goals instead of getting boggled down in manual work. This is what it means to be truly empowered by technology. It's not about working hard, it's about working smarter.

Every great journey starts with a single step. And this is where you begin. Look around and find the one task that takes up most of your time and energy. That's the one which is constantly slowing you down. Next is to automate. Find a simple no code tool that automates just that one task.

Remember, even a small change can make a huge difference in your day-to-day work life. Use the new freed up time to concentrate on what truly matters. Growing your business, thinking clearly, and building momentum. This is how you turn overwhelm into progress. One step at a time. It's simple process but can completely transform the way you work.

This journey about automation isn't just about solving a problem today. It's about unlocking a long-term success. By offering routine task, you will feel less stress like the person meditating here. Your mind will be clearer and you will have space to think strategically. When everyone on your team is free from repetitive task, they can work together, create more and also e and effectively.

And finally, it's a long-term focus. You will be able to set and achieve bigger goals knowing that small daily tasks are taken care of automatically. This is where the real power of no code AI comes in. It transform your workflow so that you can build a more sustainable and successful future for your business.

Now that you have seen power of automation, your next move is simple. Firstly, what are the tasks that currently slowing you down every day? Think about it. Think about things you do over and over again that feels repetitive. And second is to look at those task and ask what can you automate? And second is to look at those task and ask which of these can I automate? Are there simple tools that can handle these workflows? And finally, don't feel like you have to do everything at once.

Start with just one small single chain and build from there. Pick one task and one tool to begin. Remember, the key here is to empower yourself, not just to move, but to truly focus on what truly matters. Let's dive into powerful tools that can help you transform your workflow. At the center of your workflow, there are different areas where automation can make a huge impact.

For example, customer support tools that in this category helps you automate email responses and handles routine customer queries. So you spend less time on repetitive communication in sales and CRM. These are the tools streamlining leading tracking and follow-ups making sure you never miss an opportunity and are nurtured to relationship effortlessly.

With content creation tools, you can automate social media posts and campaigns, keeping you blend actively without the daily hassle of planning and scheduling. Workflow integration tools let you connect apps and process easily, syncing information across all your tools to reduce manual work and increase accuracy.

Finally, voice agents give you the power to create AI powered call assistance that handle customer support, inquiries, appointments, and instantly improving services and freeing up your time. Together, these tools complete ecosystem that help you shift your focus from repetitive task to strategic thinking, creativity, and growth.

Now, let's start with a versatile tool called Lindy AI. Lindy is a fantastic example of a non-code AI agent that can act as your personal assistance, particularly in customer

support and sales. Lindi can automate customer emails, drafting replies and tracking your inbox to handle routine communication. It also seamlessly handles meetings, scheduling, and follow-ups by automatically joining calls, transcribing them, and updating your CRM with key insights and action items.

This frees up your sales team to focus on building relationships and closing deals, not the manual data entry. Next we have a tool that is a cornstone for no core automation that is Zapier. Think of Zapier as a central nervous system for all your applications. Allows you to automate workflows by connecting thousands of different apps.

It even provides AI powered solutions to help you build your workflows faster and more efficiently. By automating these connections, Zapia saves time to boost productivity across the entire team. For example, you can create Zap that automatically adds a new lead from a social media to your CRM, then sends them a personalized welcome email and finally notifies your sales team in Slack.

All this in one single automated sequence. Similar to Zap year but with a more visual approach is make.com. Think of make as your digital playground for building automation. Make is all about visual workflow automation. Make is all about visual workflow automation. Instead of just a list of steps, you can see your entire workflow as a map, which makes it much easy to understand and build complex sequences.

It also has a strong AI integration, allowing you to incorporate powerful AI capabilities directly into your workflow. And of course, just not other tools and category. It connects multiple apps so you can seamlessly transfer information and trigger actions across all your favorite platforms. Make is perfect for everyone who wants to see their entire automation process laid out visually before they even build it.

For those who want more control, we have Node N. This is an open-source no code and low code platform that gives you power to automate almost anything. What makes node stand out is its opensource nature which allows you immensely flexible scalability and

while offering extensive app integration to connect with your entire tech stack. Finally, let's talk about future of customer interaction with synth flow.

This tool is great example of no code AI platform. Synth allows you to create incredible realistic humanlike AI assistance that can handle both inbound and outbound phone calls. These agents are designed to have natural humanlike conversations and offers seamless integration and scalability to grow with your business.

It's a powerful way to automate your call center or sales team and provide a great customer support. All these platforms offer powerful automation capabilities without needing technical knowledge. Whether you're handling customer queries, tracking emails, or managing team task, these tools can save you every week. Start exploring them today using the free plans and trials.

So if you want to look at Lindy AI, this is the interface you'll be getting. At first you need to enter basic details of your Google account or you can use any kind of email. So there is an option here to try it for free. Let's click on that. And we have a call if you want to book and know about Lindy.

And there is a free trial that you can take and decide if you want to use Lindy AI. So this is the interface you'll be getting. And you can see we can create personalized website, customer support, email, outbound calls, lead genen, meeting recording and also we can use lending car detachments here. And we can also create any kind of agent from scratch using new agent.

So let's see one of the pre-built agents. Let's say customer support email. You just give it a prompt. So after some time you will see an option here called build agent which is already ready. You can just click on it. And here you get an option of selecting what kind of triggers should be there for you to start this process.

So either it can be any kind of Gmail or any Google calendar activities. Also there are options of performing an action search knowledge base entering a loop condition and enter agent step. So you can build an entire agent here based on these options and after that you can test and turn it on.

So it get connected to your Gmail account and helps you automate the task. So coming to Zap year when you open the website you will see what are the current plans. If you want it for a free trial there is a current plan which is available for free. There is a professional one too and you can just click on next.

Your basic Gmail details would be required. So when you log in with your Gmail, here are the few questions they'll ask to personalize your entire interface. So you can go ahead and fill them. So this is the main interface of Zapier and there are options of automating tables, interfaces, chatboards and canvas. So you can explore.

There are few templates also which you can use it. So if you go to the interface option, you'll have a create option here. So you can just click on it and you can either start from scratch or export it from a any kind of template and then you can click on create interface. So from here you can build the interface from scratch where the pages are different and communicative.

What used to take hours of work together now you can just start within a 2 to 3 minutes and you'll have an entire interface for you to run. So this is the main interface where you can create multiple agents and you can start from scratch or again export it from a template given below. So when you click on create scenario it means a agent and you can add multiple apps to connect them easily.

So you have Google sheets flow control openai also inbuilt. So you just need to click on them, fill the necessity details and connect one app after the other. You can do things like replying for each comment without even putting in efforts. You have other options at what interval this command should run or agent should run. Demo four.

So for node N which is an agentic AI builder you have different kinds of plans which includes a starter pro and enterprises and it is not free to use you can trial it for free but then you have to pay initially demo file. So the last demo is synth AI which automatically answers your calls and reply to any kind of customer service which is required.

So here you can see an option called create an agent. So you can start and you can either browse from templates or again start from scratch. So you can see many other options which include lead qualification seller utility scout property reconnector. So let's go with a seller call which is lead qualification and here you can see you can give a particular name for an agent and you can give a image also so that it displays you can use AI models which includes chat GPT 4.

01 01 mini and also five which is the latest version and you can fix the time zones which is e you can set it to either Indian European based on your necessity you can add your knowledge bases you can choose what kind of customeral vocabulary you should be using and filter the words and then we have use realistic filler words like you can use words like m and a also it feels like a human so you can just configure this to your email or even calls and then you will have a voice assistant without even receiving one single call. You'll be able to get customers and even provide customer support. And with that, we have come to the end of our full course. Now, if you have any doubts or questions, ask them in the comment section below. Our team of experts will reply to you as soon as possible. Thank you and keep learning with Simply Learn.